

# Mass Exit in the Market for Governance

## A Behavioral Foundation for Theories of State Development\*

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ABSTRACT. State development theories posit that political fragmentation lets subjects exit to other authorities, spurring jurisdictional competition that drives institutional development, yet behavioral evidence remains scarce. I develop a theory of exit under horizontal and vertical authority and test it in Tokugawa Japan's distinctive setting. Across nearly 300 domains, villages sought concessions from home rulers by fleeing en masse or appealing through neighboring lords or to the central authority. I exploit domain conversions to central rule, which brought central authority closer to neighboring villages, as a natural experiment. Difference-in-differences estimates a 3.8-percentage-point increase after conversions in the probability of exit among nearby villages under other lords. Grid-year panel data show proximity to another ruler's capital predicted more exit but not confrontation. Proximity to neighboring capitals predicted lower reported tax rates where neighboring domains could absorb more migrants. These findings suggest fragmentation helps translate overlapping jurisdictions into popular checks on elite power.

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# 1. INTRODUCTION

Many influential theories of state development rest on a common premise: political fragmentation gives subjects access to alternative authorities, and that access disciplines extraction and shapes institutions. This premise carries the central causal weight of leading explanations for the Great Divergence (Cox 2017; Fernández-Villaverde et al. 2023; Jones 2003; Mokyr 2017); for accounts linking Europe’s polycentric structure to constrained taxation (Abramson 2017; Dincecco and Wang 2018; Sng and Moriguchi 2014; Stasavage 2016); for fiscal-federalism arguments that competing jurisdictions tame Leviathan (Qian and Weingast 1997; Rodden 2003; Tiebout 1956; Weingast 1995); and for recent theories arguing that the modern territorial state system cartelized the *market for governance* by restricting subject mobility (Acharya and Lee 2018; 2023). Each case shares the same basic mechanism: fragmentation offers subjects credible exit options, and the threat or use of exit forces rulers to compete or cooperate. Yet the behavioral foundation of this chain has yet to be directly demonstrated: whether access to authority outside the home jurisdiction shaped how subjects resisted. I test this link.

Mass exit has served as a form of resistance across many historical contexts. In medieval Europe, migration to towns offered serfs a way to resist their lords (Adas 1981, 217). In precolonial Java and Burma, cultivators escaped excessive demands by transferring their allegiance to rival patrons (Adas 1981, 229–231). Gandhi described an Indian village whose inhabitants began leaving in protest against their prince, prompting him to withdraw his command (Gandhi and Parel 1997, 95). In Tokugawa Japan, villages fled collectively into neighboring domains and petitioned neighboring rulers for permission to remain (Moore 1993, 255).

A common setting for such resistance combined horizontal divisions among neighboring local rulers with a vertical hierarchy under a common superior, as in Languedoc’s lordships and towns under the early modern French crown (Gerring et al. 2011, 405) and county jurisdictions within Qing China’s provincial and imperial hierarchy (Sng and Moriguchi 2014, 446–447).<sup>[3]</sup> I develop a theory of mass exit in such environments.

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<sup>[3]</sup>Comparable arrangements appeared in territories of the Holy Roman Empire under the emperor (Cantoni, Mohr, and Weigand 2024), princely states under the British Raj (Iyer 2010), and emirates

Subjects can move themselves or their claims beyond the home ruler’s exclusive control by seeking protection from a neighboring ruler or intervention by the common superior. The routes can overlap when a neighboring authority shelters subjects or mediates a dispute that reaches the superior. Easier access to another authority should lead more communities to choose exit. When neighboring jurisdictions offer sufficiently attractive livelihoods, rulers should lower taxes to retain cultivators. These tax concessions should reduce, but not eliminate, the increase in exit.<sup>[4]</sup>

Tokugawa Japan (1615–1868) combined these two dimensions of authority within a common political order. Nearly 300 feudal domains exercised local autonomy without sovereignty, each with its own legal code, tax system, and border controls, under a central authority, the Tokugawa shogunate. The shogunate retained supra-domain arbitration, attainer power, direct peasant-appeal authority, and power to assume direct rule (Moore 1993; Ravina 1999; White 1995). I use *exit through alternative authority* for collective flight into a neighboring domain (*chōsan*), a petition routed through a neighboring lord, or an appeal over the home lord’s head to the shogunate (*osso*). These routes created different forms of leverage, but each moved villagers or their claim beyond the home lord’s exclusive control.<sup>[5]</sup>

This setting helps overcome two obstacles to testing the argument. First, open rebellion leaves abundant documentary traces, whereas exit is often subtle and covert. Limited records have therefore led most studies to infer exit indirectly rather than observe it at the community level (Bailey 2014, 76). Newly digitized records of peasant resistance compiled by Aoki (1975) document both direct confrontation and exit through alternative authority at the village level across the entire Tokugawa period. Second, comparisons between entire regions, such as Europe and China, conflate differences along

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of northern Nigeria under colonial supervision (Gerring et al. 2011, 399). These settings varied in local autonomy and subjects’ access to other authorities.

<sup>[4]</sup>I show these results formally through a model of tax competition in [Supporting Information §A](#).

<sup>[5]</sup>Tokugawa mass exit operated as active resistance rather than permanent relocation: peasants presented demands to home-domain officials while seeking permission from neighboring rulers to cross the border, with some documented cases recording return once concessions were granted. Moore (1993, p. 255), drawing on Borton’s catalog, recorded 106 such mass desertions across the Tokugawa period as evidence of “the solidarity of the Japanese village”; my catalog substantially extends this count and covers the full archipelago.

many dimensions and provide little within-case variation for examining subjects' choices. Within Tokugawa's common political order, villages varied widely in access to rival rulers, while intense inter-domain competition for agricultural labor made mass migration a credible strategy. The post-1615 peace also sharply weakened the recurrent war-finance pressures that [Rosenthal and Wong \(2011\)](#) identify as overwhelming the exit-discipline mechanism in early modern Europe, reducing a major competing pressure on extraction ([Moore 1993](#)).

The empirical analysis begins with two behavioral tests and then examines a downstream fiscal trace. First, I employ a difference-in-differences analysis exploiting domain conversions to direct shogunal rule (*tenryo*), which installed a shogunal intendant (*daikan*) at the former capital and expanded surrounding villages' access to the shogunate. After direct rule began, nearby villages governed by other lords recorded 3.8 percentage points more exit but no detectable change in direct confrontation, and event studies show no differential pre-trends. Because episode destinations are unobserved, I interpret this pattern as evidence consistent with improved access to outside authority, not as a definitive stand-alone causal estimate. A grid-year panel tracking changes as nearby capitals were established, abolished, or relocated finds the same pattern: easier access to another active capital predicts more exit but no more direct confrontation.

Second, among villages that resisted, greater geographic exposure was associated with choosing exit over direct confrontation. I measure each village's exposure as the ratio of its distance to the home-domain capital to its distance to the nearest active non-home capital. This ratio is a spatial proxy for access to alternative authority, not a direct measure of competition, bargaining, or another ruler's willingness to receive migrants. I use *governance conflict* for the condition in which villages under one ruler can credibly move themselves or their claims under another. Its two components are access to alternative authority and an outside option worth the costs of invoking it. Geographic exposure proxies for access; in the fiscal analysis, neighboring domains' capacity to absorb labor captures the attractiveness of migration as an outside option. Across all recorded resistance, this exit-*share* result holds robustly under home-neighbor dyad fixed effects. Decomposing the ratio into its two absolute distances shows that foreign-capital

proximity is the larger and more robust component. Home-capital remoteness is positive but less precise under spatial inference.

As a downstream fiscal trace, a late-period domain cross-section associates greater geographic exposure with lower reported tax rates, especially where neighboring domains had greater capacity to absorb labor.

The paper makes three core contributions. First, it provides direct, micro-level evidence for a central behavioral premise running through four traditions in political economy, each resting on the same link between access to outside authority and exit: the Great Divergence argument that ruler competition for mobile subjects checked extraction and fueled European growth (Jones 2003; Mokyr 2017); state-formation accounts in which exit forced ruler bargains and built representation (Dincecco and Wang 2018; Stasavage 2016); fiscal-federalism claims that competition among jurisdictions tames Leviathan through taxpayer and labor mobility (Rodden 2003; Tiebout 1956; Weingast 1995); and the cartel theory of the territorial state, in which modern borders are erected specifically to suppress subject exit (Acharya and Lee 2018; 2023). The evidence links better access to outside authority with communities' use of exit, supplying behavioral evidence for the first link in this shared causal chain.

Second, the paper brings exit from the background to the center of the study of state development. In Hirschman's (1970) terms, the dominant analytical traditions are built around voice: revolution and rebellion drive regime change (Acemoglu and Robinson 2001; Skocpol et al. 1982), collective mobilization makes states (Hintze 1975; Tilly 1978), and the threat of revolt disciplines extraction (Besley and Persson 2009; Fearon 2011). When exit appears, it shapes incentives (Besley and Persson 2009; Fearon 2011) or long-run sorting (Jones 2003) without being directly observed. Historians of everyday resistance have long documented otherwise: exit often predominated over revolt (Adas 1981; Bailey 2014; Camp 2004; Diouf 2014; Kolchin 1990; Scott 1976; 1985; Sreenivasan 2004). This paper documents exit directly as a strategy of resistance alternative to open rebellion.

Third, the paper develops a general account of jurisdictional competition in fragmented but hierarchically ordered polities. Existing work tends to isolate either horizontal mobility or vertical oversight. Studies of horizontal exit emphasize individual preference-

sorting (Peterson 1992; Tiebout 1956), flight beyond state reach (Carneiro 1970; Herbst 2000; Pruetz 2024; Scott 2009), or indirect evidence from jurisdictional havens, movement restrictions, and tax-base responses (Guardado and Franco-Vivanco 2025; Kleven et al. 2020; Matranga and Natkhov 2025). Other strands show how exit can spur voice (Hirschman 1993; Pfaff and Kim 2003; Schoppa 2022) or how governments induce emigration (Lueders 2025; Michel, Miller, and Peters 2023). This paper identifies how horizontal and vertical authority interact: another jurisdiction can shelter subjects and threaten the home ruler’s tax base, while a superior authority can enforce the resulting claim against that ruler. Tokugawa Japan provides community-level evidence of this mechanism by distinguishing exit through neighboring jurisdictions from appeal to the shogunate.<sup>[6]</sup>

## 2. FRAGMENTATION, EXIT, AND STATE DEVELOPMENT

Theories of state development identify war and exit as two forces that constrain rulers, but direct tests have focused on war. Financing armies required rulers to extract men, money, and supplies from holders of capital, land, and labor. The resulting bargains produced tax bureaus, representative assemblies, and eventually citizenship (Tilly 1985; 1992). Research now specifies when war built states rather than destroyed them (Dincecco and Wang 2018; Spruyt 2017). Exit operates through a different constraint: divided authority gives subjects alternatives, so excessive extraction can cost rulers their tax base or expose them to outside intervention. This mechanism underpins arguments about the Great Divergence, fiscal federalism, and the territorial state system. The argument below specifies the institutional environment, the actors and their choices, and the expected consequences for resistance and taxation.

### 2.1. The Institutional Environment

The setting contains local jurisdictions that govern their own subjects within a wider hierarchy. Local rulers control taxation and administration, while a superior authority

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<sup>[6]</sup>Moore (1993, p. 230) characterized the late Tokugawa order as marked by “vertical and horizontal fissures... perhaps equally important,” foregrounding the horizontal axis without operationalizing it; this paper supplies the community-level measurement.

can arbitrate disputes and sanction them. I call this arrangement *hegemonically ordered fragmentation*. It encompasses feudal, imperial, federal, and other layered systems that combine local autonomy with central authority (Gerring et al. 2011, 377–382, 405–407). Subjects can approach another local ruler horizontally or the superior authority vertically. The jurisdictions need not be sovereign or military rivals. The argument requires that local rulers retain enough autonomy to compete for subjects, subjects can reach an alternative authority, and the superior authority can sanction local rulers.

Three features determine whether those options are usable: the relation between local autonomy and superior authority, the cultural and legal costs of changing jurisdictions, and the geographic availability of alternatives. Hierarchy determines whether another ruler can only receive subjects or can also seek enforcement from a common superior. Cultural and legal distance determines what subjects give up by leaving. Geography determines how far they must travel and how many authorities they can plausibly approach. Early modern Europe contained both sovereign states without a common superior and layered arrangements in which principalities, cities, and estates remained under imperial or royal authority; movements between them could entail changes in language, religion, or law. Qing administration placed local jurisdictions more uniformly within a single sovereign hierarchy. These arrangements therefore offered different combinations of horizontal alternatives and vertical oversight. As the costs of reaching and using another authority fall, exit becomes more credible, giving the home ruler reason to moderate extraction before losing cultivators or facing intervention.

Tokugawa Japan combined nearly 300 separately governed domains in an archipelago smaller than France with a common central authority, language, and broader legal and religious order. Domains governed locally but held territory at the shogunate's pleasure; they varied in size, wealth, and villages' proximity to rival rulers. Hierarchy was not unique to Japan: European polities also combined horizontal and vertical authority, but their component units varied more in sovereignty and in the cultural and legal costs of movement. Tokugawa's advantage is that a common shogunal order kept vertical authority and the broader cultural and legal setting relatively stable while villages differed in the number and proximity of separately governed authorities they could approach.

Cross-domain migration was prohibited, and domains could not leave the hierarchy, making this a demanding test of subjects' ability to use outside authority.

The post-1615 settlement prevented recurrent inter-domain warfare for most of the next two and a half centuries and sharply weakened the war-finance pressures that dominated early modern Europe. Domains still maintained military forces and responded to foreign threats, and the shogunate fought limited domestic campaigns. Those demands did not reproduce sustained competition among sovereign states; competition over subjects remained the principal form of inter-domain rivalry. In Europe, by contrast, fragmentation supplied both subjects' exit options and rulers' military rivals. Exit could moderate taxes in peaceful conditions, but endemic warfare overwhelmed this restraint (Rosenthal and Wong 2011, 169, 184). Cross-regional comparisons also conflate differences in law, kinship, religion, bureaucratic recruitment, and geography between Europe and China. Europe's fragmentation itself arose partly from medieval ecclesiastical competition (Grzymala-Busse 2023). These comparisons cannot by themselves isolate the exit channel.

Within Tokugawa Japan, low cultural and legal distance makes exit depend more on the geography of authority than on assimilation costs. Rival centers vary in number and proximity across villages and years, while the hierarchy, shared institutional order, and exposure to recurrent warfare remain comparatively stable. This variation reveals subjects' behavioral responses to alternative authority; it does not isolate the long-run institutional consequences of exit.

## **2.2. Rulers and Village Communities**

The central relationship is between a ruler who depends on cultivators for revenue and village communities that bear extraction collectively. Raising taxes increases receipts from those who comply but can induce others to resist. A neighboring ruler supplies an alternative whose value depends on taxation, opportunities to sustain a livelihood, and access to protection or mediation. The home ruler therefore has reason to consider both villagers' responses and the terms available elsewhere. Proximity alone is insufficient: an accessible neighbor must offer something worth the costs of approaching it.

Villagers seek to protect their livelihoods and secure concessions from the authority that taxes them. Their alternatives vary with the cost of moving people or presenting claims beyond the home jurisdiction. Distance, forgone income, and continued access to land can make exit costly even when another authority is nearby.

Tilly's account recognizes mobility as a constraint but confines its treatment to elites and regional trajectories. Merchants and financiers in Europe's capital-intensive zones could "escape with their capital to another business site if the ruler's demands became unbearable," forcing rulers in Genoa, Amsterdam, and the Hanseatic cities to offer favorable terms (Tilly 1992, 53). In coercion-intensive zones, rulers and landlords instead fixed peasants in place. Russia's Assembly Code of 1649 and Prussian serfdom bound cultivators to the land because departure would have undermined extraction (Tilly 1992, 22–23, 141). Tilly thus treats mobility as a difference between regional trajectories and peasant exit as an option rulers foreclosed, without analyzing horizontal exit as a strategy of popular resistance (Tilly 1992, 187). The question here is whether communities with usable outside options act differently from those whose demands remain under their own ruler's control.

The superior authority supplies a further source of leverage. It can hear petitions, investigate local administration, and threaten a ruler's tenure or delegated powers. Local autonomy thus coexists with vertical oversight. In Tokugawa Japan, domains set taxes, administered justice, and controlled their borders. The shogunate arbitrated disputes among lords, heard peasant petitions over their heads, converted domains to direct rule, and stripped lords of their territories. Invoking a neighboring lord began horizontally through shelter or mediation; if that lord carried the dispute upward, it also activated vertical enforcement. Villagers could instead appeal directly to the shogunate, invoking vertical oversight from the outset. Kurushima documents villagers on shogunal land whose direct petition against a taxation scheme was examined and accepted (Kurushima 2002, 185–198). In the 1642 Aizu uprising, several thousand villagers instead crossed into neighboring domains, publicly petitioned the bakufu, and threatened permanent resettlement. Their lord canceled arrears partly to forestall intervention, and the unrest contributed to his subsequent attainder without constituting its formal charge (Vlastos

1986, 35–38). The first route was vertical from the outset; the second combined a threat to the home ruler’s tax base with exposure to superior intervention.

Petitions also gave the center information for oversight. The 1633 rules permitted appeal to higher authority after district-level remedies failed, helping the center monitor crop failures and misconduct by local officials. Direct appeals subsequently established precedents for accepted, if formally unlawful, political action and provided a recurring means of resolving tax disputes (Vlastos 1986, 42–43, 55–59). In 1720, the shogunate formalized this monitoring through petition boxes whose submissions went directly to the shogun, were investigated, and remained part of governance through the end of the period (Sng and Moriguchi 2014, 8). Central oversight could therefore reinforce villagers’ leverage even though the center retained authority over them. Hegemonic ordering changes rather than eliminates jurisdictional competition: horizontal alternatives coexist with vertical enforcement, and Tokugawa Japan supplies evidence for this broader class of fragmented polities.

### 2.3. Exit and Confrontation

Rulers set taxes knowing that villagers can compare the demands at home with the alternatives available elsewhere. Villagers then decide whether to accept those demands or seek relief. Ordinary petitioning can precede resistance: a village first asks its own ruler for a concession. When that petition fails to change the terms, villagers can comply, confront the home ruler directly, or invoke an outside authority. The distinction between the latter two choices adapts Hirschman’s contrast between withdrawing from an authority and pressing it to change (Hirschman 1970, 30). Studies of peasant resistance likewise distinguish flight and other forms of avoidance from direct confrontation (Adas 1981, 217–219; Scott 1976, 65). Here, these choices occur within a polity containing multiple centers and levels of authority.

*Direct confrontation* presses demands against the home ruler through coercive petition, property destruction, or uprising. Its prospects depend on villagers’ ability to mobilize together. *Exit through alternative authority* moves villagers or their claim beyond the home ruler’s exclusive control: they may flee collectively, petition through a neighbor-

ing ruler, or appeal directly to the superior authority. Compliance preserves income after taxes without the immediate costs of resistance. Confrontation offers concessions through collective pressure at home. Exit offers outside protection or intervention at the cost of reaching and using another authority. Villages with similar grievances can therefore choose different responses because their outside options differ. The distinction applies to actions rather than entire disputes, which can contain both exit and confrontation.

Exit need not end in permanent resettlement. Continued absence deprives the home ruler of cultivators and revenue, while outside shelter or mediation makes the threat of staying away more credible. Villagers also bear income and land-access costs while absent, giving them reason to return if the ruler grants sufficient concessions. When the exit takes the form of an appeal, the claim rather than the population leaves the home ruler's exclusive control. In either case, outside involvement can change the terms on which villagers remain under, or return to, their own ruler.

These choices can substitute at one moment and complement one another later. Departure can weaken simultaneous confrontation by removing participants, raising the opportunity cost of mobilization, or lowering confidence that collective action will attract enough support (Sellars 2019). It can strengthen subsequent action when migrants transmit information or organizational support (Pfaff and Kim 2003), when departure increases the leverage of those who remain (Karadja and Prawitz n.d.), or when an outside option backs further demands (Clark, Golder, and Golder 2017; Hirschman 1993). Which relationship prevails depends on whether resistance requires participants' presence at home or can draw on support created after they leave.

Tokugawa resistance fits the condition for contemporaneous substitution: both exit through alternative authority and direct confrontation drew on village organization, and their mass forms competed for physical participation. Where outside options are attractive, easier access to another authority should therefore shift the immediate composition of resistance toward exit. Outside recognition, shelter, or intervention after exit could nevertheless increase pressure later in the dispute. The 1853 Morioka records show end-run appeals and confrontation at different points in a broader cycle of resistance (Aoki

1975). Substitution at one moment therefore does not rule out the complementary use of both strategies over time.

## 2.4. Expected Outcomes

I establish two testable implications: a behavioral implication that better access to outside authority encourages exit, and a fiscal implication that it restrains taxation when neighboring domains offer sufficiently attractive alternatives. A formalized version of the argument appears in [Supporting Information \(SI\) §A](#).

The behavioral implication has two parts. First, better access to outside authority should increase exit before taxes adjust. Making another authority easier to approach or improving the protection it offers makes exit more attractive relative to compliance. Direct confrontation should not increase if the change leaves the resources and grievances behind it unchanged ([SI Proposition 1](#)). Domain conversions to central rule provide an empirical instance: installing a nearby central representative made appeals more accessible to villages still governed by surrounding lords.

Second, better access to outside authority should continue to encourage exit even when rulers can respond. When outside authority is less costly to invoke, more villages can use exit. Home rulers can moderate taxes to retain them, but doing so reduces revenue from villages that would have complied anyway. This revenue cost limits optimal tax concessions, so rulers' responses do not fully offset the increase in exit. The formal analysis establishes this result over a broad parameter range ([SI Proposition 2](#)). Where the outside option is attractive, better access should also increase exit's share of resistance: more villages have a feasible exit option, while fewer remain available for simultaneous confrontation. The incidence of exit and its share among resisting villages are distinct implications. I test these predictions using geographic exposure, measured by villages' relative distances to home and neighboring capitals, as a proxy for access to outside authority.

The fiscal implication is that better access to outside authority should restrain taxation when the outside option is sufficiently attractive. Easier exit can make rulers lower taxes to retain cultivators, but it can also reduce the pool available for direct

confrontation. When the outside option offers a sufficiently attractive livelihood, the retention incentive dominates and rulers should lower extraction. If that option is poor, relief from confrontation can instead permit higher taxes (SI Proposition 3). I test this implication using geographic exposure and neighboring domains' capacity to absorb migrants: exposure should be associated with lower reported tax rates especially where that capacity is greater. The argument concerns constraints on extraction and choices of resistance; long-run institutional development is a further consequence that the Tokugawa evidence does not establish.

### 3. EMPIRICAL EVIDENCE

The empirical analysis first examines whether improved access to outside authority changes the incidence and composition of resistance, then tests whether geographic exposure is associated with lower taxation where neighboring domains can absorb migrants.

#### 3.1. Setting: Tokugawa Japan

During the Tokugawa period (1603–1868), nearly 300 domains, each held by a lord (*daimyō*), governed Japan. Figure 1(a) maps the 264 capitals active in 1868, and Figure 1(b) tracks the number active as domains were created, merged, transferred, or absorbed into shogunal land. These domains originated in territories controlled by local warriors during the Warring States period and, earlier, in private estates known as *shoen*. Although subordinate to the shogunate, each domain maintained its own government and military forces and set its own tax rates (Ravina 1999, 1000).

Capital-centered fiscal administration links domain fragmentation to village choices. Domains shared an administrative structure centered on the castle town (Hall 1955), while the village contract system (*murauke*) made villages responsible to the *daimyō* for tax collection (Saxonhouse 1995; Walthall 1997, 89; White 1995). Domain capitals housed the officials who received petitions, decided on shelter and mediation, and kept records that could reach Edo (Hall 1955; Ravina 1999). Crossing a border placed villagers beyond their lord's immediate reach; reaching an administrative center made their grievance

visible to another authority and could open a path to Edo. A neighboring domain governed from a distant castle town offered less access than one governed from nearby.

The village contract made extraction a collective obligation administered through identifiable offices. When a tax dispute arose, the village addressed the same institutional system that collected its dues: home officials received its demands, an outside capital could decide shelter or mediation, and Edo could receive the resulting record. The relevant distinction was therefore not simply whether individual households moved. It was whether the village kept its dispute within the home domain or placed it before an outside authority.

The long peace further connects the setting to the argument. Domains fought no major wars against one another until the shogunate’s collapse in the 1860s.<sup>[7]</sup> Domain rivalry therefore operated through competition for subjects and the “everyday forms of resistance” it enabled (Scott 1985), rather than through the warfare central to conventional state-building accounts (Hintze 1975; Tilly 1978; Weber 1919). Peace did not eliminate rivalry. It shifted rivalry toward competition for cultivators, shelter of fugitives by neighboring domains, and shogunal resolution of disputes between jurisdictions.

This separation makes the Tokugawa case useful for studying exit apart from war finance. In Europe, fragmentation simultaneously expanded subjects’ outside options and rulers’ exposure to military rivals, so mobility and wartime extraction operated together. Here, access to alternative rulers varied within a domain system in which inter-domain war remained absent for most of the period. A rise in exit near outside authority can therefore be studied without treating military mobilization as the source of the change.

### 3.2. Observed Forms of Exit and Direct Confrontation

The empirical analysis maps the distinction in §2.3 onto Aoki’s (1975) five recorded tactics: coercive appeals (*gōso*), destruction (*uchikowashi*), rebellion (*hōki*), desertion

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<sup>[7]</sup>The Shimabara Rebellion of 1637–38 was an anti-Christian and anti-Matsukura uprising suppressed by a shogunal coalition, not unrestricted inter-*daimyō* war. The Chōshū War (1864–1866) similarly pitted the shogunate against Chōshū, and the Boshin War (1868–1869) accompanied regime collapse. Institutional peace held between Shimabara and the 1860s. The tax data cover the period’s end, but tax policies remained stable even in the final years (Smith 1958).

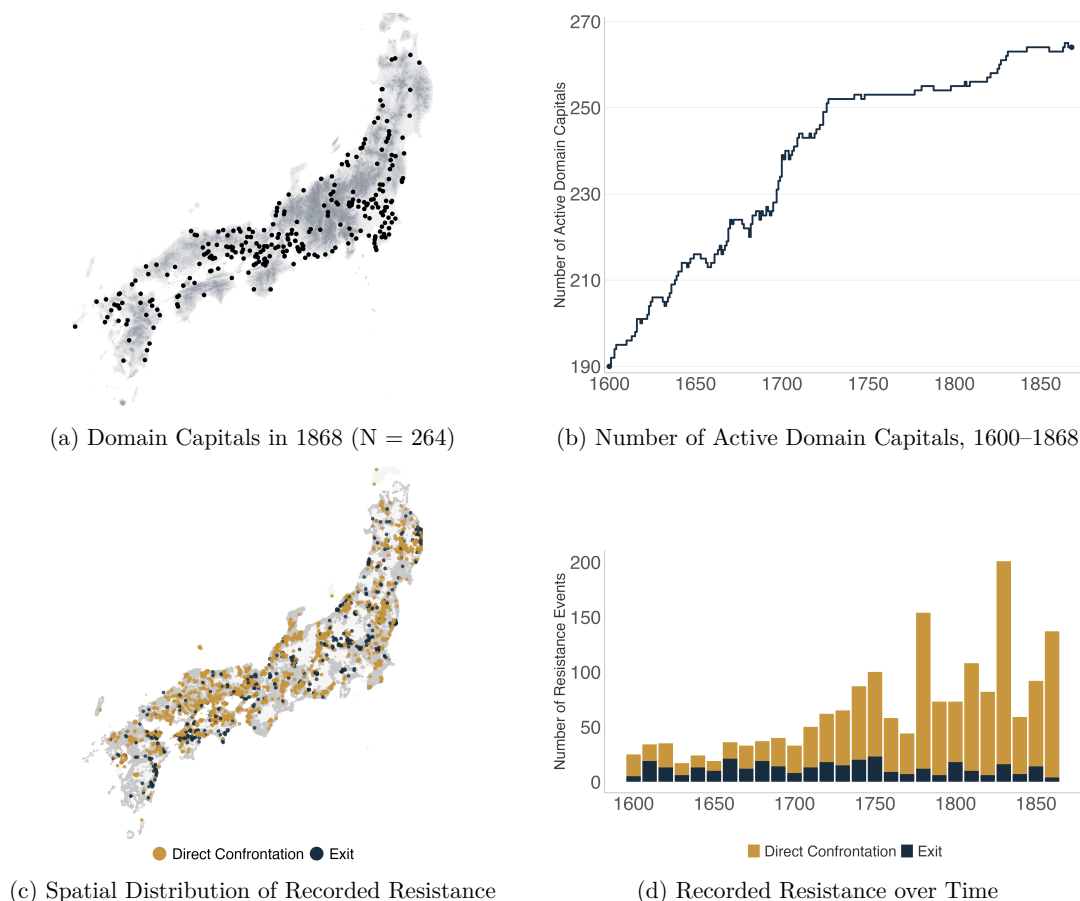


Figure 1: The Tokugawa Domain System and the Record of Peasant Resistance

**Note:** Tokugawa Japan was divided into roughly 300 domains, each governed from a castle town that served as its capital and housed the officials who received petitions. (a) Locations of the 264 capitals active in 1868, plotted over terrain; the archipelago runs about 1,900 km southwest to northeast. (b) The number of active capitals each year, 1600–1868, which changed as the shogunate created, merged, transferred, and abolished domains. (c) Villages recorded as resisting, colored by whether they exited (left the domain or petitioned an outside authority) or confronted their lord directly; gray points are villages present through the end of the period with no recorded resistance. (d) Recorded episodes per decade.

(chōsan), and end-run appeals (osso).<sup>[8]</sup> The first three constitute *direct confrontation*; the last two constitute *exit through alternative authority*.<sup>[9]</sup> Figure 1(c) maps these incidents, and Figure 1(d) plots their frequency over time.

<sup>[8]</sup> Later studies follow this classification. White (1995) analyzes protest repertoires and their evolution; Vlastos (1986) treats end-run appeals as a distinct form that dominated the mid-Tokugawa period; Walthall (1997) translates primary documents from all five categories.

<sup>[9]</sup> “Direct confrontation” is more accurate than “fight” because coercive appeals were ritualized and often only implicitly violent.

The grouped exit outcome contains three institutional routes. They differ in who moved, which authority received the grievance, and how pressure reached the home lord.

In *physical relocation* (*chōsan*), villagers collectively crossed into a neighboring domain. The receiving lord then decided whether to shelter them or return them to the home domain. Shelter removed the villagers and the tax obligations assessed to their community from the home lord’s immediate control, threatening his revenue directly. The receiving lord’s protection made departure sustainable while the two sides bargained.

A second route involved a *neighbor-routed appeal*. Its leverage came from intervention rather than fiscal loss alone. Villagers placed themselves under a rival lord’s protection and enlisted him in the dispute. The receiving lord could shelter them, transmit their demands, and mediate with the home lord. His involvement publicized the home lord’s failure to govern and converted a village grievance into a dispute between jurisdictions.

Villagers could also make a *direct appeal to the shogunate* (*osso*), placing the claim, rather than necessarily the entire village, outside the home domain. Representatives bypassed the home lord and addressed the shogunate, invoking hierarchical oversight rather than competition for subjects. Ordinary petitions to the home ruler are excluded from the analysis and are not part of the *osso* category used here. The shogunate could thereby discipline the home lord without requiring a rival lord to compete for the villagers.

These routes imply different relationships with geographic exposure. Physical relocation and neighbor-routed appeal require a reachable rival ruler willing to receive villagers. Direct appeal to Edo does not: even a village far from rival capitals could send representatives to the realm’s capital. The pooled exit measure therefore combines routes that exposure should and should not predict.

Aoki’s incident descriptions provide destination or addressee information for 176 of 328 unique exit events (Table A.9). Among events with this information, 53 of 54 desertions carry a **foreign** label, while 112 of 120 end-run appeals carry an **edo** label. These descriptive labels include both physical locations and institutional addressees, so they do not map directly onto neighboring domains, *daikansho*, or the shogunate.<sup>[10]</sup> The

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<sup>[10]</sup>Record checks find home authorities and internal locations under **foreign**, and Osaka, Kyoto, and home lords’ Edo residences under **edo**. The table reports the labels as coded from Aoki’s descriptions.

institutional route therefore remains uncertain for some end-run appeals. The desertion-only analyses isolate horizontal exit by observed action rather than inferred destination, and the findings persist under this restriction (SI §C.5, §D.2.8, and §D.2.9).

The three forms of direct confrontation differed in intensity (Aoki 1975). Coercive appeals were mass marches on domain offices or castles to present demands backed by an implicit threat of violence. Crowds of thousands wore straw raincoats and carried bamboo spears as symbols of collective resolve (Walthall 1997, 131–132). Destruction targeted the property of officials, merchants, or wealthy villagers associated with oppressive policies. Organized groups “smashed” (*uchikowashi*) houses and storerooms (Walthall 1997, 89–130). Rebellion was armed uprising against domain authority, the rarest form at roughly 1% of recorded incidents (White 1995, 142). All three sometimes won concessions but often met severe repression.

Following the action-level distinction in §2.3, a dispute containing both tactics contributes separate event records. A neighboring lord’s shelter or mediation does not turn villagers’ exit into direct confrontation; no sampled exit episode involved armed intervention by another domain. Direct confrontation denotes villagers’ coercive action against home authorities.

Aoki prints peasant uprisings separately from urban disturbances, *murakata sōdō*, and province-wide petitions. The project comparison ends with the uprising chronology; all 294 unambiguous event-page matches, including 115 in the analytic sample, fall there, so excluding the separately printed categories removes no observation (Table A.6).

Household tenure within included entries is less certain. Removing all nineteen incidents mentioning tenants, landlords, or *komae* smallholders leaves the preferred village-choice estimate positive and statistically significant (Table A.7). I therefore interpret the estimates as behavior by recorded village groups, not a landholder-only effect.

### 3.2.1. Why Non-Compliance Was Consequential

Exit publicized governance failure more readily than direct confrontation. The Buke Shohatto (Laws of Military Houses) required domains to be “well-governed,” and failure gave the shogunate grounds to confiscate, reduce, or transfer a lord’s territory (Steele, Paik, and Tanaka 2017, 3–5). Direct confrontation usually remained internal, allowing

the daimyō to contain information or control its reporting. Exit brought villagers or their claims before an authority beyond the home ruler’s exclusive control, exposing the dispute to outside scrutiny. A fleeing village thus exposed its lord to rivals and the shogunate, making attainder risk harder to suppress (Steele, Paik, and Tanaka 2017). This publicity gave daimyō strong incentives to moderate extraction where villages could reach outside authority. Mass flight was accordingly “what the warrior class feared most” (Kuromasa 1927, 69) and was prohibited by strict decree.

### **3.2.2. Landholding, Tax Liability, and the Cost of Exit**

The fiscal mechanism applies most directly to registered landholding households. Seventeenth-century cadastral surveys in shogunal territory matched parcels to cultivating stem families, assessed output and tax, and protected household claims conditional on payment (Nakabayashi et al. 2020, 158, 175). Peasants could use, buy, sell, rent, and inherit land; partible inheritance was rare, and the main family generally inherited all or most land (Kumon 2025; 2026, 158, 160; Nakabayashi 2025). A durable cultivation claim and recurring fiscal duty were thus two sides of the same institution.

Tenure nevertheless varied regionally. Less productive areas retained joint village ownership, covering about 30% of assessed national output, under which villages assigned and sometimes reallocated parcels (Nakabayashi et al. 2020, 175–176). Kaga combined corporate repartition with inheritable and transferable household use rights (Brown 1993, 94–95, 107–108). Land access was therefore place-specific—attached either to a registered household or to membership in a landholding village. Prolonged departure could weaken practical access, and permanent resettlement abandoned a valuable asset, but the sources do not show that border crossing automatically extinguished title. The 1873 Land Tax Reform standardized and transformed these prior claims rather than creating them from nothing (Nakabayashi et al. 2020, 176–177).

The costs on each side played different strategic roles. Outside shelter and possible absorption enabled villagers to remain away if the home lord refused to concede, making continued loss of cultivators and tax payments a feasible sanction rather than an empty threat. Villagers nevertheless lost income, incurred shelter costs, and risked weakening access to household- or village-held land while away. Once the lord granted sufficient

relief, returning allowed villagers to recover those benefits while restoring his tax base. Continued departure could therefore sustain the threat, while sufficient concessions made return attractive. Kaga officials followed this pattern by waiving arrears, offering rice, promising to correct abuses, and calling cultivators home (Brown 1993, 138). Nobeoka likewise granted the departing villagers' demands before they returned (Steele, Paik, and Tanaka 2017, 356).

### **3.2.3. Exit as Deliberate Protest**

Desertion and end-run appeals both depended on outside authority, and villagers sometimes combined them by petitioning while others sheltered abroad (Hosaka 2002; Miyazaki 1995). Receiving authorities supplied shelter, guards, mediation, or recognition. Continued shelter deprived the home lord of cultivators, publicized the grievance, and preserved possible absorption; return after concessions was therefore a bargaining outcome, not evidence that departure lacked force (Miyazaki 1995; Vlastos 1986). Seven documented exits or attempts ended in return, although the sources do not permit a general estimate of episode duration, a systematic reconstruction of what villagers paid while away, or an estimate of how common return was (Table A.8).

Receiving lords had two reasons not to collude in automatic return. Absorbing cultivators increased their revenue, while surrendering them benefited a rival (Miyazaki 1995). Reporting misgovernment could also prompt shogunal sanctions, whereas quiet return yielded no comparable gain. This asymmetry is the obstacle a ruler cartel must overcome (Acharya and Lee 2018).

### **3.2.4. How Exit Worked: Illustrative Cases**

Three episodes of peasant resistance show how neighboring rulers made exit effective through shelter, mediation, and possible escalation to the shogunate.

In Nobeoka (1690), 1,418 peasants fled oppressive taxation and floods to Takanabe. Its officials sheltered them, mediated, guarded representatives, and helped escalate stalled negotiations to Edo. After ten months, Nobeoka granted all demands, including tax relief, without punishment, and the villagers returned (Steele, Paik, and Tanaka 2017, 356).

The shogunate transferred the Nobeoka lord to a much smaller domain the next year (Ishikawa 1981).

In Yamashirodani (1837), border villages left Tokushima for Imabari. The Imabari lord sent guards, mediated, and threatened to absorb them if Tokushima rejected their petition (Okamura 2009). Absorption would have increased Imabari’s revenue (Miyazaki 1995).

In Matsuyama (1741), famine-stricken villages fled to neighboring Ōzu and requested migration. Their lord accepted most demands, punished his chief minister rather than the villagers who led the departure, and the villagers returned (Nakagawa 1980).

Together, these cases show how outside authority turned departure into bargaining through shelter, mediation, and possible absorption. Customary legitimacy and ritual shaped the available repertoire, while relative costs influenced which tactic villages used (Vlastos 1986; Walthall 1997; White 1995).

### **3.3. Exposure and Exit Occurrence**

Two designs examine whether recorded exit occurs more often under greater geographic exposure: a conversion-based difference-in-differences design and an observational grid-year panel.

#### **3.3.1. Conversion-Based Test: Tenryō Conversion**

This design associates conversion with 3.8 percentage points more recorded exit among nearby villages governed by surrounding lords and no detectable average change in their direct confrontation. I use the Tokugawa shogunate’s abolition of domains and conversion of their territories into direct shogunal land (*tenryo*), an institutional change that introduced a powerful nearby authority.

Conversion was a general instrument of shogunal lordship, not a policy directed at neighboring villagers. Daimyō governed fiefs as invested vassals: the shogunate could revoke, reduce, or transfer their holdings, but could not tax the inhabitants while the territory remained daimyō land. When a ruling house failed to produce an heir or a lord lost his tenure through misconduct or political sanction, direct administration shifted the fiscal rights to the shogunate, installed its own officials, and preserved its discretion to re-

enfeoff the territory later. This explains why the same disposition followed heterogeneous internal triggers. Attainder and transfer also fell disproportionately on smaller *fudai* houses whose tenure the shogunate could alter most readily, rather than on domains selected because neighboring peasants had become unusually mobilized (Kimura, Fujino, and Murakami 2015; Kodama and Kitajima 1989; Ravina 1999, 17–21).

A tenryō conversion opened a credible appeals route to the shogunate. A shogunal intendant (*daikan*) installed at the former domain capital allowed nearby villagers to petition locally rather than travel to Edo. Conversion also set a recent precedent for absorbing adjacent territory, while contiguity with shogunal land lowered the administrative cost of further attainder. The Buke Shohatto’s “well-governed” mandate (Section 3.2) had already underwritten the seizure of the Arima daimyō after the 1690 Nobeoka mass exit. Conversion thus made exit-driven petitions cheaper to file and more likely to expose surrounding lords to seizure.

These conversions test the short-run implication before home rulers adjust their tax rates. A nearby shogunal appeals route lowers the effective cost of exit and should raise exit, but not direct confrontation, in villages of neighboring domains. Because internal disorder sometimes triggered conversion, the null on nearby confrontation also tests whether unrest simply spilled across domain borders.

This selection process does not itself violate the difference-in-differences design. The shogunate selected a focal daimyō’s tenure for conversion, whereas the outcomes exclude that domain and count resistance only in villages belonging to surrounding lords. Succession failure, punishment, or fiscal reorganization inside the converted domain threatens identification only if it reflects a regional shock that independently changes subsequent resistance in those foreign villages. Focal-domain fixed effects absorb persistent differences around capitals, year fixed effects absorb shocks common to all domains, and the pre-conversion hazard estimates show that lagged nearby exit and confrontation do not jointly predict conversion timing ( $p = 0.344$ ; SI C.13). Excluding conversions connected to documented unrest also leaves the exit estimate positive and confrontation null (Table A.19). These checks do not require every conversion to have the same cause;

they test whether the causes generate the specific cross-border outcome trend that would confound the design.

Aoki (1975) catalogs 7,664 incidents of peasant resistance from local histories, official documents, and diaries. Each observation records the date, home domain, nearest neighboring domain,<sup>[11]</sup> and resistance mode, which I classify as exit or direct confrontation under Section 3.2. I exclude 574 peaceful petitions and inter-village resource or boundary disputes.

The catalog identifies 2,790 sites. I georeference 2,571 of them using several sources and exclude those whose locations remain imprecise.<sup>[12]</sup> The resulting data contain 910 resistance episodes involving 2,571 villages and 2,741 tactic choices (Table A.5, SI B). Because large mass exits were more likely than small flights to enter the historical record, unrecorded minor cases are unlikely to account for the main results. Capital locations and the creation and abolition dates of all 294 domains come from Kimura, Fujino, and Murakami (2015) and Kodama and Kitajima (1989) (Table A.10, SI B); I assign branches (shi-han) to their main domains (hon-pan).

I construct an annual domain-year panel for 1615–1868. For each domain-year, I count exit and confrontation events in villages of other domains within  $r \in \{30\text{km}, 40\text{km}, 50\text{km}\}$  of the capital, the zone exposed to the new exit opportunity. Figure 2 illustrates the design.

Consider a domain abolished in year  $t$  and converted to tenryo. The outcomes do not come from villages inside that converted territory. They come from villages that remain governed by surrounding lords but lie within radius  $r$  of the converted capital. For the treated domain, the design compares resistance in those foreign villages during the five years before conversion with resistance during the five years after a shogunal intendant arrives. It then compares that before-and-after change with the contemporaneous change in foreign villages surrounding domains that never converted. The converted capital sup-

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<sup>[11]</sup>I used Kodama and Kitajima (1989) to identify the neighboring domains existing at the time of each resistance event. The nearest foreign domain is identified using time-varying domain existence data: for each event, only domains active at that date are potential alternative authorities.

<sup>[12]</sup>These sources include Honda, Natsume, and Nemoto (2022), the geocoding API provided by the Ministry of Land, Infrastructure, Transport and Tourism of Japan, and the Historical Location Name Comprehensive Database provided by the Center for Open Data in the Humanities.

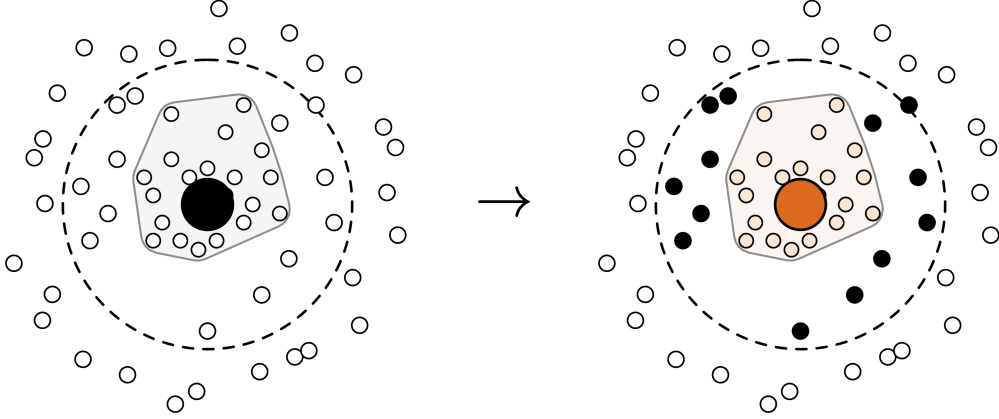


Figure 2: DiD Design: Tenryō Conversion

**Note:** Stylized illustration. The large central circle is the focal capital, the gray polygon is its territory, small circles are villages, and the dashed circle is the analysis radius  $r$ . Left: before conversion, villages of surrounding lords within  $r$  lack a nearby shogunal authority. Right: abolition converts the focal capital and territory to direct shogunal rule (*tenryō*, orange); black circles identify surrounding foreign villages inside  $r$ , while pale circles inside the gray territory are excluded from the outcomes and open circles fall outside the treated foreign-village set. No village-to-territory arrows are shown because episode destinations are unobserved. The DiD compares changes in exit and confrontation in surrounding foreign villages with changes around never-converted domains.

plies the new outside authority; villages in neighboring domains supply the measured exit and confrontation outcomes. Domain fixed effects absorb persistent differences around focal capitals, and year fixed effects absorb shocks common to all domains. The coefficient estimates whether resistance around a converted capital changed more at conversion than resistance around never-converted capitals observed in those same years.

I exclude four abolitions that transferred territory directly to another domain rather than to *tenryō*. The 55 treated domains come from 79 dissolutions across all 294 domains and are listed by year and stated reason in [SI C.1](#) ([Kimura, Fujino, and Murakami 2015](#); [Kodama and Kitajima 1989](#)). I restrict converted domains to five years before and after their first conversion, retaining strict pre-conversion years and post-conversion years under direct shogunal rule. Never-converted controls enter only for the same calendar years. I estimate a domain-year linear probability difference-in-differences model with domain and year fixed effects:

$$\text{NonCompliance}_{j,t} = \beta_1 (\text{Converted}_j \times \text{Post}_{j,t}) + \varphi_j + \omega_t + \varepsilon_{j,t}$$

Here,  $\text{NonCompliance}_{j,t} \in \{\text{Exit}_{j,t}, \text{Confront}_{j,t}\}$  indicates any nearby exit or confrontation within  $r$  kilometers of domain  $j$ 's capital in year  $t$ .  $\text{Converted}_j$  identifies ever-converted domains, and  $\text{Post}_{j,t}$  identifies years after first conversion under direct shogunal rule.

Destination misclassification motivates the first checks. The design cannot observe the destination of every exit. Sources usually record where villagers came from but rarely where or how they traveled, so proximity identifies converted territory as a plausible outside option rather than a confirmed destination. [Figure 2](#) accordingly depicts exposure within the analysis radius without village-to-territory arrows. At the tighter 40km and 30km radii, the exit coefficients remain positive at 0.0262 ( $p = 0.025$ ) and 0.0163 ( $p = 0.038$ ), respectively, while nearby confrontation remains near zero at every radius. The exit estimate also remains positive when the outcome is restricted to desertion ([SI C.5](#)). A general problem of geographically mismeasured unrest would not readily produce all three patterns.

Conversion endogeneity motivates a separate exclusion exercise. A page-level local-source and Aoki chronology audit identifies three cases clearly connected to peasant unrest or petitioning—Saijō (1665), Shimabara (1668), and Hōjō (1712)—and two mixed cases, Numata (1681) and Ōmi Komuro (1788). Removing the three clear cases increases the exit estimate from 0.0385 to 0.0421 (two-way cluster  $p = 0.0067$ ); removing all five yields 0.0397 ( $p = 0.0130$ ). A maximal screen also removes five conversions preceded within ten years by an Aoki event in the same named domain or province, even though the histories do not identify those events as causes; the exit estimate is 0.0443 ( $p = 0.0117$ ). The confrontation placebo remains near zero in all three samples ([Table A.19](#)). These screens show that the result does not depend on cases with documented or potential links to unrest, although domain histories cannot establish the absence of unrecorded unrest.

I completed a case-by-case treatment-status audit of all 55 baseline cases using local source volumes. Twenty establish direct shogunal administration at the recorded date, twenty-one are partial reductions, family partitions, transfers, or otherwise noncomparable events, ten leave the intervening administrator unstated, and four use the wrong date. Correcting Takayanagi to 1739, Kōfu to 1724, and Sagara to 1787 yields a strict 23-

case screen; Numazu is excluded because no transition occurred in 1724. The exit estimate remains positive at 0.0385 (cluster  $p = 0.0903$ ; Conley  $p = 0.0338$ ). An ambiguity-inclusive corrected screen of 33 cases yields 0.0310 (cluster  $p = 0.0682$ ; Conley  $p = 0.0340$ ). After the three clear unrest cases are also removed, the corresponding estimates are 0.0447 and 0.0366 (Table A.18). The strict subset also produces a positive confrontation estimate, whereas confrontation remains statistically indistinguishable from zero in the ambiguity-inclusive screen. The audit therefore preserves the positive exit association but shows that the conversion design’s exit-specific contrast depends on the broader defensible treatment envelope; the grid and village-choice designs carry the tactic-specific inference.

Table 1 reports the main result. At 50km, tenryo conversion is associated with a 0.0385 increase in nearby exit probability ( $p = 0.010$ ), or 3.8 percentage points. This difference is roughly 59% of the control group’s pre-conversion mean of 0.065. By contrast, the confrontation estimate is only 0.0076 ( $p = 0.532$ ). The baseline treatment universe therefore shows more exit without a detectable parallel change in direct confrontation.

| Outcome:                     | Any Nearby Exit |            |             | Any Nearby Confrontation |          |          |
|------------------------------|-----------------|------------|-------------|--------------------------|----------|----------|
| Sample Radius:               | 30km            | 40km       | 50km        | 30km                     | 40km     | 50km     |
| <i>Variables</i>             |                 |            |             |                          |          |          |
| Converted $\times$ Post      | 0.0163          | 0.0262     | 0.0385      | 0.0084                   | 0.0033   | 0.0076   |
| Cluster SE                   | (0.0078)**      | (0.0116)** | (0.0147)*** | (0.0093)                 | (0.0092) | (0.0122) |
| Conley SE (70km)             | (0.0084)*       | (0.0107)** | (0.0131)*** | (0.0104)                 | (0.0108) | (0.0136) |
| <i>Controls</i>              |                 |            |             |                          |          |          |
| Domain FE                    | Yes             | Yes        | Yes         | Yes                      | Yes      | Yes      |
| Year FE                      | Yes             | Yes        | Yes         | Yes                      | Yes      | Yes      |
| <i>Fit Statistics</i>        |                 |            |             |                          |          |          |
| Num. Obs.                    | 33,877          | 33,877     | 33,877      | 33,877                   | 33,877   | 33,877   |
| Tenryo-Converted/All Domains | 55/294          | 55/294     | 55/294      | 55/294                   | 55/294   | 55/294   |
| R <sup>2</sup>               | 0.038           | 0.052      | 0.062       | 0.078                    | 0.101    | 0.136    |
| Mean Dep. Var.               | 0.010           | 0.018      | 0.026       | 0.020                    | 0.031    | 0.044    |

Table 1: Domain-to-Tenryo Conversion and Nearby Exit

**Note:** Linear probability model (LPM) estimates from an annual domain-year panel. “Nearby” denotes resistance in villages of *other* domains within the specified radius. Converted domains are observed from five years before to five years after conversion and matched to never-converted controls in the same calendar years. SEs in parentheses are (i) two-way clustered by domain and year and (ii) Conley (70km) on capital coordinates; stars use the corresponding estimator’s  $p$ -value. Signif. Codes: .\*\*\*: 0.01, .\*\*: 0.05, .\*: 0.1.

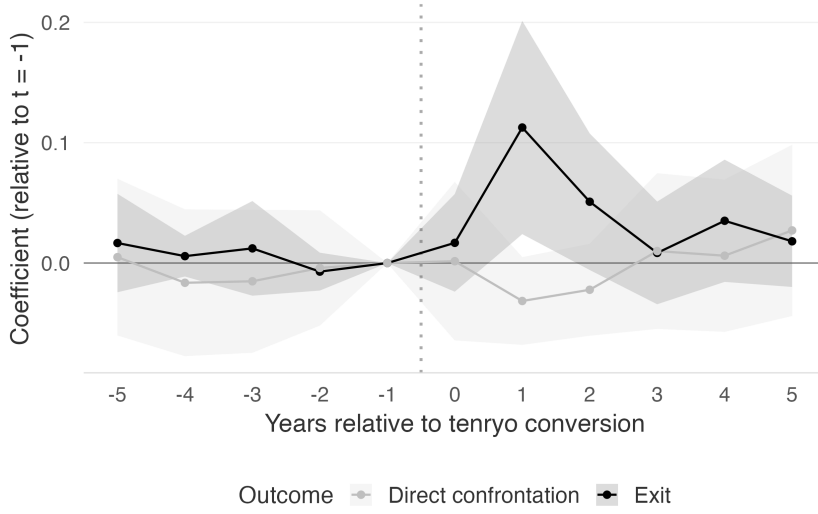


Figure 3: Tenryo-Conversion Event-Study Coefficients

**Note:** Yearly event-study coefficients for nearby exit (black) and direct confrontation (light gray), with  $s = -1$  as the reference. Shaded bands show 95% confidence intervals; the horizontal line marks zero and the vertical dotted line marks conversion between  $s = -1$  and  $s = 0$ . “Nearby” denotes resistance in villages of *other* domains within 50km of the focal capital. Standard errors are two-way clustered by domain and year.

The conversion-timing checks begin with the assumption that converted and never-converted domains would have followed parallel resistance trends absent conversion. I assess this assumption with an event study:

$$\text{NonCompliance}_{j,t} = \sum_{s \neq -1} \beta_s (\text{Converted}_j \times \mathbf{1}[t - t_j^* = s]) + \varphi_j + \omega_t + \varepsilon_{j,t}$$

where  $\text{Converted}_j$  denotes tenryo-converted domains,  $t_j^*$  is the first conversion year, and  $s = -1$  is the reference period. In [Figure 3](#), the pre-treatment coefficients are individually small and jointly insignificant ( $F = 1.26$ ,  $p = 0.283$ ; confrontation:  $F = 0.18$ ,  $p = 0.947$ ). After conversion, exit rises at  $t = 1$  ( $\hat{\beta}_1 = 0.113$ ,  $p < 0.05$ ) and remains elevated, while confrontation stays flat.

Three diagnostics bound the design’s identifying assumptions. The [Rambachan and Roth \(2023\)](#) relative-magnitudes breakdown is  $\overline{M}^* = 0.20$ ; under larger departures the estimate remains positive but confidence intervals include zero ([SI C.9](#), [SI C.10](#)). Lagged nearby exit and confrontation do not jointly predict conversion timing ( $p = 0.344$ ; [SI C.13](#)). Clean treated-versus-never-treated comparisons receive 94.5% of the Goodman-Bacon weight, and the Sun–Abraham estimator yields an exit ATT of 0.0413

while confrontation remains null (SI C.7, SI C.6). Exclusions for nearby conversions and potentially exposed controls preserve a positive exit coefficient (SI C.11).

Spatial, functional-form, and randomization checks lead to the same qualitative pattern. Conley bandwidth sweeps, count models, and two permutation schemes produce a positive exit estimate and a near-zero confrontation estimate (SI C.14, SI C.15, SI C.4, SI C.12). The association is concentrated in the pre-1750 cohort, which contains 49 of the 55 conversions, when attainder remained especially credible (Ravina 1999; White 1995); the later cohorts are too small to support comparable inference (SI C.8).

### 3.3.2. General Test: Grid-Year Panel

The conversion design estimates the response to a specific improvement in access to shogunal authority. The grid-year panel broadens the comparison: greater geographic exposure is associated with more recorded exit and no detectable increase in confrontation as nearby capitals are established, abolished, or relocated.

I measure location-level exposure with a *distance ratio*:

$$\rho_{i,t}^{\text{grid}} := \frac{d_{i,t}^1}{d_{i,t}^2}$$

where  $d_{i,t}^1$  and  $d_{i,t}^2$  are straight-line distances from location  $i$  to the nearest and second-nearest active capitals in year  $t$ . The ratio approaches one when a new capital appears closer than the prior second-nearest or when the nearest capital disappears or moves farther away. Values near zero indicate one dominant capital; values near one indicate two equally accessible rulers. Section 3.4.1 defines the village-level analog.

A village-year panel would be preferable, but domain affiliation is observed only at the end of the Tokugawa period. The governing domain of most non-resisting villages is therefore unknown in earlier years. I address this limitation by assigning villages to fixed 5km hexagons and computing the grid exposure ratio  $\rho_{i,t}^{\text{grid}}$  from the two nearest active capitals without using domain affiliation. The 46,086-village late-Tokugawa *Kyūdaka–Kyūryō Torishirabe Chō* catalog defines populated cells (Section 3.4). The resulting panel contains 2,225,802 cell-years across 8,763 cells, with annual counts of exit and confrontation in each hexagon.

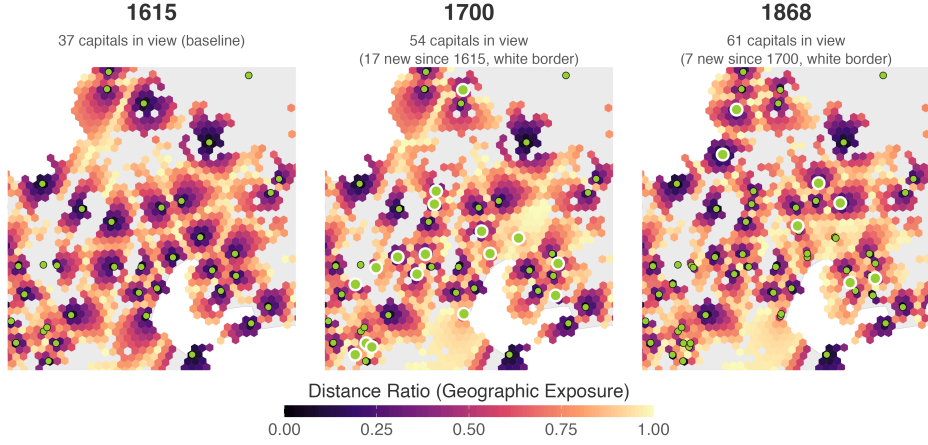


Figure 4: Time-Varying Distance Ratio across 5km Hexagonal Grid: Kinki Region

**Note:** Each panel maps  $\rho_{i,t}^{\text{grid}}$  (nearest-capital distance / second-nearest-capital distance) in a  $\approx 200\text{km} \times 200\text{km}$  window of central Japan (lat 34.4–36.2°N, lon 135.5–137.6° E), covering Kyoto, Osaka, Nagoya, and Lake Biwa. This region has the most cells whose ratios changed across the three years. Yellowgreen circles mark active capitals; white borders mark capitals added since the previous snapshot. Yellow indicates greater geographic exposure and purple a dominant nearest capital. Changes reflect domain creation and dissolution.

The estimating equation is

$$\text{NonCompliance}_{i,t} = \beta_3 \rho_{i,t}^{\text{grid}} + \lambda_i + \eta_t + \varepsilon_{i,t}$$

with hexagon and year fixed effects, where  $\text{NonCompliance}_{i,t}$  is the exit or confrontation count. Identification comes from capital dissolutions, transfers, and reestablishments that reorder the nearest authorities. Figure 4 maps  $\rho_{i,t}^{\text{grid}}$  in three years: broad spatial patterns persist, while domain creation and dissolution change exposure locally. Hexagon and year fixed effects absorb time-invariant geography and common shocks.

Table 2 shows that the distance ratio is associated with exit but not confrontation, whereas remoteness from any authority predicts neither. In columns (1)–(3), a one-unit increase in the ratio is associated with a 0.035-percentage-point increase in cell-year exit probability. The estimate is significant at 5% with both cell-and-year clustering ( $p = 0.048$ ) and Conley inference ( $p = 0.037$ ). The smaller positive estimate for confrontation is not statistically significant. Because the baseline exit probability is only 0.022%, a one-standard-deviation increase in the ratio ( $\text{SD} = 0.26$ ) corresponds to roughly 40% of that mean.

| Outcome:              | Grid exposure ratio ( $\rho_{i,t}^{\text{grid}}$ ) |            |                        | Closest-capital dist. ( $d_{i,t}^1$ , km) |            |                        |
|-----------------------|----------------------------------------------------|------------|------------------------|-------------------------------------------|------------|------------------------|
|                       | (1)                                                | (2)        | (3)                    | (4)                                       | (5)        | (6)                    |
|                       | Any exit                                           | Any confr. | Any confr.<br>(strict) | Any exit                                  | Any confr. | Any confr.<br>(strict) |
| Coefficient           | 0.00035                                            | 0.00026    | −0.00012               | 0.00000                                   | 0.00000    | 0.00000                |
| Cluster SE            | (0.00017)**                                        | (0.00019)  | (0.00011)              | (0.00000)*                                | (0.00000)  | (0.00000)              |
| Conley SE (10km)      | (0.00017)**                                        | (0.00023)  | (0.00012)              | (0.00000)                                 | (0.00000)  | (0.00000)              |
| <i>Controls</i>       |                                                    |            |                        |                                           |            |                        |
| Grid FE               | Yes                                                | Yes        | Yes                    | Yes                                       | Yes        | Yes                    |
| Year FE               | Yes                                                | Yes        | Yes                    | Yes                                       | Yes        | Yes                    |
| <i>Fit Statistics</i> |                                                    |            |                        |                                           |            |                        |
| Num.Obs.              | 2,225,802                                          | 2,225,802  | 2,225,802              | 2,225,802                                 | 2,225,802  | 2,225,802              |
| R <sup>2</sup>        | 0.006                                              | 0.007      | 0.006                  | 0.006                                     | 0.007      | 0.006                  |
| Mean Dep. Var.        | 0.00022                                            | 0.00064    | 0.00026                | 0.00022                                   | 0.00064    | 0.00026                |

Table 2: Geographic Exposure Predicts Recorded Exit

**Note:** LPMs on any-event indicators at the 5km hexagon-year level for all 8,763 cells. Columns (1)–(3) use the annual grid exposure ratio  $\rho_{i,t}^{\text{grid}} = \frac{d_{i,t}^1}{d_{i,t}^2}$  from the two closest active capitals. Columns (4)–(6) use distance to the closest active capital  $d_{i,t}^1$  (km), which measures remoteness without the second-nearest capital. “Confrontation (strict)” includes only destruction and uprising. All models include grid and year FE. SEs are (i) two-way clustered by cell and year and (ii) Conley (10km). See [SI D.2.11](#) for ever-active cells. Signif. Codes: .\*\*\*: 0.01, .\*\*: 0.05, .\*: 0.1.

Columns (4)–(6) distinguish geographic exposure from simple remoteness by using distance to the closest active capital  $d_{i,t}^1$  and discarding the second-nearest. The exit coefficient is small and marginal with clustered standard errors ( $p = 0.092$ ) but null with Conley standard errors ( $p = 0.162$ ); confrontation is also null. The contrast between the two sets of columns points to the relative geometry of competing rulers, rather than distance from the nearest ruler, as the relevant condition.

Across alternative samples and measures, the exit coefficient remains positive, although inference is weaker in some specifications. This holds among the 1,466 ever-active cells, in Poisson and Negative Binomial fixed-effects models, for desertion alone, with terrain-weighted least-cost-path distances, and when *daikansho* enter the choice set ([SI §§D.2.9–D.2.12](#) and [D.2.14](#)).

### 3.4. Exposure Shifts the Form of Non-Compliance Toward Exit

The preceding tests ask whether exit occurrence is higher under greater exposure. This section conditions on recorded non-compliance and asks which tactic a resisting village chose: exit or direct confrontation. It estimates composition within the resisting sample, not whether exposure makes a village resist. Separating these margins matters because more exit events could reflect more disputes rather than substitution between tactics.

The village and tax analyses use 46,086 villages compiled primarily from the *Kyū-daka–Kyūryō Torishirabe Chō* (Ledgers of Former Assessments and Territories, hereafter LFAT), a Meiji land census that records each village’s home domain at the end of the Tokugawa period. Village locations come from [Honda, Natsume, and Nemoto \(2022\)](#). When several domains taxed one village, I assign the village to the domain holding the largest share of its assessed productivity (*kokudaka*, a rice-denominated index of aggregate agricultural output rather than literal rice harvest alone). The data cover virtually all arable land in Japan except high-altitude, non-cultivable regions ([Figure 1\(a\)–\(b\)](#); [Figure 6](#)).

#### 3.4.1. Measuring Exposure to Alternative Authority

I measure each village’s exposure with  $\rho_i$ , the distance to its own lord’s capital divided by the distance to the nearest other lord’s capital. Unlike the grid exposure ratio  $\rho_{i,t}^{\text{grid}}$  in [Section 3.3.2](#), which uses the two nearest active capitals,  $\rho_i$  anchors on the village’s known home domain. Proximity to an outside authority lowered exit costs by facilitating the intervention on which mass exit depended ([Section 3.2](#)): guards or mediators could protect fleeing peasants, villagers could more easily reach neighboring capitals or shogunal offices, and authorities could more readily transfer allegiance. The measure is:

$$\rho_i := \frac{d_i^{\text{HC}}}{d_i^{\text{FC}}}$$

Read the ratio as relative access. A village 10km from its home capital and 20km from the nearest foreign capital has  $\rho_i = 0.5$ ; reversing those distances gives  $\rho_i = 2$ . Higher values place outside authority closer relative to home authority. Because the ratio rises when the home capital is farther away or the foreign capital is closer, Models 6 and 7

later separate the two distances. Here,  $d_i^{\text{HC}}$  is the straight-line distance from village  $i$  to its home-domain capital and  $d_i^{\text{FC}}$  is the distance to the nearest active non-home capital at the time of resistance. The data do not reconstruct village-specific historical roads, passes, checkpoints, guards, or realized travel routes, so the ratio measures geographic access to administrative centers rather than each village’s exact travel cost. I calculate both distances from the georeferenced resistance data and the time-varying capital panel in [Section 3.3.1](#). Active capitals include existing daimyō seats and the former seats of domains converted to tenryo, where a shogunal intendant (*daikan*) replaced the daimyō ([Vlastos 1986, 49](#)). The latter remain in the choice set throughout the tenryo period.<sup>[13]</sup>

### 3.4.2. Empirical Strategy: Home-Neighbor Dyad Fixed Effects

The home-neighbor dyad design compares otherwise similar resisting villages facing the same pair of rulers. Each observation is a village involved in recorded resistance in [Aoki \(1975\)](#), and the outcome indicates whether it used exit or direct confrontation. [Figure 5](#) illustrates the comparison.

Home-neighbor dyad fixed effects isolate within-pair variation in the distance ratio. Each dyad is an ordered pair of home and nearest-rival domains (for example, A–B and A–C in [Figure 5](#)). Dyad fixed effects absorb time-invariant home- and neighbor-domain characteristics, including persistent differences in coercive capacity, ruler ideology, agricultural endowments, and administrative quality, as well as stable pair-specific cross-border ties, transit corridors, and border controls. They remove persistent features shared by villages with the same home lord and nearest rival; they do not make those villages identical or absorb village geography. Within an A–B dyad, the estimate compares A villages whose nearest rival is B but whose relative position between the two capitals differs, as shown within each dashed circle in [Figure 5](#). The controls account for the measured local geography listed below, and decade fixed effects absorb period-wide shocks.

The main specification is:

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<sup>[13]</sup>In some cases, the shogunate may have administered converted territory from a nearby *daikansho* (a shogunal intendant’s district office) rather than from the former castle town. To address this, [SI D.2.13](#), [SI D.2.14](#), and [SI D.3.1](#) replicate all main analyses with the nearest capitals redefined to include *daikansho* locations.

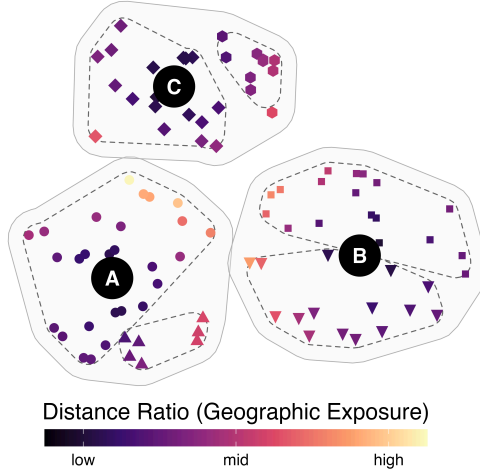


Figure 5: How Home-Neighbor Dyad Fixed Effects Isolate Within-Pair Variation

**Note:** A schematic, not data. Three domains (A, B, C) are drawn with their capitals as black circles and their villages as shapes shaded by the distance ratio  $\rho_i$ , the village's distance to its own lord's capital divided by its distance to the nearest other lord's capital. Dark shading marks villages sitting close to their own capital and far from any rival, light shading the reverse. A village's home domain and its nearest rival together define a dyad, shown by the dashed circles. Including a fixed effect for each dyad discards comparisons across dyads and retains only comparisons among villages facing the same pair of lords, so the estimate cannot be driven by differences between regions or between particular lords.

$$\text{Exit}_i = \beta_2 \ln(\rho_i) + X_i' \gamma_2 + \underbrace{\varphi_{j,k}}_{\text{home-neighbor dyad FE}} + \underbrace{\omega_t}_{\text{decade FE}} + \varepsilon_i$$

Here,  $i$  indexes a village-event, while  $j$  and  $k$  index its home and nearest-rival domains and  $t$  its decade.  $\text{Exit}_i$  equals one for exit and zero for direct confrontation. The logged  $\rho_i$  uses the capitals active at event  $i$ .<sup>[14]</sup>  $X_i$  contains latitude, longitude, their interaction, logged elevation and ruggedness, logged distance to the nearer of Edo or Kyoto, and logged wet-rice suitability.<sup>[15]</sup> The dyad fixed effects  $\varphi_{j,k}$  absorb time-invariant features of each ordered home-rival pair, while  $\omega_t$  absorbs decade-specific shocks.

### 3.4.3. Geographic Exposure Predicts Exit Share

Table 3 shows that exposure predicts exit rather than confrontation conditional on resistance. The coefficients are positive throughout and statistically significant in the

<sup>[14]</sup>The logged ratio constrains the coefficients on the two logged distances to equal magnitude with opposite signs. Linear hypothesis tests find no evidence against this restriction (SI D.3.5).

<sup>[15]</sup>Wet rice suitability is measured as potential output under irrigation and medium input, based on the Caloric Suitability Index developed by Galor and Özak (2016).

| Outcome:                         | Exit <sub>i</sub> |             |             |             |             |                |            |
|----------------------------------|-------------------|-------------|-------------|-------------|-------------|----------------|------------|
|                                  | Distance Ratio    |             |             |             |             | Abs. Distances |            |
| Models:                          | (1)               | (2)         | (3)         | (4)         | (5)         | (6)            | (7)        |
| <i>Variables</i>                 |                   |             |             |             |             |                |            |
| Distance ratio ( $\ln(\rho_i)$ ) | 0.0458            | 0.0519      | 0.0201      | 0.0212      | 0.0244      |                |            |
| Cluster SE                       | (0.0154)***       | (0.0141)*** | (0.0071)*** | (0.0072)*** | (0.0068)*** |                |            |
| Conley SE (10km)                 | (0.0117)***       | (0.0093)*** | (0.0078)*** | (0.0078)*** | (0.0085)*** |                |            |
| $\ln(d_i^{\text{HC}})$           |                   |             |             |             |             | 0.0160         | 0.0203     |
| Cluster SE                       |                   |             |             |             |             | (0.0104)       | (0.0083)** |
| Conley SE (10km)                 |                   |             |             |             |             | (0.0106)       | (0.0113)*  |
| $\ln(d_i^{\text{FC}})$           |                   |             |             |             |             | −0.0306        | −0.0309    |
| Cluster SE                       |                   |             |             |             |             | (0.0129)**     | (0.0135)** |
| Conley SE (10km)                 |                   |             |             |             |             | (0.0138)**     | (0.0148)** |
| <i>Controls</i>                  |                   |             |             |             |             |                |            |
| Decade FE                        | No                | Yes         | Yes         | Yes         | Yes         | Yes            | Yes        |
| Home Domain FE                   | No                | No          | Yes         | Yes         | No          | Yes            | No         |
| Neighbor Domain FE               | No                | No          | Yes         | Yes         | No          | Yes            | No         |
| Home-Nbr Dyad FE                 | No                | No          | No          | No          | Yes         | No             | Yes        |
| Geography                        | No                | No          | No          | Yes         | Yes         | Yes            | Yes        |
| <i>Fit Statistics</i>            |                   |             |             |             |             |                |            |
| Num.Obs.                         | 2,739             | 2,739       | 2,739       | 2,739       | 2,739       | 2,739          | 2,739      |
| R2                               | 0.028             | 0.225       | 0.621       | 0.626       | 0.678       | 0.627          | 0.678      |
| Mean Dep. Var.                   | 0.246             | 0.246       | 0.246       | 0.246       | 0.246       | 0.246          | 0.246      |

Table 3: Exposure to Alternative Authority Predicts Exit (Exit versus Confrontation)

**Note:** OLS estimates of each village’s exit-versus-confrontation choice conditional on resisting. Models (1)–(5) use the log distance ratio; Models (6)–(7) replace it with the two absolute distances ( $d_i^{\text{HC}}$ : home capital;  $d_i^{\text{FC}}$ : nearest foreign capital). Full results in Table A.32. SEs in parentheses: (i) clustered on home, nearest-neighbor, decade; (ii) Conley (10 km) on village coordinates; stars reflect that estimator’s  $p$ -value. Signif. Codes: \*\*\*: 0.01, \*\*: 0.05, \*: 0.1.

preferred controlled specifications. In Model 5, with home-neighbor dyad fixed effects and full controls, a one-standard-deviation increase in the log distance ratio ( $\approx 1.58$ ) raises exit probability by 3.9 percentage points, or 16% of the 24.6% baseline rate. The less restrictive home- and neighbor-FE specification (Model 4) yields a statistically indistinguishable 3.4-point estimate.

Models 6 and 7 separate the two distances because the ratio combines two distinct accounts. Greater distance from the home capital could increase exit through weaker administration at the periphery; shorter distance to the foreign capital instead increases access to an outside authority. In the preferred dyadic specification (Model 7), the home-capital coefficient is 0.0203 (cluster  $p = 0.022$ ; Conley  $p = 0.073$ ), while the foreign-capital coefficient is  $-0.0309$  (cluster  $p = 0.031$ ; Conley  $p = 0.037$ ). The two coefficients are statistically consistent with the ratio’s equal-and-opposite restriction ( $p = 0.924$ ;

SI D.3.5). A one-standard-deviation decrease in  $\ln(d_i^{\text{FC}})$  ( $\approx 0.92$ ) corresponds to 2.8 percentage points more exit, or 11% of the baseline rate. Foreign-capital proximity is therefore the larger and more robust component, although the home-capital estimate does not permit ruling out weaker peripheral administration.

Measurement and sample composition motivate one set of robustness checks. The preferred exit coefficient remains positive with alternative productivity and political-distance controls (SI D.2.1), after excluding repeat-resistance villages (SI D.2.2), and when dropping cross-island dyads whose straight-line routes cross sea straits (SI D.2.4). It also remains positive when adding shogunal intendant offices (*daikansho*) to the measured choice set (SI D.2.13).

Restricting exit to desertion retains 58–70% of the pooled point estimates across all three behavioral designs. The 50km conversion estimate is 0.0266 (cluster  $p = 0.031$ ; Conley  $p = 0.027$ ), the preferred village-choice coefficient is 0.0141 (cluster  $p = 0.051$ ; Conley  $p = 0.041$ ), and the grid-year coefficient is 0.000241 (cluster  $p = 0.082$ ; Conley  $p = 0.048$ ). These estimates isolate horizontal exit by observed action without inferring unrecorded destinations (SI §C.5, §D.2.8, and §D.2.9).

Alternative explanations motivate the other checks. Controlling for distance to the nearest domain border changes the distance-ratio coefficient little, while border distance itself is not associated with exit choice (SI D.3.2). A sequential choice decomposition merges the resisting sample with the comprehensive village ledger. It finds no detectable association between exposure and resistance rather than compliance in the main specification and yields an exit-versus-confrontation estimate similar to Model 5. A symmetric time-averaged exposure measure produces a small negative first-step estimate while preserving the positive composition estimate (SI D.3.3). The evidence therefore distinguishes resistance incidence from tactic composition rather than treating compliance as absent.

### 3.5. Exposure and Constrained Taxation

If viable exit disciplines rulers, exposed domains should tax less to retain their population. The economic sequence begins with competition over scarce labor. A neighboring ruler able to employ additional cultivators has reason to receive them, making the village's

exit option credible. The home ruler then anticipates losing both cultivators and the tax base they support and moderates extraction. When neighbors lack the capacity to absorb labor, geographic access alone creates a weaker threat. Systematic tax data survive only as a late-period cross-section, so this section seeks a suggestive fiscal trace rather than a causal estimate: a pooled relationship between exposure and taxation cannot distinguish exit from state-capacity decay. Neighbor characteristics provide a sharper test. The exit mechanism predicts a negative tax gradient only where neighboring domains are attractive enough to absorb migrants. Generic peripheral state-capacity decay instead predicts lower taxation where home administration is weak, regardless of neighboring labor demand, and therefore predicts no such asymmetry. I measure domain exposure by averaging the village distance ratio within each domain. Figure 6 maps the log ratio (panel a) and reported effective tax rates (panel b).

### 3.5.1. Empirical Strategy

Capital locations could be endogenous if rulers placed capitals or exchanged border villages to deter exit. Most capitals, however, were inherited from the Sengoku period (Hall 1955) and fixed by the 1615 one-castle-per-domain edict (*ikkoku ichijō rei*); later

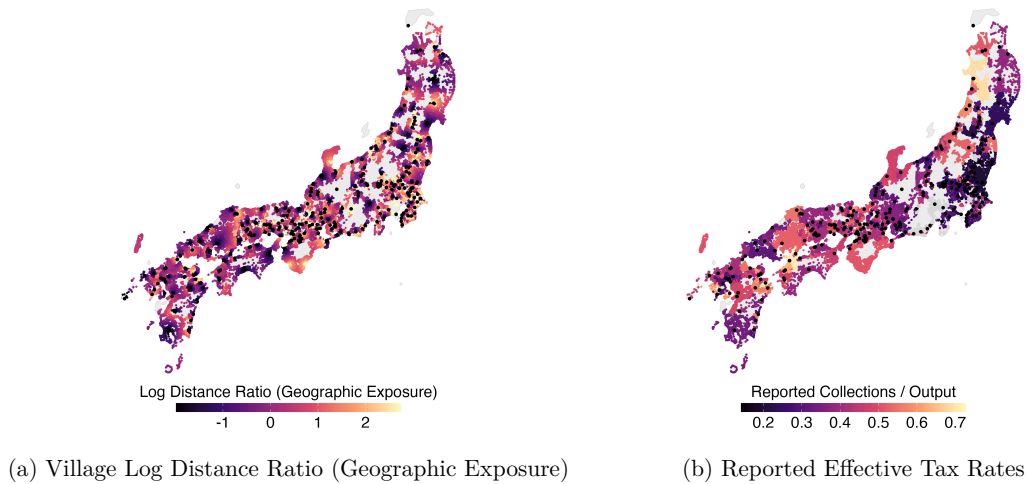


Figure 6: Geographic Exposure and Tax Rates Across Villages

**Note:** Panel (a) plots village locations shaded by the log distance ratio  $\ln(\rho_i) = \ln(d_i^{\text{HC}}/d_i^{\text{FC}})$ , where HC is the home-domain capital and FC is the nearest foreign-domain capital; higher values indicate greater geographic exposure. Panel (b) shades those villages by their domain's reported collections divided by reported output, the effective tax-rate measure used in the analysis. Gray points denote villages in domains without tax data, and black dots mark domain capitals.

relocations required shogunal approval and were rare. The shogunate also placed fudai vassals strategically and kept tozama lords on the periphery (Ravina 1999). I therefore control for each daimyō’s alliance status.<sup>[16]</sup> This also absorbs political ties that could affect both shogunal approval of relocations or village transfers and extractive capacity. Any deliberate adjustment to deter exit would reduce exposure and thus attenuate the negative tax relationship.

The home-capital distance could also proxy for domain scale, peripheral state capacity (Sng and Moriguchi 2014), or village development. Controls therefore include domain population,<sup>[17]</sup> agricultural productivity, territory size, the *gōshi system* (lower-ranking samurai serving as village officials, which raised monitoring costs), and the share of villages named “shinden” (new rice field). Tax-delivery costs are unlikely to drive the result because villages used nearby storage facilities rather than domain capitals.<sup>[18]</sup> Tokugawa institutions also limited local agents’ authority (Section 3.1). Residual confounding by scale or coercive capacity should attenuate the estimate because larger domains could field larger armies and enforce higher rates.

The outcome is the domain effective tax rate from Kodama and Kitajima (1989) (pp. 425–431): *shunodaka* (reported collections in rice equivalents) divided by *uchidaka* (the domain’s internal record of actual production, distinct from its public *kokudaka* assessment). This ratio measures the share of reported output collected.<sup>[19]</sup> The Meiji government compiled these data in 1869–1870 from each domain’s average production and collections over the preceding five years. Averaging reduces harvest noise, while widespread *jōmen-hō* (fixed-rate taxation) kept many domains’ rates administratively stable in the final Tokugawa decades (Iinuma 1974, 20–24; Smith 1958).

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<sup>[16]</sup>Daimyō are classified according to their relationship with the shogunate: shinpan (relatives), fudai (hereditary vassals who fought for the shogunate at Sekigahara in 1600), and tozama (outside lords who became allies only after the battle).

<sup>[17]</sup>Population is measured at the end of the Tokugawa period, so variables that use it should be read as late-period proxies rather than time-invariant characteristics.

<sup>[18]</sup>Fukui Prefecture (1994: 335–346) documents this practice in Fukui Domain.

<sup>[19]</sup>The measure is reported collections relative to reported output, not the ruler’s chosen extraction rate. It absorbs accounting conventions, regional assessment traditions, incentives when reporting to the new Meiji government, and late-period fiscal disturbances. I therefore treat the tax results as associational and emphasize neighbor heterogeneity rather than the pooled coefficient.

Because *shunodaka* and *uchidaka* cover 1865–69, they measure a late-period steady state rather than annual change. This window remains informative: *jōmen-hō* fixed nominal rates across decades (Smith 1958), limiting year-to-year variation, while the fiscal argument concerns long-run extraction rather than fiscal shocks. Mid-Tokugawa tax scatters in Smith (1958) are qualitatively consistent with the late-period cross-section, but comprehensive mid-period domain data are unavailable. Tax rates also varied within domains, yet systematic village-level data do not survive. The analysis therefore estimates domain-average relationships and cannot test discriminatory village-level tax schedules.<sup>[20]</sup>

I exclude non-daimyō territories, including shogunal *goryō* and shrine- or temple-administered *jisharyō*, because they lack comprehensive fiscal data and had limited tax autonomy (Murakami 1996).  $\rho_j$  averages the village distance ratio from Section 3.4.1 within each domain using end-period capitals.

### 3.5.2. The Tax–Exposure Link Binds Where Neighbors Attract Labor

The pooled benchmark regresses domain tax rates on exposure and controls:

$$\tau_j = \alpha_4 + \beta_4 \ln(\rho_j) + X_j' \gamma_4 + \varepsilon_j$$

Here  $\tau_j$  is domain  $j$ 's tax rate,  $\rho_j$  is its average village distance ratio, and  $X_j$  contains three control groups. Core controls are logged domain size and wet-rice suitability,<sup>[21]</sup> the share of new-farmland villages, shogunal alliance, and the gōshi system. Geographic controls are logged elevation and ruggedness and the monthly shares of rainy and snowy days.<sup>[22]</sup> Neighbor controls are weighted means of adjacent domains' territory, population, rice productivity, new farmlands, family ties to the focal domain, and shogunal alliance; weights count the focal villages for which each domain is the nearest neighbor.

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<sup>[20]</sup>Public goods could in principle substitute for tax relief, making tax rates an incomplete measure of oppression. Yet public works required corvée labor (*bueki*) from villages and thus added to their burdens, while petitions and resistance records center on excessive taxes and unfair assessments rather than missing infrastructure (Aoki 1975).

<sup>[21]</sup>The wet-rice measure is an agro-climatic potential-yield surface (Galor and Özak 2016) rather than an observed annual output series; I use it only as a proxy for labor-absorption capacity.

<sup>[22]</sup>Weather variables represent proportion of rainy/snowy days per month, imputed as weighted averages using daily weather data from 40 locations, with distance to domain capital as weight. Data from [Historical Weather Database](#) by Professor Minoru Yoshimura.

Table 4 reports a negative cross-sectional association between exposure and reported tax rates. Across Models 1–4, a one-standard-deviation increase in the log distance ratio ( $SD = 1.34$ ) corresponds to a 1.5–3.2 percentage-point decrease. The association remains negative with the full set of controls and in a pure spatial-lag model, although it is only marginal in the fully controlled spatial-lag model (Anselin and Rey 2014) (Models 5–6). Its direction matches Steele, Paik, and Tanaka (2017), who find that collective desertions, but not the broader category of insurrection, predict lower taxes at the domain-period level.

Several checks assess whether model choice or omitted variables account for the pooled association. The sign remains negative with alternative covariates, including kokudaka, samurai share, and population relative to wet-rice suitability (SI E.2.1). Applying the sensitivity analysis of Diegert, Masten, and Poirier (2022) shows that, when exposure- and outcome-side confounding are bounded equally, both SD ratios must reach about 0.554 before the identified set admits a nonnegative coefficient (SI E.2.2). Double/debiased machine learning, following Chernozhukov et al. (2018), tempers the result: ridge nuisance models retain negative estimates, but flexible stacked and random-forest specifications produce estimates near zero with wide confidence intervals (SI E.2.3). The sign also remains negative when daikansho enter the choice set (SI D.3.1). Thus the pooled OLS association is stable across conventional specifications but is not uniformly corroborated by flexible DML; the neighbor-heterogeneity test below asks whether the OLS pattern reflects exit rather than state-capacity decay.

To test the exit channel, I interact exposure with  $\hat{y}_{-j}$ , a discrete measure of neighboring domains’ outside-option quality:

$$\tau_j = \alpha_5 + \beta_5 [\ln(\rho_j) \times \hat{y}_{-j}] + \bar{\beta}_5 \ln(\rho_j) + \tilde{\beta}_5 \hat{y}_{-j} + X_j' \gamma_5 + \varepsilon_j$$

I proxy outside-option quality with neighboring domains’ demand for labor. Labor-scarce rulers attracted workers through land redistribution, tools and rations, irrigation, and other public goods (Hyugano 1957; Miyazaki 1995), with stronger incentives under greater scarcity. Persistent shortages in Kantō led domains to adopt peasant recruitment policies (*iri-byakushō seisaku*) offering tax exemptions, food and seed grants, and other settlement inducements (Gorai 1950).

| Outcome:                         | $\tau_j$    |             |            |            |                      |                      |
|----------------------------------|-------------|-------------|------------|------------|----------------------|----------------------|
| Models:                          | (1)         | (2)         | (3)        | (4)        | (5)                  | (6)                  |
| <i>Variables</i>                 |             |             |            |            |                      |                      |
| Distance ratio ( $\ln(\rho_j)$ ) | -0.0241     | -0.0203     | -0.0110    | -0.0112    | -0.0140              | -0.0094              |
| HC SE                            | (0.0053)*** | (0.0058)*** | (0.0055)** | (0.0055)** | (0.0047)***          | (0.0051)*            |
| Conley SE (70km)                 | (0.0066)*** | (0.0067)*** | (0.0056)** | (0.0055)** | (0.0058)**           | (0.0050)*            |
| <i>Controls</i>                  |             |             |            |            |                      |                      |
| Core                             | No          | Yes         | Yes        | Yes        | No                   | Yes                  |
| Geography                        | No          | No          | Yes        | Yes        | No                   | Yes                  |
| Neighbors                        | No          | No          | No         | Yes        | No                   | Yes                  |
| $\tau_j$ Spatial Lag             | No          | No          | No         | No         | Yes                  | Yes                  |
| <i>Fit Statistics</i>            |             |             |            |            |                      |                      |
| Num.Obs.                         | 232         | 230         | 230        | 230        | 230                  | 230                  |
| R2                               | 0.074       | 0.141       | 0.370      | 0.389      | 0.312 <sup>[a]</sup> | 0.439 <sup>[a]</sup> |
| Mean Dep. Var.                   | 0.392       | 0.394       | 0.394      | 0.394      | 0.394                | 0.394                |

Table 4: Pooled Benchmark: Domain-Level Geographic Exposure and Tax Rates

**Note:** OLS estimates (ML in Models 5–6) of reported effective tax rates (collections divided by reported output) on the log domain-average distance ratio. Full results in Table A.52. SEs in parentheses: (i) HC1 (ML SEs in 5–6); (ii) Conley (70 km) on capital coordinates, computed via two-step filtered-residual approximation (Anselin and Rey 2014, pp. 34–36); stars reflect that estimator’s  $p$ -value. Signif. Codes: \*\*\*: 0.01, \*\*: 0.05, \*: 0.1. <sup>[a]</sup> Nagelkerke pseudo-R2

The moderator operationalizes this labor-absorption capacity.  $\hat{y}_{-j}$  equals “Low” when neighbors’ weighted average labor-to-productivity ratio is above its median and “High” otherwise.<sup>[23]</sup> I weight each neighbor by the number of domain  $j$  villages for which its capital is nearest and divide its population by potential wet-rice yield from Galor and Özak (2016).<sup>[24]</sup> High  $\hat{y}_{-j}$  thus denotes neighbors with greater capacity and incentives to absorb migrants. The interaction absorbs neighbors’ average rice yield and territory size, which I drop from  $X_j$ .

The interaction probes the retention mechanism. Exposure should induce the home ruler to moderate extraction most strongly when a labor-scarce neighbor can credibly absorb departing cultivators. A countervailing possibility is that departure reduces the

<sup>[23]</sup> Hainmueller, Mummolo, and Xu (2019) show that continuous multiplicative-interaction models rest on a linear-interaction assumption that frequently fails in practice, and recommend replacing the continuous moderator with a discrete one precisely for settings like this, in which the moderating relationship is inherently nonlinear.

<sup>[24]</sup> I use the potential yield measure under irrigated conditions with medium inputs.

number of villagers available for simultaneous confrontation, weakening pressure on the home ruler when the neighboring jurisdiction is unattractive. The interaction does not identify this channel or imply that exit and confrontation are substitutes across a longer sequence. Its observable prediction is that the negative tax gradient should be strongest where absorption is credible.

Figure 7 shows the predicted asymmetry in every specification. The negative tax gradient is concentrated among domains with high  $\hat{y}_{-j}$ , whose neighbors had low labor-to-potential-yield ratios and strong demand for migrants. With geographic controls (Models 3 and 4), the association is statistically significant only for this group; estimates for domains facing labor-abundant neighbors are small and statistically insignificant. The pattern supports the retention argument: the negative gradient attenuates when the outside option is weak. It does not establish that exposure raises taxes among domains facing unattractive neighbors, the stronger implication that would follow if confrontation relief fully dominated. The interaction sharpens the comparison but does not eliminate domain- or region-level confounding, so the asymmetry remains associational rather than a causal estimate of ruler moderation. The same asymmetry appears when population density proxies for outside-option quality (SI E.3.3) and when the neighbor set comprises all domains within 100 km. A Hainmueller-Mummolo-Xu kernel-smoothed plot using the continuous moderator yields a negative association through the middle of the support but is flat rather than monotonically sharpening (SI E.3.4).

### 3.5.3. Additional Analyses

Neither simple remoteness nor border proximity explains the tax pattern. Adding average logged home-capital distance, the predictor implied by state-capacity decay (Sng and Moriguchi 2014), leaves the exposure coefficient negative (SI E.2.4); replacing the ratio with remoteness produces no significant negative association once geographic controls enter (SI E.3.1). Distance to the nearest border also does not predict taxes (SI E.3.2),<sup>[25]</sup> matching the village result in Section 3.4. The relevant exposure is to rival authority.

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<sup>[25]</sup>The boundary data were constructed following the procedure described in (Honda, Natsume, and Nemoto 2022).

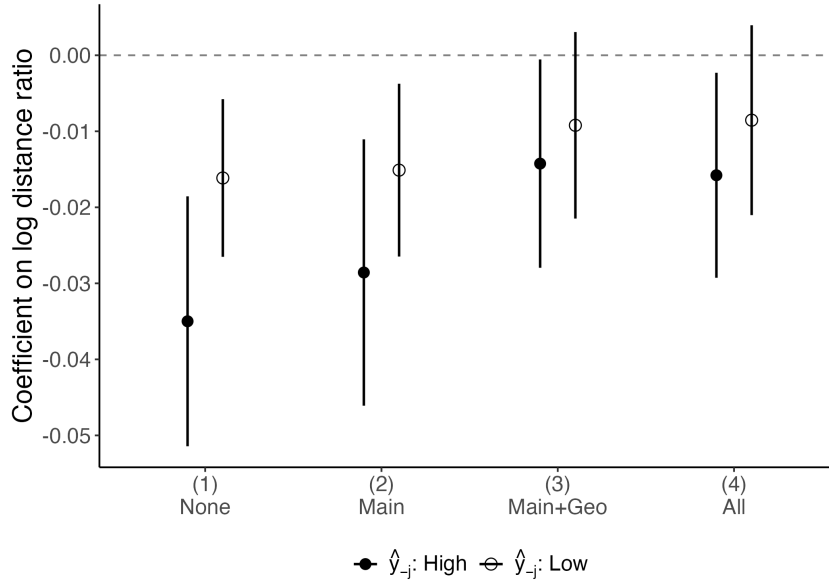


Figure 7: The Tax-Exposure Link Strengthens with Outside-Option Quality

**Note:** Group-specific OLS slopes on  $\ln(\rho_j)$ . Filled points marked High denote below-median neighbor labor-to-potential-yield ratios, hence greater capacity and incentive to absorb migrants; open points marked Low denote above-median ratios. Lines are 95% heteroskedasticity-consistent confidence intervals. Controls are cumulative: (1) none; (2) domain; (3) +geography; (4) +neighbor. Full table in [Table A.53](#).

## 4. CONCLUSION

The four designs recover successive links between outside access, exit, tactic choice, and extraction. Tenryō conversion and the grid-year panel associate improved outside access with more recorded exit. Among resisting villages, proximity to a rival capital supplied the more robust component of the shift toward exit. Exposed domains reported lower taxes when neighbors offered attractive outside options, an associational pattern consistent with ruler moderation. Across distinct units and sources of variation, the evidence depicts exit as bargaining sustained by shelter, mediation, and recognition. Reachable alternative authority made that strategy viable and placed rulers' tax bases at risk.

The paper develops a general theory of hegemonically ordered fragmentation: credible local alternatives can discipline rulers even when those rulers remain subordinate to a superior authority. Tokugawa supplies evidence for the theory's behavioral mechanism under demanding conditions because migration was prohibited and domains held territory at the shogunate's pleasure. A neighboring lord could protect and mediate, while the shogunate could sanction the home lord. The evidence is strongest on village behavior,

with taxation as a downstream trace; it does not show that these effects accumulated into sovereign states.

The same strategic structure suggests two applications beyond the case. In civil wars with rebel governance, civilians may shift support and resources between governments and insurgent authorities; rival protection can constrain predation only where another armed actor can govern and protect those who move ([Kalyvas 2006](#)). In competition for foreign investment, firms rather than villages are mobile, and governments face discipline when another jurisdiction offers a credible productive location ([Acharya and Lee 2023](#); [Kleven et al. 2020](#)). The Tokugawa designs identify neither relationship. They show why both analogies require an alternative authority that mobile actors can reach and reasonably expect to receive them.

## **DATA AVAILABILITY STATEMENT**

Replication data and code will be deposited in the International Organization Dataverse (<https://dataverse.harvard.edu/dataverse/IOJ>) upon conditional acceptance. The release will include the documented intermediate geographic inputs, scripts, and codebook needed to reproduce the reported tables and figures.

## **ETHICAL STATEMENT**

This research uses publicly available historical records and archival data on Tokugawa-era Japan and does not involve human subjects. No IRB review was required.

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## SUPPORTING INFORMATION

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## A. FORMAL ANALYSIS

### A.1. Setup

Consider two neighboring domains  $j \in \{1, 2\}$ . All primitives are common knowledge. Each ruler governs a continuum of villages of measure one and simultaneously chooses a common domain-level tax component  $\tau_j \in [0, 1]$ . A single rate matches the domain-average empirical estimand but excludes schedules keyed to observed village exit costs. Rulers cannot make side payments. The ruler equilibrium is Nash, and an enforceable tax cartel lies outside the model.

The timing is: (i) rulers set taxes; (ii) villages observe both rates and may petition their home ruler; (iii) conditional on ordinary petitioning not changing the rate, each village  $i_j \in [0, 1]$  chooses  $a_{i_j} \in \{\text{comply}, \text{fight}, \text{exit}\}$ ; and (iv) outside shelter or mediation may follow exit, after which the home ruler can concede and induce return. Here, *fight* is the model's compact label for direct confrontation. Let  $\mu_j^a$  denote the share choosing action  $a$ , and let each village in domain  $j$  produce yield  $y_j > 0$ .

A compliant village receives  $y_j(1 - \tau_j) + \varepsilon_{i_j}$ , where  $\varepsilon_{i_j} \sim \text{Uniform}[-\delta, \delta]$  and  $\delta > 0$ . A fight succeeds with probability  $\mu_j^{\text{fight}}$ , yields  $y_j$  if successful and zero otherwise, and costs  $k > 0$ . Exit yields the neighboring domain's net-of-tax income less village-specific exit cost:  $y_{-j}(1 - \tau_{-j}) - c_{i_j}$ , where  $c_{i_j} \in \{c_L, c_H\}$  and  $c_H > c_L > 0$ . The share  $\theta_j \in [0, 1]$  with cost  $c_L$  measures geographic exposure to a rival capital; the remaining  $1 - \theta_j$  face cost  $c_H$ .

Exit is a temporary bargaining strategy. A departing village bears  $c_{i_j}$ , removes labor and payments from the home tax base, and activates outside shelter or mediation. The model assumes that a post-exit concession matches the outside option and induces return. It does not derive the earlier petition failure, concession size, exit duration, or division of bargaining surplus.

Village utility is

$$u_{i_j}(a_{i_j}; \tau_j, \tau_{-j}) = \begin{cases} y_j(1 - \tau_j) + \varepsilon_{i_j} & \text{if } a_{i_j} = \text{comply} \\ y_j \mu_j^{\text{fight}} - k & \text{if } a_{i_j} = \text{fight} \\ y_{-j}(1 - \tau_{-j}) - c_{i_j} & \text{if } a_{i_j} = \text{exit} \end{cases}.$$

Ruler  $j$  collects taxes only from compliant villages and maximizes

$$u_j(\tau_j, \tau_{-j}) = [1 - \mu_j^{\text{fight}} - \mu_j^{\text{exit}}] y_j \tau_j = \mu_j^{\text{comply}} y_j \tau_j.$$

Before stating any proposition, the following assumption partitions resistance strategies: high-exit-cost villages choose between compliance and fight, whereas low-exit-cost villages choose between compliance and exit.

ASSUMPTION 1: For each  $j \in \{1, 2\}$ ,

$$\frac{k - c_L}{\delta - c_L} > \frac{y_j(1 - \theta_j)}{2\delta} \quad \text{and} \quad c_H > y_{-j} + \delta.$$

The results maintain the following regularity and interiority conditions:  $\delta > c_L$ ;  $2\delta > y_j(1 - \theta_j)$ ;  $0 < \mu_j^{\text{fight}}, \mu_j^{\text{exit}} < 1$ ;  $0 < \tau_j^* < 1$ ;  $0 < p_j < 1$  and  $0 < p_j + q_j < 1$ ; and  $P_j > 0$ , for both domains. These conditions ensure positive denominators, interior village cutoffs, and interior ruler best responses.

## A.2. Equilibrium Results

PROPOSITION 1 (Resistance Composition under Neighboring-Option Changes):

Under [Assumption 1](#) and interior village cutoffs, for fixed tax rates  $(\tau_j, \tau_{-j}) \in (0, 1)^2$ :

- (i)  $\frac{\partial \mu_j^{\text{exit}}}{\partial y_{-j}} > 0$  and  $\frac{\partial \mu_j^{\text{fight}}}{\partial y_{-j}} = 0$ ; (ii)  $\frac{\partial \mu_j^{\text{exit}}}{\partial \tau_{-j}} < 0$  and  $\frac{\partial \mu_j^{\text{fight}}}{\partial \tau_{-j}} = 0$ .

Only the exit payoff contains neighboring-domain quantities. The proposition is therefore a fixed-tax outside-option comparative static that motivates, but does not structurally reproduce, the conversion-based test: conversion is not a primitive in the model, and the empirical design does not identify a change in  $y_{-j}$  or  $\tau_{-j}$ . Its zero fight derivatives are not a general claim: comparison-based grievances or outside resources, shelter, and political pressure could instead raise confrontation. The empirical null concerns average incidence and does not rule out those channels in particular episodes.

PROPOSITION 2 (Geographic Exposure and Exit): Under [Assumption 1](#) and the stated regularity conditions: (a) whenever  $y_{-j} > \kappa_j^*$ , the equilibrium exit share among non-compliant villages is strictly increasing in  $\theta_j$ ; and (b) the equilibrium exit rate is strictly increasing in  $\theta_j$ .

Greater exposure expands the low-cost exit pool and contracts the high-cost fight pool. The ruler's tax response offsets only part of the resulting increase in exit because further tax

reductions eventually sacrifice more revenue per compliant village than they recover through retention.

PROPOSITION 3 (Governance Conflict and Equilibrium Tax Rate): Under [Assumption 1](#) and the stated regularity conditions, for each  $j \in \{1, 2\}$  there exists a unique threshold  $\kappa_j^* > 0$  such that  $\frac{\partial \tau_j^*}{\partial \theta_j} < 0$  if and only if  $y_{-j} > \kappa_j^*$ .

Within the model's single-action benchmark, exposure tightens the exit constraint but also shrinks the contemporaneous pool available for direct confrontation. An attractive neighbor makes the retention force dominate, so the ruler lowers taxes; below the threshold, confrontation relief can instead permit higher taxes. This event-level abstraction does not rule out the complementary use of exit and confrontation across a multi-stage protest sequence.

### A.3. Equilibrium and Proofs

Under [Assumption 1](#), the resistance shares are

$$\mu_j^{\text{fight}}(\tau_j) = \frac{(1 - \theta_j)[\delta - k - y_j(1 - \tau_j)]}{2\delta - y_j(1 - \theta_j)},$$

$$\mu_j^{\text{exit}}(\tau_j, \tau_{-j}) = \frac{\theta_j}{2\delta} [y_{-j}(1 - \tau_{-j}) - c_L - y_j(1 - \tau_j) + \delta].$$

For the fight share, a high-cost village fights when  $\varepsilon_{i_j} \leq y_j \mu_j^{\text{fight}} - k - y_j(1 - \tau_j)$ . With the uniform distribution clipped at zero and one, the induced map on  $[0, 1 - \theta_j]$  has slope zero on clipped portions and  $(1 - \theta_j) \frac{y_j}{2\delta} < 1$  on its interior. It is a contraction with a unique fixed point. Interiority places that point at the expression above, so all-fight and no-fight corners are not additional equilibria. The exit expression follows from the low-cost village's exit-versus-compliance cutoff.

### A.3.1. Proof of Proposition 1

PROOF: Fix  $j$  and hold  $(\tau_j, \tau_{-j})$  fixed. Differentiating the resistance-share expressions above gives

$$\frac{\partial \mu_j^{\text{exit}}}{\partial y_{-j}} = \theta_j \frac{1 - \tau_{-j}}{2\delta} > 0, \quad \frac{\partial \mu_j^{\text{exit}}}{\partial \tau_{-j}} = -\theta_j \frac{y_{-j}}{2\delta} < 0.$$

The fight expression contains neither  $y_{-j}$  nor  $\tau_{-j}$ , so both corresponding derivatives are zero.

### A.3.2. Equilibrium Tax Rates

Define

$$\begin{aligned} A_j &:= \frac{(1 - \theta_j)y_j}{2\delta - y_j(1 - \theta_j)} + \frac{\theta_j y_j}{2\delta}, \\ B_j &:= \frac{(1 - \theta_j)(\delta - y_j - k)}{2\delta - y_j(1 - \theta_j)} + \frac{\theta_j}{2\delta} [y_{-j} - c_L - y_j + \delta], \\ C_j &:= \frac{\theta_j y_{-j}}{2\delta}. \end{aligned}$$

Then total resistance is  $B_j + A_j \tau_j - C_j \tau_{-j}$ , and ruler utility is

$$u_j(\tau_j, \tau_{-j}) = y_j \tau_j [1 - B_j + C_j \tau_{-j} - A_j \tau_j].$$

Because  $\frac{\partial^2 u_j}{\partial \tau_j^2} = -2y_j A_j < 0$ , the unique best response is

$$\tau_j = p_j + q_j \tau_{-j}, \quad p_j = \frac{1 - B_j}{2A_j}, \quad q_j = \frac{C_j}{2A_j}.$$

Moreover,  $A_j \geq \theta_j \frac{y_j}{2\delta}$  implies  $q_j \leq \frac{y_{-j}}{2y_j}$  and hence  $q_1 q_2 \leq \frac{1}{4} < 1$ . The two best responses therefore intersect once and yield

$$\tau_j^* = \frac{p_j + q_j p_{-j}}{1 - q_j q_{-j}}.$$

### A.3.3. Proof of Proposition 2

PROOF: At fixed tax rates,  $\frac{\partial \mu_j^{\text{exit}}}{\partial \theta_j} = \frac{\mu_j^{\text{exit}}}{\theta_j} > 0$ . Writing  $D_j = 2\delta - y_j(1 - \theta_j) > 0$  and  $N_j = \delta - k - y_j(1 - \tau_j) > 0$ , differentiation gives  $\frac{\partial \mu_j^{\text{fight}}}{\partial \theta_j} = -2\delta \frac{N_j}{D_j^2} < 0$ .

First suppose  $y_{-j} \leq \kappa_j^*$ . Proposition 3 gives  $\frac{d\tau_j^*}{d\theta_j} \geq 0$ , and the other ruler's affine best response gives  $\frac{d\tau_{-j}^*}{d\theta_j} = q_{-j} \frac{d\tau_j^*}{d\theta_j} \geq 0$ . Let  $W_j = y_j - y_{-j}q_{-j}$ . Since  $A_{-j} \geq \theta_{-j} \frac{y_{-j}}{2\delta}$ ,  $W_j \geq \frac{y_j}{2} > 0$ . For  $E_j = y_{-j}(1 - \tau_{-j}^*) - c_L - y_j(1 - \tau_j^*) + \delta$ , therefore,  $\frac{dE_j}{d\theta_j} = W_j \frac{d\tau_j^*}{d\theta_j} \geq 0$ , and

$$\frac{d\mu_j^{*,\text{exit}}}{d\theta_j} = \frac{E_j}{2\delta} + \theta_j \frac{W_j}{2\delta} \cdot \frac{d\tau_j^*}{d\theta_j} > 0.$$

Now suppose  $y_{-j} > \kappa_j^*$ , so  $\frac{d\tau_j^*}{d\theta_j} < 0$ . At the optimum, compliance equals  $A_j \tau_j^*$ . Because  $A'_j < 0$ ,

$$\frac{d(\mu_j^{\text{fight}} + \mu_j^{\text{exit}})}{d\theta_j} = -A'_j \tau_j^* - A_j \frac{d\tau_j^*}{d\theta_j} > 0.$$

Meanwhile,

$$\frac{d\mu_j^{\text{fight}}}{d\theta_j} = \frac{[-2\delta N_j + (1 - \theta_j)y_j D_j \frac{d\tau_j^*}{d\theta_j}]}{D_j^2} < 0.$$

Thus exit increases because total resistance increases while fight decreases, proving part (b). For part (a), the quotient-rule numerator for the exit share is  $\left(\frac{d\mu_j^{*,\text{exit}}}{d\theta_j}\right) \mu_j^{*,\text{fight}} - \mu_j^{*,\text{exit}} \left(\frac{d\mu_j^{*,\text{fight}}}{d\theta_j}\right) > 0$ .

### A.3.4. Proof of Proposition 3

PROOF: Let  $\Omega_j := y_{-j}$ . Define the  $\Omega_j$ -free part of  $B_j$  and its associated intercept as

$$\tilde{B}_j := \frac{(1 - \theta_j)(\delta - y_j - k)}{2\delta - y_j(1 - \theta_j)} + \frac{\theta_j}{2\delta}(\delta - c_L - y_j), \quad \tilde{p}_j := \frac{1 - \tilde{B}_j}{2A_j}.$$

Then  $B_j = \tilde{B}_j + \theta_j \frac{\Omega_j}{2\delta}$  and  $p_j = \tilde{p}_j - q_j$ . Since

$$A'_j = \frac{y_j}{2\delta} - \frac{2\delta y_j}{[2\delta - y_j(1 - \theta_j)]^2} \leq 0,$$

$\frac{\partial q_j}{\partial \theta_j} = \Omega_j K_j > 0$ , where  $K_j = \frac{A_j - \theta_j A'_j}{4\delta A_j^2} > 0$ . Let  $P_j := \frac{\partial \tilde{p}_j}{\partial \theta_j} > 0$  and  $R_j := \theta_j \frac{q_{-j}}{4\delta A_j} \geq 0$ .

The sign of  $\frac{\partial \tau_j^*}{\partial \theta_j}$  is the sign of

$$\Xi_j(\Omega_j) = (1 - R_j \Omega_j) P_j + \Omega_j K_j [p_{-j} + q_{-j} \tilde{p}_j - 1].$$

This expression is affine in  $\Omega_j$  with a strictly negative slope under best-response interiority, and  $\Xi_j(0) = P_j > 0$ . It therefore crosses zero exactly once, at

$$\kappa_j^* := \frac{P_j}{[R_j P_j - K_j(p_{-j} + q_{-j} \tilde{p}_j - 1)]} > 0.$$

Hence  $\frac{\partial \tau_j^*}{\partial \theta_j} < 0$  exactly when  $y_{-j} > \kappa_j^*$ .

### A.4. Non-Vacuousness

The maintained conditions hold on a non-empty open set. One example is  $\delta = 10$ ,  $y_1 = y_2 = 18$ ,  $k = \frac{7}{2}$ ,  $c_L = \frac{1}{2}$ ,  $c_H = 29$ , and  $\theta_1 = \theta_2 = \frac{5}{6}$ . At these values,  $A_j = \frac{63}{68}$ ,  $B_j = \frac{77}{272}$ ,  $C_j = \frac{3}{4}$ ,  $p_j = \frac{65}{168}$ ,  $q_j = \frac{17}{42}$ , and  $\tau_j^* = \frac{13}{20}$ . The equilibrium resistance shares are  $\mu_j^{\text{fight}} = \frac{1}{510}$  and  $\mu_j^{\text{exit}} = \frac{19}{48}$ , while  $P_j = \frac{2}{21} > 0$  and  $\kappa_j^* \approx 8.49 < y_{-j} = 18$ . Every maintained inequality is strict, so the admissible parameter set contains an open neighborhood of this vector.

## B. DESCRIPTIVE STATISTICS

### B.1. Village Resistance Data

|                          | Unique | Missing % | Mean     | SD      | Min     | Median   | Max      | Histogram                                                                             | Source                                                              |
|--------------------------|--------|-----------|----------|---------|---------|----------|----------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| <b>Basic</b>             |        |           |          |         |         |          |          |                                                                                       |                                                                     |
| Home Ruler               | 168    | 0         | —        | —       | —       | —        | —        | —                                                                                     | <a href="#">Aoki (1975)</a>                                         |
| Nbr Ruler                | 224    | 0         | —        | —       | —       | —        | —        | —                                                                                     | <a href="#">Aoki (1975)</a>                                         |
| Year                     | 218    | 0         | 1,777.3  | 59.0    | 1,615.0 | 1,782.0  | 1,867.0  |    | <a href="#">Aoki (1975)</a>                                         |
| Decade                   | 26     | 0         | 1,772.7  | 59.2    | 1,610.0 | 1,780.0  | 1,860.0  |    | <a href="#">Aoki (1975)</a>                                         |
| <b>Outcome</b>           |        |           |          |         |         |          |          |                                                                                       |                                                                     |
| Exit                     | 2      | 0         | 0.2      | 0.4     | 0.0     | 0.0      | 1.0      |    | <a href="#">Aoki (1975)</a>                                         |
| Direct Confrontation     | 2      | 0         | 0.8      | 0.4     | 0.0     | 1.0      | 1.0      |    | <a href="#">Aoki (1975)</a>                                         |
| <b>Predictor</b>         |        |           |          |         |         |          |          |                                                                                       |                                                                     |
| $\rho_i$                 | 2,529  | 0         | 7.1      | 61.3    | 0.0     | 1.0      | 2,792.0  |    | —                                                                   |
| $\ln(\rho_i)$            | 2,529  | 0         | 0.2      | 1.5     | -5.6    | 0.0      | 7.9      |    | —                                                                   |
| Dist. to Home Cap        | 2,517  | 0         | 45.8     | 85.5    | 0.0     | 21.2     | 1,096.0  |   | —                                                                   |
| Dist. to Nbr Cap         | 2,508  | 0         | 24.0     | 29.7    | 0.0     | 16.4     | 494.5    |  | —                                                                   |
| $\rho_i^{LCP}$           | 168    | 0         | 7.2      | 28.1    | 0.2     | 1.7      | 601.7    |  | —                                                                   |
| <b>Controls</b>          |        |           |          |         |         |          |          |                                                                                       |                                                                     |
| Latitude                 | 2,458  | 0         | 35.7     | 2.0     | 27.7    | 35.4     | 42.0     |  | <a href="#">Honda, Natsume, and Nemoto (2022)</a> and other sources |
| Longitude                | 2,445  | 0         | 136.5    | 3.3     | 128.9   | 136.6    | 142.0    |  | <a href="#">Honda, Natsume, and Nemoto (2022)</a> and other sources |
| Dist. to Edo             | 2,484  | 0         | 413.6    | 235.6   | 29.4    | 413.2    | 1,356.8  |  | —                                                                   |
| Dist. to Edo or Kyoto    | 2,484  | 0         | 234.0    | 152.4   | 0.0     | 191.1    | 1,041.7  |  | —                                                                   |
| Elevation                | 103    | 0         | 19.9     | 20.9    | 0.0     | 12.0     | 108.0    |  | Geospatial Information Authority of Japan                           |
| Ruggedness               | 146    | 0         | 3.7      | 3.8     | 0.0     | 2.5      | 22.6     |  | —                                                                   |
| Wet-Rice Suitability     | 716    | 0         | 18,549.9 | 3,417.3 | 0.0     | 19,003.2 | 22,570.0 |  | <a href="#">Galor and Özak (2016)</a>                               |
| Avg. Caloric Suitability | 1,133  | 0         | 2,221.3  | 242.7   | 941.1   | 2,297.5  | 2,531.2  |  | <a href="#">Galor and Özak (2016)</a>                               |

Table A.5: Summary Statistics of the Village Resistance Data ( $N = 2,741$ )

## B.2. Source-Category and Actor Audit

| Source section                    | Printed pages | PDF pages | OCR start | OCR end | Used in audit |
|-----------------------------------|---------------|-----------|-----------|---------|---------------|
| Peasant uprisings                 | 7–364         | 9–366     | 51        | 23323   | Yes           |
| Urban disturbances                | 365–410       | 367–412   | 23324     | 25841   | No            |
| Village conflicts (murakata sodo) | 411–620       | 413–622   | 25842     | 36731   | No            |
| Province-wide petitions           | 621–636       | 623–638   | 36732     | 37490   | No            |

Table A.6: Source-Section Boundary in Aoki’s Chronology

**Note:** Aoki prints separate chronologies. The project comparison script reads OCR lines 51–23,324 (lines[50:23324] in Python’s zero-based indexing); the last included line is the heading of the next section, so substantive comparison content ends at line 23,323. The separately printed *murakata sōdō* chronology begins at PDF p. 413 (printed p. 411; OCR line 25,842) and is outside that window. Page boundaries were checked against the PDF’s form-feed markers and title pages. This audit establishes exclusion of the separate source category, not the household tenure status of every actor named within included peasant-uprising entries. Source: (Aoki 1975).

| Sample                        | Estimate | Cluster $p$ | Conley $p$ | Events | Rows |
|-------------------------------|----------|-------------|------------|--------|------|
| Baseline                      | 0.0244   | 0.0013      | 0.0041     | 909    | 2739 |
| Exclude tenure-keyword events | 0.0218   | 0.0008      | 0.0113     | 890    | 2672 |

Table A.7: Village-Choice Sensitivity to Tenure-Related Narratives

**Note:** The restricted sample removes entire resistance events whose demand narrative contains *kosaku* (tenant), *jinushi* (landlord), or *komae* (smallholder). This excludes nineteen events and sixty-seven village rows, only one of which is coded as exit. Both rows use the preferred home-neighbor dyad and decade fixed effects with the main geographic controls. Cluster inference is three-way by home ruler, neighboring ruler, and decade; Conley inference uses a 10 km cutoff. The screen is a conservative content proxy, not a household-level tenure classification.

## B.3. Recorded Return after Exit

| ID   | Year | Home | Village | Recorded end                                   | Aoki p. |
|------|------|------|---------|------------------------------------------------|---------|
| 809  | 1759 | 松山   | 久妙寺     | Returned after reconciliation                  | 123     |
| 2822 | 1788 | 宇和島  | 惣川村西    | Required to return; later punishment           | 166     |
| 801  | 1800 | 松山   | 波万      | Persuaded and returned that day                | 181     |
| 2823 | 1813 | 宇和島  | 岩木      | Domain persuasion; return and imprisonment     | 196     |
| 2922 | 1818 | 吉田   | 俵津蒲     | Returned on the twenty-fifth                   | 204     |
| 2924 | 1820 | 吉田   | 奥野川     | Stopped or persuaded; return and repression    | 207     |
| 2824 | 1827 | 宇和島  | 富野川     | Persuaded to return; later exile or relocation | 218     |

Table A.8: Exit Events Explicitly Recording Return

**Note:** These seven analytic-sample narratives explicitly record return after flight or attempted flight. They establish that some recorded exits were temporary, but they do not identify a representative return rate because Aoki did not code duration or final disposition systematically. Only ID 809 states that return followed reconciliation; the other entries record persuasion, an order to return, administrative transfer, or subsequent punishment. Source: (Aoki 1975).

## B.4. Audit of Destination Codes

| Tactic            | Foreign-coded | Edo-coded | Kyoto-coded | Uncoded | Total |
|-------------------|---------------|-----------|-------------|---------|-------|
| Desertion         | 53            | 1         | 0           | 126     | 180   |
| End-run appeal    | 7             | 112       | 1           | 20      | 140   |
| Unclassified exit | 2             | 0         | 0           | 6       | 8     |
| Total             | 62            | 113       | 1           | 152     | 328   |

Table A.9: Destination and Addressee Information from Aoki’s Incident Descriptions

**Note:** Counts cover the 1615–1868 sample and collapse repeated village records to one event using `revolt_id`. Destination and addressee information was extracted from Aoki’s incident descriptions. The labels include both physical locations and institutional addressees: record checks find home authorities or internal locations under `foreign`, and Osaka, Kyoto, or a lord’s Edo residence under `edo`. The table reports these descriptive labels as coded; they do not map directly onto neighboring domains, daikansho, or the shogunate. At the village-record level, 471 of 765 sample records have a nonmissing code. Source: (Aoki 1975).

## B.5. Domain Data

|                       | Unique | NA % | Mean        | SD          | Min      | Median      | Max          | Hist.                                                                                 | Source                                              |
|-----------------------|--------|------|-------------|-------------|----------|-------------|--------------|---------------------------------------------------------------------------------------|-----------------------------------------------------|
| <b>Outcome</b>        |        |      |             |             |          |             |              |                                                                                       |                                                     |
| Exits                 | 23     | 0    | 2.2         | 5.5         | 0.0      | 0.0         | 37.0         |    | Aoki (1975)                                         |
| Direct Confrontations | 39     | 0    | 7.0         | 14.4        | 0.0      | 1.0         | 164.0        |    | Aoki (1975)                                         |
| Resistance Events     | 45     | 0    | 9.2         | 17.7        | 0.0      | 2.0         | 201.0        |    | Aoki (1975)                                         |
| Exit Rate             | 56     | 40   | 0.3         | 0.4         | 0.0      | 0.1         | 1.0          |    | Aoki (1975)                                         |
| Tax Rate ( $\tau_j$ ) | 256    | 7    | 0.4         | 0.1         | 0.1      | 0.4         | 0.7          |    | Kodama and Kijima (1989)                            |
| <b>Predictor</b>      |        |      |             |             |          |             |              |                                                                                       |                                                     |
| $\rho_j$              | 241    | 12   | 7.3         | 15.3        | 0.1      | 2.2         | 138.5        |    | —                                                   |
| $\ln(\rho_j)$         | 241    | 12   | 1.0         | 1.3         | -2.4     | 0.8         | 4.9          |    | —                                                   |
| Dist. to Home Cap     | 241    | 12   | 49.0        | 66.7        | 2.5      | 22.4        | 442.5        |    | —                                                   |
| Dist. to Nbr Cap      | 242    | 12   | 16.1        | 10.8        | 3.9      | 12.5        | 80.2         |    | —                                                   |
| $\rho_j^{LCP}$        | 242    | 12   | 239.0       | 1,988.8     | 0.1      | 2.7         | 24,956.3     |    | —                                                   |
| <b>Controls: Main</b> |        |      |             |             |          |             |              |                                                                                       |                                                     |
| Wet-Rice Suitability  | 242    | 12   | 3,011,700.3 | 4,937,787.1 | 48,040.8 | 1,232,259.1 | 41,226,133.2 |  | Galor and Özak (2016)                               |
| Territory Size        | 258    | 6    | 811.0       | 1,785.9     | 9.5      | 257.1       | 16,262.8     |  | Honda, Natsume, and Nemoto (2022) and other sources |
| Population            | 255    | 7    | 87,781.0    | 159,087.3   | 2,466.0  | 30,833.0    | 1,065,910.0  |  | LFAT                                                |
| Pop. Density          | 249    | 9    | 166.5       | 125.1       | 11.9     | 150.2       | 1,285.0      |  | —                                                   |
| # New Land            | 32     | 12   | 7.8         | 27.9        | 0.0      | 1.0         | 232.0        |  | —                                                   |
| Shogunate Kin (0/1)   | 2      | 0    | 0.6         | 0.5         | 0.0      | 1.0         | 1.0          |  | Kimura, Fujino, and Murakami (2015)                 |
| Goushi System (0/1)   | 2      | 0    | 0.1         | 0.3         | 0.0      | 0.0         | 1.0          |  | Kimura, Fujino, and Murakami (2015)                 |

Table A.10: Summary Statistics of the Domain Data ( $N = 275$ )

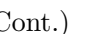
|                              | Unique | Missing % | Mean        | SD          | Min       | Median      | Max          | Histogram                                                                             | Source                                              |
|------------------------------|--------|-----------|-------------|-------------|-----------|-------------|--------------|---------------------------------------------------------------------------------------|-----------------------------------------------------|
| <b>Controls: Geography</b>   |        |           |             |             |           |             |              |                                                                                       |                                                     |
| Latitude                     | 265    | 1         | 35.5        | 1.7         | 31.6      | 35.3        | 41.4         |    | Kimura, Fujino, and Murakami (2015)                 |
| Longitude                    | 265    | 1         | 136.5       | 3.3         | 128.8     | 136.6       | 141.5        |    | Kimura, Fujino, and Murakami (2015)                 |
| Mean Elevation               | 242    | 12        | 16.1        | 13.6        | 2.6       | 12.0        | 88.4         |    | —                                                   |
| Mean TRI                     | 240    | 12        | 3.2         | 1.8         | 0.2       | 2.9         | 8.9          |    | —                                                   |
| Dist. to Edo or Kyoto        | 242    | 12        | 188.5       | 148.9       | 25.2      | 135.0       | 671.3        |    | —                                                   |
| % Rainy Days                 | 251    | 3         | 0.1         | 0.0         | 0.1       | 0.1         | 0.2          |    | Yoshimura HWDB <sup>[26]</sup>                      |
| % Snowy Days                 | 251    | 3         | 0.0         | 0.0         | 0.0       | 0.0         | 0.1          |    | Yoshimura HWDB                                      |
| <b>Controls: Neighbor</b>    |        |           |             |             |           |             |              |                                                                                       |                                                     |
| Prop. Same Fam.              | 65     | 12        | 0.1         | 0.2         | 0.0       | 0.0         | 1.0          |    | Kimura, Fujino, and Murakami (2015)                 |
| % New Farmland               | 78     | 12        | 32.9        | 69.2        | 0.0       | 7.5         | 421.0        |    | —                                                   |
| Avg. Wet-Rice Suitability    | 242    | 12        | 2,135,287.7 | 3,139,273.3 | 288,682.9 | 1,350,670.1 | 41,057,071.8 |  | Galor and Özak (2016)                               |
| Avg. Territory Size          | 242    | 12        | 433.8       | 602.5       | 12.4      | 278.0       | 6,926.3      |  | Honda, Natsume, and Nemoto (2022) and other sources |
| Avg. Population              | 239    | 13        | 56,178.5    | 84,147.0    | 6,264.6   | 31,723.2    | 1,061,486.4  |  | LFAT                                                |
| Avg. Pop. Density            | 239    | 13        | 175.5       | 107.1       | 39.9      | 161.0       | 1,028.7      |  | —                                                   |
| Avg. Pop. Density (Prod.)    | 239    | 13        | 0.0         | 0.0         | 0.0       | 0.0         | 0.2          |  | —                                                   |
| <b>Controls: Alternative</b> |        |           |             |             |           |             |              |                                                                                       |                                                     |
| Samurai Pop.                 | 230    | 16        | 70,945.3    | 134,463.8   | 614.0     | 21,519.5    | 896,808.0    |  | Kimura, Fujino, and Murakami (2015)                 |
| # Villages                   | 170    | 6         | 196.5       | 304.5       | 1.0       | 82.5        | 2,243.0      |  | LFAT                                                |
| Dist. to Edo                 | 242    | 12        | 370.8       | 244.7       | 42.2      | 309.4       | 1,039.5      |  | —                                                   |

Table A.11: Summary Statistics of the Domain Data (Cont.)

<sup>[26]</sup>Historical Weather Database made available online by Minoru Yoshimura: <http://tk2-202-10627.vs.sakura.ne.jp/>

## C. EVENT STUDY ROBUSTNESS

### C.1. Tenryo-Conversion Universe

The treatment set is constructed from the event seed `analysis/data/domain/domain_events.csv`, itself built from (Kimura, Fujino, and Murakami 2015) and (Kodama and Kitajima 1989). Across the 1615–1868 panel window, these sources record 79 domain dissolutions. Of these, 75 (95%) were followed by direct shogunal administration (tenryō); the remaining four passed directly to another daimyō and are excluded from the treatment group per the rule stated in Section 3.3.1. The 75 dissolve-to-tenryō events correspond to 72 unique domains, three of which experienced a second dissolve-to-tenryō event in a later spell. Restricting to domains active in the panel and retaining each domain’s first qualifying dissolution drops four domains whose earliest in-panel dissolution was a daimyō-to-daimyō transfer (their later tenryō conversion occurs outside the matched-control window). These filters yield the 55 treated domains listed in Table A.12.

The list spans the full sample window, with 49 cases concentrated before 1750 and 6 after. The event seed supplied a nonblank reason for 40 of 55 cases, none with a volume or page citation. The local audit in SI C.2 now supplies a page-level disposition for all 55, fills the fifteen blank formal reasons, and corrects Kaibara. It confirms Nagamine, Nariwa, and Tamanawa as conversions, while identifying noncomparable events, ambiguous intervening administrations, and four date errors. The table is therefore both a typology and an audit trail for the baseline coding; the sensitivity analysis does not presume that every row is cleanly comparable. The cohort split in SI C.8 confirms that the baseline exit response loads on the pre-1750 sub-sample that holds 49 of the 55 provisional cases.

| No. | Year | Domain                 | Province   | Provisional Reason                   |
|-----|------|------------------------|------------|--------------------------------------|
| 1   | 1617 | Itabashi (板橋藩)         | Shimotsuke | transfer                             |
| 2   | 1618 | Nagamine (長峰藩)         | Echigo     | transfer                             |
| 3   | 1619 | Musashi Komuro (武蔵小室藩) | Musashi    | attainder (Ina house)                |
| 4   | 1632 | Ōmori (大森藩)            | Ōmi        | kokudaka halved                      |
| 5   | 1639 | Fukōzu (深溝藩)           | Mikawa     | residence moved                      |
| 6   | 1642 | Nariwa (成羽藩)           | Bitchū     | tenryō after transfer                |
| 7   | 1645 | Fuchū (府中藩)            | Hitachi    | no heir                              |
| 8   | 1650 | Kaibara (柏原藩)          | Tanba      | family dispute (oie sōdō)            |
| 9   | 1653 | Toyooka (豊岡藩)          | Tajima     | no heir                              |
| 10  | 1656 | Funai (府内藩)            | Bungo      | no heir                              |
| 11  | 1657 | Marugame (丸亀藩)         | Sanuki     | no heir                              |
| 12  | 1658 | Kakegawa (掛川藩)         | Tōtōmi     | no heir                              |
| 13  | 1658 | Yashima (矢島藩)          | Dewa       | early-period abolition               |
| 14  | 1660 | Sakura (佐倉藩)           | Shimōsa    | misconduct                           |
| 15  | 1665 | Mibu (壬生藩)             | Shimotsuke | no heir                              |
| 16  | 1665 | Saijō (西条藩)            | Iyo        | misconduct (negligence)              |
| 17  | 1666 | Miyazu (宮津藩)           | Tango      | family dispute (oie sōdō)            |
| 18  | 1668 | Shimabara (島原藩)        | Hizen      | misgovernment                        |
| 19  | 1670 | Harima Shingū (播磨新宮藩)  | Harima     | no heir                              |
| 20  | 1671 | Ichinoseki (一関藩)       | Mutsu      | Date affair                          |
| 21  | 1676 | Asō (麻生藩)              | Hitachi    | no heir                              |
| 22  | 1677 | Omigawa (小見川藩)         | Shimōsa    | no heir (sub-daimyō)                 |
| 23  | 1678 | Yamazaki (山崎藩)         | Harima     | no heir                              |
| 24  | 1679 | Kururi (久留里藩)          | Kazusa     | insanity                             |
| 25  | 1679 | Niwase (庭瀬藩)           | Bitchū     | no heir                              |
| 26  | 1681 | Numata (沼田藩)           | Kōzuke     | misgovernment (illicit construction) |
| 27  | 1681 | Takada (高田藩)           | Echigo     | Echigo sōdō                          |
| 28  | 1683 | Tatebayashi (館林藩)      | Kōzuke     | political (Tsunayoshi enthronement)  |

Table A.12: The 55 Tenryō-Conversion Events Used as Treatment in Table 1 (Part 1 of 2)

**Note:** Cases are drawn from (Kimura, Fujino, and Murakami 2015) and (Kodama and Kitajima 1989) via the event seed in `analysis/data/domain/domain_events.csv`. The universe of abolitions recorded in the file is 79; removing the 4 cases coded as direct transfers to another daimyō leaves 75 provisional dissolve-to-tenryō events. Restricting to domains active in 1615–1868 and retaining the first qualifying dissolution per domain yields the 55 rows shown. The reason field is an audit aid, not a completed typology: only 40 source rows have a nonblank reason, none has a volume or page citation, and web verification uncovered corrections and several reductions, partitions, or temporary transfers that require source-volume review. Pre-1750 cohort: 49 cases; post-1750 cohort: 6 cases.

| No. | Year | Domain                   | Province   | Provisional Reason                |
|-----|------|--------------------------|------------|-----------------------------------|
| 29  | 1684 | Sanuki (佐貫藩)             | Kazusa     | misconduct                        |
| 30  | 1687 | Fukumoto (福本藩)           | Harima     | partition-abolition               |
| 31  | 1687 | Karasuyama (烏山藩)         | Shimotsuke | violation of bakufu law           |
| 32  | 1687 | Sōmi (沢海藩)               | Echigo     | territory confiscated             |
| 33  | 1689 | Fukami (深見藩)             | Sagami     | attainder (Genroku 2)             |
| 34  | 1689 | Kitami (喜多見藩)            | Musashi    | attainder (Genroku 2)             |
| 35  | 1692 | Hachiman (八幡藩)           | Mino       | no heir                           |
| 36  | 1692 | Takayama (高山藩)           | Hida       | political (Kanamori transfer)     |
| 37  | 1693 | Matsuyama-Bitchū (備中松山藩) | Bitchū     | no heir                           |
| 38  | 1698 | Fukuyama (福山藩)           | Bingo      | no heir                           |
| 39  | 1698 | Sano (佐野藩)               | Shimotsuke | political (Hotta transfer)        |
| 40  | 1698 | Yoshii (吉井藩)             | Kōzuke     | political (Hotta transfer)        |
| 41  | 1701 | Akō (赤穂藩)                | Harima     | Asano incident (violence)         |
| 42  | 1702 | Nagashima (長島藩)          | Ise        | violence                          |
| 43  | 1703 | Tamanawa (玉縄藩)           | Sagami     | transfer to Ōtaki                 |
| 44  | 1704 | Kōfu (甲府藩)               | Kai        | political (Ienobu enthronement)   |
| 45  | 1704 | Yamura (谷村藩)             | Kai        | political (linked to Kōfu)        |
| 46  | 1712 | Hōjō (北条藩)               | Awa        | misgovernment (Man'goku uprising) |
| 47  | 1724 | Numazu (沼津藩)             | Suruga     | political (Mizuno transfer)       |
| 48  | 1725 | Matsumoto (松本藩)          | Shinano    | insanity (assault in Edo castle)  |
| 49  | 1746 | Takayanagi (高柳藩)         | Echigo     | transfer to Mikusa                |
| 50  | 1767 | Obata (小幡藩)              | Kōzuke     | Meiwa incident (Yamagata Daini)   |
| 51  | 1786 | Sagara (相良藩)             | Tōtōmi     | political (Tanuma fall)           |
| 52  | 1788 | Ōmi Komuro (近江小室藩)       | Ōmi        | territory confiscated             |
| 53  | 1807 | Matsumae (松前藩)           | Ezo        | Ezo defence (temporary seizure)   |
| 54  | 1866 | Hamada (浜田藩)             | Iwami      | Chōshū War defeat                 |
| 55  | 1868 | Kaifuchi (貝淵藩)           | Kazusa     | Boshin War (pro-bakufu)           |

Table A.13: The 55 Tenryō-Conversion Events Used as Treatment in Table 1 (Part 2 of 2)

**Note:** Continuation of Table A.12. Same source definitions apply.

## C.2. Local-Source Treatment Audit

The local *Hanshi Daijiten* and *Hanshi Sōran* volumes supply page-level dispositions for all 55 baseline rows. Twenty are retained at their recorded date, twenty-one are excluded as noncomparable, ten are ambiguous because the intervening administrator is unstated, and four have the wrong date. Three date errors are correctable: Takayanagi to 1739, Kōfu to 1724, and Sagara to 1787; Numazu is excluded because the source records no transition in 1724. The audit also fills every formerly blank reason, corrects Kaibara, and confirms Nagamine, Nariwa, and Tamanawa as conversions.

The source-stated formal causes comprise 22 succession, partition, or incapacity cases; 18 misconduct, misgovernment, or political-sanction cases; 10 administrative or strategic reorganizations; 2 wartime or regime-change cases; and 3 coding errors or non-events. No source states fiscal distress or peasant unrest as the formal cause of a baseline row. Potential links to unrest are recorded separately as a cross-cutting evidentiary screen rather than inferred from event proximity. [Table A.15](#) reports this cause-by-disposition cross-tab and retains the full 55-row audit; [Table A.18](#) reports strict, ambiguity-inclusive, and combined source-plus-unrest screens.

| Grouped formal cause                             | All rows  | Retain    | Exclude   | Ambig.    | Wrong date |
|--------------------------------------------------|-----------|-----------|-----------|-----------|------------|
| Succession, partition, or incapacity             | 22        | 6         | 9         | 6         | 1          |
| Misconduct, misgovernment, or political sanction | 18        | 9         | 5         | 3         | 1          |
| Administrative or strategic reorganization       | 10        | 5         | 3         | 1         | 1          |
| Wartime defeat or regime change                  | 2         | 0         | 2         | 0         | 0          |
| Coding error or non-event                        | 3         | 0         | 2         | 0         | 1          |
| Fiscal distress                                  | 0         | 0         | 0         | 0         | 0          |
| Peasant unrest or petitioning                    | 0         | 0         | 0         | 0         | 0          |
| <b>Total</b>                                     | <b>55</b> | <b>20</b> | <b>21</b> | <b>10</b> | <b>4</b>   |

Table A.14: Succession and Sanction Account for 40 of 55 Recorded Conversion Rows

**Note:** Cause category groups each source-stated formal reason independently of source disposition. Political succession is grouped with succession. Violence paired with mental derangement is grouped with sanction because violence is the recorded precipitating act. Strategic military transfer is grouped with administrative or strategic reorganization. Fiscal distress and peasant unrest or petitioning remain explicit zero-count categories because neither appears as a source-stated formal cause. The row-level audit records potentially relevant unrest separately rather than inferring its absence from a short domain history.

| Domain        | Year    | Cause group              | Formal reason                        | Disposition                                                               | Treatment | Local source                                                                                      |
|---------------|---------|--------------------------|--------------------------------------|---------------------------------------------------------------------------|-----------|---------------------------------------------------------------------------------------------------|
| Itabashi      | 1617    | Administrative/strategic | Administrative transfer              | Most land transferred to another lord; no whole-domain tenryo conversion  | Exclude   | Hanshi Daijiten v.2 p.194 (PDF 210); Hanshi Soran p.102 (PDF 104)                                 |
| Nagamine      | 1618    | Administrative/strategic | Administrative relocation            | Former Nagamine holding became tenryo after transfer to Nagaoka           | Retain    | Hanshi Daijiten v.3 p.105 (PDF 119)                                                               |
| Musashimuro   | Ko-1619 | Succession/household     | No heir                              | House attained but 1,180 koku regranted to the lord's brother             | Exclude   | Hanshi Daijiten v.2 p.632 (PDF 648)                                                               |
| Omori         | 1632    | Succession/household     | Succession restriction               | Heir retained 5,000 koku; only half returned temporarily to bakufu        | Exclude   | Hanshi Daijiten v.5 p.30 (PDF 46); Hanshi Soran pp.258-259 (PDF 260-261)                          |
| Fukozu        | 1639    | Succession/household     | Family partition and relocation      | Brother received 5,000 koku including Fukozu and domain moved to Nakajima | Exclude   | Hanshi Daijiten v.4 p.314 (PDF 330)                                                               |
| Nariwa        | 1642    | Administrative/strategic | Administrative transfer              | Former Nariwa holding became tenryo after transfer to Bitchu Matsuyama    | Retain    | Hanshi Daijiten v.6 p.255 (PDF 271)                                                               |
| Fuchu         | 1645    | Succession/household     | No heir                              | Domain temporarily abolished; intervening administration not identified   | Ambiguous | Hanshi Daijiten v.2 p.39 (PDF 55)                                                                 |
| Kaibara       | 1650    | Succession/household     | No heir                              | Full confiscation and conversion to tenryo                                | Retain    | Hanshi Daijiten v.5 p.429 (PDF 445); Hanshi Soran p.275 (PDF 277)                                 |
| Toyooka       | 1653    | Succession/household     | No heir                              | Former domain became bakufu land until Kyogoku entry in 1668              | Retain    | Hanshi Daijiten v.5 p.463 (PDF 479)                                                               |
| Funai         | 1656    | Succession/household     | No heir                              | House ended; intervening administration before 1658 entry not stated      | Ambiguous | Hanshi Daijiten v.7 p.402 (PDF 418)                                                               |
| Marugame      | 1657    | Succession/household     | No heir                              | Bakufu daikan governed until Kyogoku entry in 1658                        | Retain    | Hanshi Daijiten v.6 p.447 (PDF 463); chronology p.451 (PDF 467)                                   |
| Kakogawa      | 1658    | Succession/household     | No heir                              | Neighboring lord held custody until 1659; bakufu-land status not stated   | Ambiguous | Hanshi Daijiten v.4 p.177 (PDF 193)                                                               |
| Yashima       | 1658    | Succession/household     | Testamentary family partition        | Two sub-10,000-koku branches became hatamoto in 1659                      | Exclude   | Hanshi Daijiten v.1 p.389 (PDF 405)                                                               |
| Sakura        | 1660    | Misconduct/sanction      | Official misconduct                  | Lord attained; direct administration before 1661 entry not established    | Ambiguous | Hanshi Daijiten v.2 p.453 (PDF 469); Aoki printed p.33 (PDF 35)                                   |
| Mibu          | 1665    | Coding/non-event         | Domain-name error                    | No-heir abolition occurred in Minagawa, not Mibu; Mibu continued          | Exclude   | Hanshi Daijiten v.2 p.226 (PDF 242); Hanshi Soran p.114 (PDF 116); Mibu entry v.2 p.210 (PDF 226) |
| Saijo         | 1665    | Misconduct/sanction      | Official dereliction                 | Territory explicitly became bakufu land until 1670                        | Retain    | Hanshi Daijiten v.6 p.469 (PDF 485); Aoki printed p.38 (PDF 40)                                   |
| Miyazu        | 1666    | Misconduct/sanction      | Household conflict and misgovernment | Territory became bakufu land until 1669                                   | Retain    | Hanshi Daijiten v.5 p.443 (PDF 459); Aoki printed p.39 (PDF 41)                                   |
| Shimabara     | 1668    | Misconduct/sanction      | Misgovernment                        | Former domain remained public land for one year and four months           | Retain    | Hanshi Daijiten v.7 p.212 (PDF 228); Aoki printed p.40 (PDF 42)                                   |
| Harima Shingu | 1670    | Succession/household     | Succession restriction               | Brother retained the seat and 3,000 koku; domain effectively continued    | Exclude   | Hanshi Daijiten v.5 p.583 (PDF 599)                                                               |

Table A.15: Page-Level Local-Source Audit of All Conversion Rows (Part 1 of 3)

**Note:** The audit reviews every one of the 55 baseline treatment rows in the local *Hanshi Daijiten* and *Hanshi Soran*. Cause group records the grouped formal reason summarized in Table A.14; it does not determine treatment status. “Retain” requires direct shogunal administration at the coded date. “Exclude” identifies a partial reduction, family partition, transfer, or post-shogunate event rather than a comparable conversion. “Ambiguous” denotes an abolition or confiscation for which the entry does not identify the intervening administrator. A year in parentheses is the source-verified correction. Nagamine, Nariwa, and Tamanawa are retained because the cited entries confirm that their former holdings became tenryō.

| Domain      | Year | Cause group              | Formal reason                         | Disposition                                                                                      | Treatment | Local source                                                                                |
|-------------|------|--------------------------|---------------------------------------|--------------------------------------------------------------------------------------------------|-----------|---------------------------------------------------------------------------------------------|
| Ichinoseki  | 1671 | Misconduct/sanction      | Political punishment in Date dispute  | Holding returned to Sendai main domain, not direct shogunal rule                                 | Exclude   | Hanshi Daijiten v.1 p.135 (PDF 151)                                                         |
| Aso         | 1676 | Succession/household     | No heir                               | Estate confiscated but regranted two months later; direct administration not stated              | Ambiguous | Hanshi Daijiten v.2 p.110 (PDF 126)                                                         |
| Omigawa     | 1677 | Coding/non-event         | Domain-name error                     | Event occurred in Owa; Omigawa did not dissolve in 1677                                          | Exclude   | Hanshi Soran p.75 (PDF 77); Omigawa entry Hanshi Daijiten v.2 p.427 (PDF 443)               |
| Yamazaki    | 1678 | Succession/household     | No heir                               | Shiso district explicitly became tenryo                                                          | Retain    | Hanshi Daijiten v.5 p.562 (PDF 578); narrative p.557 (PDF 573)                              |
| Kururi      | 1679 | Succession/household     | Mental incapacity                     | Son retained 3,000 koku and most castle territory passed to Maebashi                             | Exclude   | Hanshi Daijiten v.2 p.530 (PDF 546); chronology p.533 (PDF 549)                             |
| Niwase      | 1679 | Succession/household     | No heir                               | Vacancy until 1683; bakufu-land status not stated                                                | Ambiguous | Hanshi Daijiten v.6 p.199 (PDF 215)                                                         |
| Numata      | 1681 | Misconduct/sanction      | Misgovernment and official failure    | Territory came under daikan administration                                                       | Retain    | Hanshi Daijiten v.2 p.257 (PDF 273); Aoki printed pp.47,49 (PDF 49,51)                      |
| Takada      | 1681 | Succession/household     | Household conflict                    | Territory became bakufu land and was surveyed in 1682                                            | Retain    | Hanshi Daijiten v.3 p.113 (PDF 129)                                                         |
| Tatebayashi | 1683 | Succession/household     | Succession termination                | Most land already partitioned to hatamoto; only residual domain reverted                         | Exclude   | Hanshi Daijiten v.2 p.280 (PDF 296); chronology p.288 (PDF 304); Aoki printed p.45 (PDF 47) |
| Samuki      | 1684 | Misconduct/sanction      | Official misconduct                   | Domain abolished until 1710; intervening direct administration not stated                        | Ambiguous | Hanshi Daijiten v.2 p.546 (PDF 562)                                                         |
| Fukumoto    | 1687 | Succession/household     | Family partition                      | Domain status ended in 1665; 1687 was a later hatamoto split                                     | Exclude   | Hanshi Daijiten v.5 p.530 (PDF 546); Hanshi Soran p.295 (PDF 297)                           |
| Karasuyama  | 1687 | Misconduct/sanction      | Succession-rule violation             | Lord removed and another daimyo entered; no whole-domain conversion                              | Exclude   | Hanshi Daijiten v.2 p.165 (PDF 181)                                                         |
| Soumi       | 1687 | Misconduct/sanction      | Misconduct                            | All territory confiscated                                                                        | Retain    | Hanshi Daijiten v.3 p.54 (PDF 68); Hanshi Soran p.154 (PDF 156)                             |
| Fukami      | 1689 | Misconduct/sanction      | Official dereliction                  | 7,800 koku confiscated but 2,200 retained                                                        | Exclude   | Hanshi Daijiten v.2 p.681 (PDF 697)                                                         |
| Kitami      | 1689 | Misconduct/sanction      | Collective punishment after violence  | Fief confiscated and lord exiled                                                                 | Retain    | Hanshi Daijiten v.2 p.675 (PDF 691)                                                         |
| Hachiman    | 1692 | Succession/household     | No heir                               | House continued elsewhere and another lord entered Hachiman in 1692                              | Exclude   | Hanshi Daijiten v.4 p.33 (PDF 49)                                                           |
| Takayama    | 1692 | Administrative/strategic | Administrative and political transfer | Entire Hida domain became bakufu territory under Hida daikan                                     | Retain    | Hanshi Daijiten v.4 p.3 (PDF 19)                                                            |
| Takahashi   | 1693 | Succession/household     | No heir                               | Bakufu ordered survey and castle custody; bakufu-land status not explicit                        | Ambiguous | Hanshi Daijiten v.6 p.243 (PDF 259)                                                         |
| Fukuyama    | 1698 | Succession/household     | No heir                               | Entire territory became bakufu land under daikan; 50,000 koku remained bakufu land after regraut | Retain    | Hanshi Daijiten v.6 p.267 (PDF 283); Aoki printed p.57 (PDF 59)                             |

Table A.16: Page-Level Local-Source Audit of All Conversion Rows (Part 2 of 3)

**Note:** Continuation of Table A.15. Page numbers are printed-page locators; PDF-page locators appear in parentheses in the source field.

| Domain     | Year | Cause group              | Formal reason                      | Disposition                                                                       | Treatment         | Local source                                                                                                   |
|------------|------|--------------------------|------------------------------------|-----------------------------------------------------------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------|
| Sano       | 1698 | Administrative/strategic | Administrative relocation          | Lord transferred to Katata; direct administration of Sano not established         | Exclude           | Hanshi Daijiten v.2 p.236 (PDF 252)                                                                            |
| Yoshii     | 1698 | Administrative/strategic | Administrative relocation          | Former territory became divided holdings rather than a whole bakufu domain        | Exclude           | Hanshi Daijiten v.2 p.336 (PDF 352)                                                                            |
| Ako        | 1701 | Misconduct/sanction      | Violence                           | Former domain became bakufu land until 1702                                       | Retain            | Hanshi Daijiten v.5 p.608 (PDF 624)                                                                            |
| Nagashima  | 1702 | Misconduct/sanction      | Violence and mental derangement    | Another lord entered in the same year; direct administration not shown            | Exclude           | Hanshi Daijiten v.4 p.385 (PDF 401)                                                                            |
| Tamanawa   | 1703 | Administrative/strategic | Administrative relocation          | Former Tamanawa holding became tenryo after transfer to Otaki                     | Retain            | Hanshi Daijiten v.2 p.683 (PDF 699); Hanshi Soran p.146 (PDF 148)                                              |
| Kofu       | 1704 | Succession/household     | Political succession               | Yanagisawa replaced Tokugawa house in 1704; Kofu became tenryo in 1724            | Wrong date (1724) | Hanshi Daijiten v.3 p.329 (PDF 345); p.330 (PDF 346)                                                           |
| Yamura     | 1704 | Administrative/strategic | Administrative relocation          | Domain ended but exact former territory's bakufu-land status is not explicit      | Ambiguous         | Hanshi Daijiten v.3 p.322 (PDF 338); Hanshi Soran p.181 (PDF 183)                                              |
| Hojo       | 1712 | Misconduct/sanction      | Misgovernment and extraction       | Territory confiscated; intervening bakufu-land status not explicit                | Ambiguous         | Hanshi Daijiten v.2 p.569 (PDF 585); chronology p.572 (PDF 588); Aoki printed p.66 (PDF 68)                    |
| Numazu     | 1724 | Coding/non-event         | Non-event                          | Territory had been under daikan administration since 1632; no 1724 conversion     | Wrong date        | Hanshi Daijiten v.4 p.115 (PDF 131); p.116 (PDF 132)                                                           |
| Matsumoto  | 1725 | Misconduct/sanction      | Violence and mental derangement    | Territory became bakufu land under two daikan for eight months                    | Retain            | Hanshi Daijiten v.3 p.451 (PDF 467); Aoki printed p.70 (PDF 72)                                                |
| Takayanagi | 1746 | Administrative/strategic | Appointment-related land exchange  | Former holding reverted in 1739; 1746 records later move to Mikusa                | Wrong date (1739) | Hanshi Daijiten v.3 p.122 (PDF 136); Hanshi Soran p.156 (PDF 158)                                              |
| Obata      | 1767 | Misconduct/sanction      | Political punishment               | Heir transferred and another lord entered Obata; no direct conversion             | Exclude           | Hanshi Daijiten v.2 p.361 (PDF 377); chronology p.367 (PDF 383)                                                |
| Sagara     | 1786 | Misconduct/sanction      | Political fall                     | Reduction occurred in 1786; remaining territory became direct bakufu land in 1787 | Wrong date (1787) | Hanshi Daijiten v.4 p.161 (PDF 177); chronology p.165 (PDF 181)                                                |
| Omi Komuro | 1788 | Misconduct/sanction      | Official misconduct and corruption | All territory confiscated                                                         | Retain            | Hanshi Daijiten v.5 p.5 (PDF 21); Hanshi Soran pp.259-260 (PDF 261-262); Aoki printed p.146 (PDF 148)          |
| Matsumae   | 1807 | Administrative/strategic | Strategic and military transfer    | Entire Matsumae-Ezo territory became bakufu land                                  | Retain            | Hanshi Daijiten v.1 p.5 (PDF 21); chronology p.16 (PDF 32)                                                     |
| Hamada     | 1866 | Wartime collapse         | Military defeat                    | Territory remained under Choshu occupation and custody, not shogunal rule         | Exclude           | Hanshi Daijiten v.6 p.119 (PDF 135); chronology p.128 (PDF 144); Aoki printed pp.291,303,305 (PDF 293,305,307) |
| Kaifuchi   | 1868 | Wartime collapse         | Boshin War allegiance              | Meiji government confiscated Jozai territory after the shogunate ended            | Exclude           | Hanshi Daijiten v.2 p.526 (PDF 542); chronology p.527 (PDF 543)                                                |

Table A.17: Page-Level Local-Source Audit of All Conversion Rows (Part 3 of 3)

**Note:** Continuation of Table A.15. The full machine-readable audit also records the event-specific unrest screen in `analysis/output/tables/tabA_conversion_source_audit.csv`.

| Specification                                  | Outcome                 | Estimate | Cluster SE | Cluster $p$ | Conley $p$ | Treated |
|------------------------------------------------|-------------------------|----------|------------|-------------|------------|---------|
| Baseline                                       | Exit                    | 0.0385   | 0.0147     | 0.0098      | 0.0033     | 55      |
| Baseline                                       | Con-<br>fronta-<br>tion | 0.0076   | 0.0122     | 0.5322      | 0.5754     | 55      |
| Initial 9-row priority screen                  | Exit                    | 0.0330   | 0.0156     | 0.0361      | 0.0215     | 46      |
| Initial 9-row priority screen                  | Con-<br>fronta-<br>tion | 0.0134   | 0.0147     | 0.3624      | 0.3912     | 46      |
| Strict, recorded dates                         | Exit                    | 0.0463   | 0.0264     | 0.0817      | 0.0178     | 20      |
| Strict, recorded dates                         | Con-<br>fronta-<br>tion | 0.0480   | 0.0236     | 0.0444      | 0.1083     | 20      |
| Strict, corrected dates                        | Exit                    | 0.0385   | 0.0225     | 0.0903      | 0.0338     | 23      |
| Strict, corrected dates                        | Con-<br>fronta-<br>tion | 0.0472   | 0.0193     | 0.0158      | 0.0589     | 23      |
| Retain + ambiguous                             | Exit                    | 0.0355   | 0.0189     | 0.0625      | 0.0223     | 30      |
| Retain + ambiguous                             | Con-<br>fronta-<br>tion | 0.0231   | 0.0187     | 0.2203      | 0.2958     | 30      |
| Retain + ambiguous, corrected                  | Exit                    | 0.0310   | 0.0168     | 0.0682      | 0.0340     | 33      |
| Retain + ambiguous, corrected                  | Con-<br>fronta-<br>tion | 0.0255   | 0.0162     | 0.1171      | 0.1825     | 33      |
| Strict, corrected; no clear unrest             | Exit                    | 0.0447   | 0.0233     | 0.0576      | 0.0098     | 21      |
| Strict, corrected; no clear unrest             | Con-<br>fronta-<br>tion | 0.0508   | 0.0205     | 0.0144      | 0.0720     | 21      |
| Retain + ambiguous, corrected; no clear unrest | Exit                    | 0.0366   | 0.0179     | 0.0433      | 0.0140     | 30      |
| Retain + ambiguous, corrected; no clear unrest | Con-<br>fronta-<br>tion | 0.0273   | 0.0178     | 0.1272      | 0.2155     | 30      |

Table A.18: Sensitivity to Local-Source Treatment-Universe Corrections

**Note:** The page-level audit covers all 55 baseline rows. The strict screen retains only entries that establish direct shogunal administration at the stated date; the corrected version adds Takayanagi (1739), Kōfu (1724), and Sagara (1787). The ambiguity-inclusive screen also retains abolitions or confiscations whose intervening administrator is unstated. The final two specifications exclude Saijō, Shimabara, and Hōjō, the three cases clearly linked to unrest or petitioning. Each row re-estimates the 50 km two-way fixed-effects specification used in Table 1. The exit coefficient remains positive throughout. In the smallest strict samples, however, the confrontation coefficient is also positive, so those rows do not by themselves identify an exit-specific response. Cluster standard errors are two-way by domain and year; Conley  $p$ -values use a 70 km cutoff.

### C.3. Potentially Unrest-Related Conversions

The available case histories are sufficient to identify a narrow set of three conversions clearly connected to peasant unrest or petitioning and a broader set that adds two mixed cases. A systematic Aoki screen adds a maximal envelope: five further conversions had an event in the same named domain or province during the preceding ten years, although the local histories do not identify those events as causes. None of the screens can certify the absence of unrest in every remaining conversion. Table A.19 therefore reports bounded exclusions rather than presenting silence in short domain chronologies as evidence that treatment timing was unrelated to local conflict.

| Specification                  | Outcome       | Estimate | Cluster SE | Cluster $p$ | Conley $p$ | Treated |
|--------------------------------|---------------|----------|------------|-------------|------------|---------|
| Baseline                       | Exit          | 0.0385   | 0.0147     | 0.0098      | 0.0033     | 55      |
| Baseline                       | Confrontation | 0.0076   | 0.0122     | 0.5322      | 0.5754     | 55      |
| Exclude 3 clear cases          | Exit          | 0.0421   | 0.0153     | 0.0067      | 0.0014     | 52      |
| Exclude 3 clear cases          | Confrontation | 0.0076   | 0.0129     | 0.5556      | 0.6020     | 52      |
| Exclude 5 clear/mixed cases    | Exit          | 0.0397   | 0.0158     | 0.0130      | 0.0013     | 50      |
| Exclude 5 clear/mixed cases    | Confrontation | 0.0059   | 0.0136     | 0.6619      | 0.6827     | 50      |
| Maximal 10-year history screen | Exit          | 0.0443   | 0.0174     | 0.0117      | 0.0010     | 45      |
| Maximal 10-year history screen | Confrontation | 0.0092   | 0.0145     | 0.5264      | 0.5643     | 45      |

Table A.19: Sensitivity to Excluding Potentially Unrest-Related Conversions

**Note:** The narrow set excludes Saijō (1665), Shimabara (1668), and Hōjō (1712), whose histories directly connect abolition to peasant unrest or petitioning. The broader set also excludes Numata (1681) and Ōmi Komuro (1788) as mixed cases. The maximal screen additionally removes Sakura, Miyazu, Tatebayashi, Matsumoto, and Hamada because Aoki records an event in the same named domain or province during the preceding ten years, even though the local histories do not identify those events as causes of conversion. The exit estimate remains positive and the confrontation estimate remains null under all three screens. Cluster standard errors are two-way by domain and year; Conley  $p$ -values use a 70 km cutoff. Silence in a short domain history is not treated as proof that no unrest occurred. Sources: (Aoki 1975; Kimura, Fujino, and Murakami 2015; Kodama and Kitajima 1989).

## C.4. Tenryo-Conversion Count Robustness

| Outcome:                     | Nearby Exit Count |           |           | Nearby Confrontation Count |          |          |
|------------------------------|-------------------|-----------|-----------|----------------------------|----------|----------|
| Radius:                      | 30km              | 40km      | 50km      | 30km                       | 40km     | 50km     |
| Converted $\times$ Post      | 0.0359            | 0.0592    | 0.0770    | 0.0057                     | 0.0103   | 0.0398   |
| Cluster SE                   | (0.0238)          | (0.0323)* | (0.0396)* | (0.0162)                   | (0.0228) | (0.0284) |
| Conley SE (70km)             | (0.0291)          | (0.0390)  | (0.0448)* | (0.0171)                   | (0.0256) | (0.0314) |
| Num. Obs.                    | 33,877            | 33,877    | 33,877    | 33,877                     | 33,877   | 33,877   |
| Domain FE                    | Yes               | Yes       | Yes       | Yes                        | Yes      | Yes      |
| Year FE                      | Yes               | Yes       | Yes       | Yes                        | Yes      | Yes      |
| Tenryo-Converted/All Domains | 55/294            | 55/294    | 55/294    | 55/294                     | 55/294   | 55/294   |

Table A.20: Count Robustness for Domain-to-Tenryo Conversions

**Note:** OLS fixed-effects estimates with event counts (rather than binary indicators) as outcomes, using the same annual domain-year panels as Table 1. Tenryo-converted domains are observed from five years before to five years after conversion and matched to never-converted controls observed in the same calendar years represented by the conversion window. Two SE rows: (i) two-way clustered on domain and year; (ii) Conley spatial standard errors with a 70 km cutoff on domain capital coordinates. Signif. Codes: \*\*\*: 0.01, \*\*: 0.05, \*: 0.1.

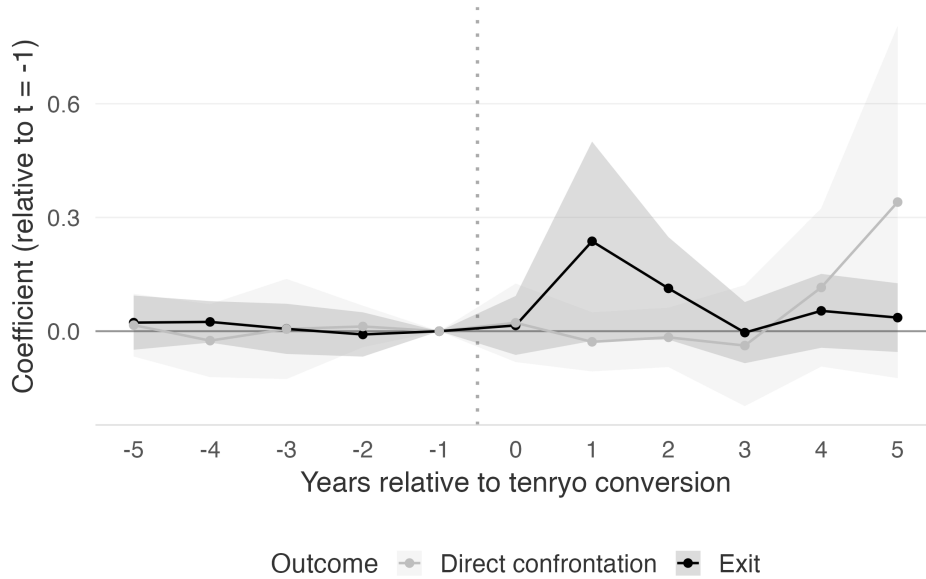


Figure A.8: OLS Count Event-Study Coefficients (50km)

**Note:** Year-by-year OLS coefficients from the specification  $\text{count}_{j,t} = \sum_{s \neq -1} \beta_s (\mathbf{1}[t - t_j^* = s]) + \varphi_j + \omega_t + \varepsilon_{j,t}$ , where the outcome is the count of nearby exit (black) or direct-confrontation (light gray) events within 50km of domain  $j$ 's capital and  $s = -1$  is the reference. Shaded bands show 95% confidence intervals; the horizontal line marks zero and the vertical dotted line marks conversion between  $s = -1$  and  $s = 0$ . Standard errors are two-way clustered by domain and year.

## C.5. Disaggregating Exit into Desertion and End-Run Appeal

Table A.21 replicates Table 1 with the exit outcome restricted to desertion only (*chōsan*), excluding end-run appeals.

| Outcome:              | Any Nearby Desertion |           |            |
|-----------------------|----------------------|-----------|------------|
| Sample Radius:        | 30km                 | 40km      | 50km       |
| <i>Variables</i>      |                      |           |            |
| Converted × Post      | 0.0114               | 0.0176    | 0.0266     |
| Cluster SE            | (0.0063)*            | (0.0101)* | (0.0122)** |
| <i>Controls</i>       |                      |           |            |
| Domain & Year FE      | Yes                  | Yes       | Yes        |
| <i>Fit Statistics</i> |                      |           |            |
| Num. Obs.             | 33,877               | 33,877    | 33,877     |
| R2                    | 0.033                | 0.044     | 0.055      |
| Mean Dep. Var.        | 0.004                | 0.006     | 0.010      |

Table A.21: Desertion-Only DiD

**Note:** Replicates Table 1 with the exit outcome restricted to desertion (*chōsan*), excluding end-run appeals. Specification matches Table 1. Two-way clustered (domain & year) standard errors. Signif. Codes: \*\*\*: 0.01, \*\*: 0.05, \*: 0.1.

## C.6. Staggered-Robust Estimation

Table A.22 compares the main TWFE pooled estimate with the interaction-weighted estimator of Sun and Abraham (2021), which is robust to heterogeneous treatment effects under staggered adoption. The Sun and Abraham aggregate ATT for exit is 0.0413 ( $p < 0.001$ ), virtually identical to the TWFE pooled estimate of 0.0385; the confrontation ATT remains insignificant at 0.0089 ( $p = 0.262$ ). The near-identical TWFE and Sun-Abraham point estimates are consistent with the Goodman-Bacon decomposition in Table A.23, in which 94.5% of the TWFE estimand weight falls on clean treated-versus-never-treated  $2 \times 2$  comparisons; the bad-comparison problem of heterogeneous-timing DiD is quantitatively small in this panel because never-treated domains make up the overwhelming majority of the sample.<sup>[27]</sup>

| Estimator            | Nearby Exit |             |           | Nearby Confrontation |          |       |
|----------------------|-------------|-------------|-----------|----------------------|----------|-------|
|                      | ATT         | SE          | $p$       | ATT                  | SE       | $p$   |
| TWFE                 | 0.0385      | (0.0147)**  | 0.01      | 0.0076               | (0.0122) | 0.532 |
| Sun & Abraham (2021) | 0.0413      | (0.0045)*** | $< 0.001$ | 0.0089               | (0.0079) | 0.262 |

Table A.22: TWFE vs. Sun and Abraham Staggered-Robust Estimates

**Note:** The TWFE row reproduces the main 50km specification from Table 1. The Sun and Abraham row reports the interaction-weighted estimator that is robust to heterogeneous treatment effects under staggered adoption. Never-converted domains serve as the comparison group. Two-way clustered (domain & year) standard errors in parentheses. Signif. Codes: .\*\*\*: 0.01, \*\*: 0.05, \*: 0.1.

<sup>[27]</sup>The Callaway and Sant'Anna (2021) estimator, another widely used staggered-robust method, is preferred on many panels but is ill-suited here because of cell sparsity: several of the 55 distinct conversion cohorts contain only one or two treated domains, which breaks the estimator's influence-function-based inference. Sun and Abraham's interaction-weighted approach, which operates within the standard fixed-effects regression framework, accommodates sparse cohorts more naturally.

## C.7. Goodman-Bacon Decomposition of the TWFE Estimator

Table A.23 reports the Goodman-Bacon (2021) decomposition of the two-way fixed-effects estimator on the balanced 50 km panel used for the tenryo DiD. The weight on clean treated-versus-never-treated  $2 \times 2$  comparisons is 94.5%; only 3.8% of the weight falls on the “bad comparison” of already-treated units serving as controls for later-treated units, and 1.7% on yet-to-be-treated comparisons. The dominant never-treated weight means that the near-identical TWFE and Sun-Abraham point estimates reported in Table A.22 are a mechanical consequence of sample composition, not a test of cohort-effect homogeneity.

| Comparison type               | Weight | Avg. $2 \times 2$ estimate |
|-------------------------------|--------|----------------------------|
| Treated vs. never-treated     | 94.5%  | 0.0101                     |
| Treated vs. already-treated   | 3.8%   | -0.0024                    |
| Treated vs. yet-to-be-treated | 1.7%   | 0.0151                     |

Table A.23: Goodman-Bacon Decomposition of the 50 km Tenryo DiD

**Note:** The TWFE-on-balanced-panel estimate is 0.0097. Never-treated units make up 82.6% of the sample. The dominant weight on treated-vs-never-treated  $2 \times 2$  comparisons means the TWFE-vs-Sun-Abraham agreement reported in Table A.22 is mechanical, not evidence of cohort-effect homogeneity.

## C.8. Cohort-Split Sun and Abraham ATT

To inspect heterogeneity directly, [Table A.24](#) re-runs the Sun and Abraham (2021) interaction-weighted estimator separately for treated domains whose tenryo conversion year falls in each of three bins: pre-1750, 1750–1799, and post-1800. The pre-1750 cohort holds 49 of the 55 conversions and delivers an aggregate ATT of 0.0443 ( $p < 0.001$ ), essentially reproducing the pooled result. The post-1800 cohort contains only three conversions but also yields a positive ATT of 0.0367 ( $p < 0.001$ ). The small 1750–1799 bin (three conversions) shows a marginally negative point estimate ( $-0.0064$ ) that is not load-bearing for the aggregate effect. The exit response is therefore concentrated in the period when the tenryo-conversion instrument has the most statistical weight, and no cohort reverses the sign of the effect.

| Cohort bin  | Cohorts | Treated units | ATT     | SE          | $p$     |
|-------------|---------|---------------|---------|-------------|---------|
| All cohorts | 44      | 55            | 0.0413  | (0.0045)*** | < 0.001 |
| pre-1750    | 38      | 49            | 0.0443  | (0.0049)*** | < 0.001 |
| 1750–1799   | 3       | 3             | −0.0064 | (0.0019)*** | 0.001   |
| post-1800   | 3       | 3             | 0.0367  | (0.0016)*** | < 0.001 |

Table A.24: Cohort-Split Sun and Abraham (2021) ATT on Nearby Exit (50 km)

**Note:** Each row re-runs the interaction-weighted estimator on the sub-sample of treated domains whose tenryo-conversion year falls in the indicated bin (plus all never-treated controls). Two-way clustered (domain and year) standard errors in parentheses. Signif. codes: .\*\*\*: 0.01, .\*\*: 0.05, .\*: 0.1.

### C.9. Rambachan-Roth Relative-Magnitudes Breakdown

Table A.25 reports the full relative-magnitudes sensitivity curve of Rambachan and Roth (2023) for the average post-conversion exit ATT at 50 km.  $\overline{M}$  is the ratio of the largest post-period deviation from parallel trends to the largest pre-period deviation. The 95% robust CI on the average post-conversion exit effect excludes zero for every  $\overline{M} \leq 0.20$ ; this is the breakdown value  $\overline{M}^*$ . The exit pre-trend joint Wald test (the basis for the relative-magnitudes scale) is reassuring in its own right:  $F = 1.26$ ,  $p = 0.283$  on four pre-period coefficients, consistent with genuinely flat pre-trends. The sensitivity exercise therefore confirms that the main exit effect withstands post-period deviations up to one-fifth of the largest pre-period deviation, on top of a pre-trend joint test that does not reject parallel trends in the first place.

| Panel A: $\Delta^{RM}$ (Relative Magnitudes) |                   |            |
|----------------------------------------------|-------------------|------------|
| $\overline{M}$                               | 95% robust CI     | Conclusion |
| 0.00                                         | [0.0108, 0.0694]  | excludes 0 |
| 0.20                                         | [0.0022, 0.0935]  | excludes 0 |
| 0.50                                         | [−0.0355, 0.1403] | includes 0 |
| 1.00                                         | [−0.1157, 0.2224] | includes 0 |
| 2.00                                         | [−0.2798, 0.3081] | includes 0 |
| Panel B: $\Delta^{SD}$ (Smoothness)          |                   |            |
| $M$                                          | 95% robust CI     | Conclusion |
| 0.0000                                       | [0.0227, 0.0773]  | excludes 0 |
| 0.0010                                       | [0.0108, 0.0855]  | excludes 0 |
| 0.0016                                       | [0.0011, 0.0929]  | excludes 0 |
| 0.0020                                       | [−0.0054, 0.0978] | includes 0 |
| 0.0030                                       | [−0.0215, 0.1098] | includes 0 |
| 0.0050                                       | [−0.0535, 0.1323] | includes 0 |
| 0.0100                                       | [−0.1453, 0.1599] | includes 0 |

Table A.25: Rambachan-Roth (2023) Sensitivity Analysis

**Note:** Panel A:  $\Delta^{RM}$  (relative magnitudes).  $\overline{M}$  bounds the maximum post-period deviation from parallel trends as a fraction of the largest pre-period deviation. Breakdown:  $\overline{M} = 0.20$ . Panel B:  $\Delta^{SD}$  (smoothness).  $M$  bounds the maximum second difference of the trend. Breakdown:  $M = 0.0016$ . Both panels report 95% robust CIs for the average post-conversion exit ATT at 50 km. Exit pre-trend joint Wald:  $F = 1.26$ ,  $p = 0.283$  (df = 4).

## C.10. Rambachan-Roth Sensitivity at Canonical Benchmarks

As a companion to Table A.25, Table A.26 reports the robust 95% confidence interval for the average post-conversion exit ATT at the canonical Rambachan and Roth (2023) benchmarks  $\bar{M} \in \{0.5, 1, 2\}$ , so that readers can read the sensitivity at the scale RR itself recommends for comparison. The CI excludes zero at  $\bar{M} \leq 0.20$ ; at  $\bar{M} = 0.5$  the CI is  $[-0.036, 0.140]$ , at  $\bar{M} = 1$  it is  $[-0.116, 0.222]$ , and at  $\bar{M} = 2$  it is  $[-0.280, 0.308]$ . Read together with the pre-period Wald test ( $F = 1.26$ ,  $p = 0.283$ , which does not reject parallel trends), the effect withstands post-period deviations up to one-fifth of the largest pre-period deviation; the  $\bar{M} = 0.5, 1, 2$  columns transparently report where that envelope widens. The smoothness restriction  $\Delta^{SD}$  tells the same story on a different scale: the CI excludes zero for second-difference bounds up to  $M = 0.0016$ , and at the empirical pre-period dispersion  $\sigma_{\text{pre}} = 0.0103$  (the standard deviation of the four pre-period event-time coefficients) the CI is  $[-0.148, 0.164]$ . The sensitivity curve is monotone, and the breakdown values  $\bar{M}^* = 0.20$  and  $M^* = 0.0016$  are the sharp envelope on how much untested post-period differential trend the sign of the effect tolerates.

| Panel A: $\Delta^{RM}$ (Relative Magnitudes) |                   |            |
|----------------------------------------------|-------------------|------------|
| $\bar{M}$                                    | 95% robust CI     | Conclusion |
| 0.00                                         | [0.0108, 0.0694]  | excludes 0 |
| 0.20                                         | [0.0022, 0.0935]  | excludes 0 |
| 0.50                                         | [-0.0355, 0.1403] | includes 0 |
| 1.00                                         | [-0.1157, 0.2224] | includes 0 |
| 2.00                                         | [-0.2798, 0.3081] | includes 0 |
| Panel B: $\Delta^{SD}$ (Smoothness)          |                   |            |
| $M$                                          | 95% robust CI     | Conclusion |
| 0.0000 (tight)                               | [0.0227, 0.0773]  | excludes 0 |
| 0.0016 (breakdown)                           | [0.0011, 0.0929]  | excludes 0 |
| $\sigma_{\text{pre}} = 0.0103$               | [-0.1484, 0.1635] | includes 0 |

Table A.26: Extended Rambachan-Roth (2023) Sensitivity Analysis

**Note:** Extension of Table A.25 that reports the full relative-magnitudes curve at the canonical benchmarks  $\bar{M} \in \{0, 0.2, 0.5, 1, 2\}$  and the smoothness restriction  $\Delta^{SD}$  at both the breakdown value ( $M = 0.0016$ ) and the empirical pre-period dispersion ( $\sigma_{\text{pre}} = 0.0103$ , the SD of the four pre-period event-time coefficients). The average post-conversion exit ATT at 50 km tolerates post-period deviations up to 20% of the largest pre-period deviation under  $\Delta^{RM}$  and up to a second-difference bound of  $M = 0.0016$  under  $\Delta^{SD}$ ; larger bounds include zero. Exit pre-trend joint Wald:  $F = 1.26$ ,  $p = 0.283$ .

### C.11. SUTVA and Spatial Interference

Because tenryo conversions cluster in time and space, some domain-years in the 50 km DiD panel fall within the treatment radius of multiple contemporaneous conversions, potentially violating the stable unit treatment value assumption (SUTVA). I bound this concern with three exclusion exercises. First, I map the overlap: 9.0% of domain-year observations have at least one other conversion within 50 km and  $\pm 5$  years. Second, I drop all converted domains whose capitals lie within 100 km of another conversion with a conversion date within 5 years (“contaminated” treated units), removing 30 of the 55 converted domains. Third, I exclude all never-converted control domains whose capitals fall within 50 km of any converted capital, as these controls may be indirectly treated (147 of 239 controls excluded). [Table A.27](#) reports the results. The exit coefficient retains its positive sign and comparable magnitude across all four samples. The point estimates range from 0.027 to 0.038; the reduction in statistical significance in the more restrictive samples reflects the loss of treated units rather than a change in the underlying estimate. The confrontation coefficient remains small and statistically insignificant throughout.

| Sample                          | Nearby Exit           |             | Nearby Confrontation |                      |
|---------------------------------|-----------------------|-------------|----------------------|----------------------|
|                                 | $\hat{\beta}$         | SE          | $\hat{\beta}$        | SE                   |
| Full sample ( $N = 33,877$ )    | 0.0385<br>(0.0131)*** | (0.0147)*** | 0.0076               | (0.0122)<br>(0.0136) |
| Drop overlap ( $N = 33,547$ )   | 0.0302<br>(0.0207)    | (0.0207)    | 0.0067               | (0.0203)<br>(0.0227) |
| Clean controls ( $N = 13,886$ ) | 0.0382<br>(0.0132)*** | (0.0155)**  | 0.0053               | (0.0125)<br>(0.0145) |
| Strictest ( $N = 13,556$ )      | 0.0270<br>(0.0213)    | (0.0216)    | 0.0031               | (0.0214)<br>(0.0249) |
| Domain FE: Yes    Year FE: Yes  |                       |             |                      |                      |

Table A.27: SUTVA / Spatial Interference Robustness

**Note:** Each row re-estimates the main 50 km DiD specification on progressively restricted samples. “Drop overlap” removes converted domains within 100 km and 5 years of another conversion. “Clean controls” removes never-converted domains within 50 km of any converted capital. “Strictest” applies both exclusions. Two SE rows are reported in parentheses: (i) two-way clustered on domain and year and (ii) Conley spatial standard errors with a 70 km cutoff. Significance stars on each SE row reflect that estimator’s  $p$ -value. Signif. Codes: .\*\*\*: 0.01, .\*: 0.05, .: 0.1.

### C.12. Permutation Test (Full-Pool Randomization)

As a companion to the narrower permutation test reported in the main text, which randomly reassigns fake conversion events among the 239 never-converted domains and serves as a panel-structure placebo, I also run the stricter Fisher-style randomization inference in which the placebo conversion events can land on any of the 294 domains, including the actual 55. This version asks the identification-oriented question: was the actual treatment assignment unusual among all possible 55-out-of-294 assignments? I draw 55 placebo conversion events from the full 294-domain pool in each of 500 iterations, with conversion-year timing drawn from the empirical distribution of the 55 actual conversion years. Each iteration re-applies the same  $[-5, +5]$  event window and matched-control-years restriction used in Table 1 and re-estimates the 50 km pooled TWFE exit coefficient. The permutation distribution is centered at zero (mean  $-0.0001$ , SD  $0.0140$ ) and the actual estimate of  $0.0385$  sits far into the right tail: the two-sided  $p$ -value is  $0.006$  and the one-sided right-tail  $p$ -value is  $0.002$ . Both improve on the panel-structure version ( $p = 0.012$ ) and confirm that the exit response is specific to the actual identity and timing of tenryo conversions rather than an artifact of the staggered-adoption panel.

| <b>Full-Pool Permutation Test (50 km, Any Nearby Exit)</b>                     |                                                      |
|--------------------------------------------------------------------------------|------------------------------------------------------|
| Actual $\hat{\beta}$ (Converted $\times$ Post)                                 | 0.0385                                               |
| Cluster SE $p$ -value (actual)                                                 | 0.0098                                               |
| Permutation iterations                                                         | 500                                                  |
| Permutation pool                                                               | All 294 domains                                      |
| Placebo events per iteration                                                   | 55 (sampled without replacement from 294)            |
| Conversion-year draw                                                           | Empirical distribution of actual 55 conversion years |
| Permutation mean $\hat{\beta}$                                                 | -0.0001                                              |
| Permutation SD                                                                 | 0.0140                                               |
| Permutation 2.5% quantile                                                      | 0.0281                                               |
| Permutation 97.5% quantile                                                     | -0.0001                                              |
| <b>Two-sided <math>p</math>-value</b> ( $ \text{perm}  \geq  \text{actual} $ ) | <b>0.0060</b>                                        |
| One-sided right-tail $p$ ( $\text{perm} \geq \text{actual}$ )                  | 0.0020                                               |

Table A.28: Full-Pool Permutation Test for the Exit Coefficient

**Note:** Permutation test in which the 55 placebo conversion events are drawn from the full 294-domain pool (rather than only the 239 never-converted controls, which is the narrower “panel-structure” placebo reported in the main text). Conversion-year timing is drawn from the empirical distribution of the 55 actual conversion years. Each iteration assigns 55 randomly selected domains as “converted,” applies the same  $[-5, +5]$  event window and matched-control-years restriction used in Table 1, and re-estimates the 50 km pooled TWFE exit coefficient. The reported two-sided  $p$ -value is the fraction of permutation estimates whose absolute value weakly exceeds the actual absolute estimate.

### C.13. Hazard Model for Conversion Timing

The F-test on pre-period event-time coefficients reported in the main text (Table 1) bounds one threat to identification: a secular differential trend between treated and control domains over the five-year window leading up to conversion. A distinct concern is whether attainer decisions responded to **recent** local unrest, making conversion timing endogenous to nearby exit or confrontation. I test this with a hazard model on the panel of 294 domains over 1615–1868. The outcome indicates conversion to tenryo in year  $t$ ; domains are censored at conversion or 1868. The covariates are lagged 5-year moving averages of nearby exit and confrontation within 50 km, plus territory size, distance to the Edo–Kyoto axis, and alliance status. Neither lagged exit intensity ( $p = 0.1447$  Cox,  $p = 0.3608$  logit) nor confrontation intensity ( $p = 0.9495$  Cox,  $p = 0.5485$  logit) predicts conversion timing; the joint Wald test fails to reject at  $p = 0.3439$ . Fudai/kin status is the one significant predictor, consistent with the historiography of attainer politics (Ravina 1999). The hazard model, pre-trend test, and permutation analysis provide complementary checks on timing.

| Panel A: Cox proportional hazards     |                            |        |        |              |
|---------------------------------------|----------------------------|--------|--------|--------------|
| Variable                              | $\hat{\beta}$              | SE     | $p$    | Hazard ratio |
| Nearby exit MA(5)                     | -5.1838                    | 3.5539 | 0.1447 | 0.0056       |
| Nearby confrontation MA(5)            | 0.1308                     | 2.0674 | 0.9495 | 1.1398       |
| log(area)                             | -0.2334                    | 0.1615 | 0.1484 | 0.7918       |
| log(dist to Edo-Kyoto axis)           | 0.1912                     | 0.2951 | 0.5171 | 1.2106       |
| Fudai/kin (vs. tozama)                | 1.3445                     | 0.5047 | 0.0077 | 3.8364       |
| Joint Wald (exit + confrontation = 0) | chi-square = 2.135, df = 2 |        |        | $p = 0.3439$ |
| Panel B: Discrete-time logit          |                            |        |        |              |
| Variable                              | $\hat{\beta}$              | SE     | $p$    | Odds ratio   |
| Nearby exit MA(5)                     | -3.1981                    | 3.4998 | 0.3608 | 0.0408       |
| Nearby confrontation MA(5)            | -1.1433                    | 1.9052 | 0.5485 | 0.3188       |
| log(area)                             | -0.1733                    | 0.1575 | 0.2712 | 0.8409       |
| log(dist to Edo-Kyoto axis)           | 0.1746                     | 0.2901 | 0.5473 | 1.1907       |
| Fudai/kin (vs. tozama)                | 1.3142                     | 0.5070 | 0.0095 | 3.7218       |

Table A.29: Hazard Models for Tenryo Conversion Timing

**Note:** Cox proportional-hazards and discrete-time logit estimates of the tenryo conversion hazard. The panel contains 51,849 domain-years across 294 domains from 1615 to 1868, with 34 tenryo-conversion events in the estimation sample (34 of the 55 conversions fall after the first-five-year MA warm-up). Nearby-exit and nearby-confrontation variables are lagged 5-year moving averages within 50 km of the focal domain’s capital. Neither lagged exit nor confrontation intensity predicts conversion timing at conventional levels under either estimator; the joint Wald test fails to reject at  $p = 0.3439$ . The only significant predictor is fudai/kin alliance status, consistent with the historiography of shogunal attainer.

### C.14. Conley Bandwidth Sweep (DiD)

The 70 km Conley cutoff used for the DiD main-text standard errors is approximately one first-order spatial neighbor distance, but is a researcher degree of freedom. Table A.30 sweeps the cutoff across five standard values (30, 50, 70, 100, 150 km). The point estimate is invariant to the inference method; the Conley SE varies modestly (0.0103 to 0.0131) with all  $p$ -values remaining below 0.005.

| Inference             | Cutoff | $\hat{\beta}$ | SE     | $p$    | $N$   |
|-----------------------|--------|---------------|--------|--------|-------|
| Cluster (domain+year) | –      | 0.0385        | 0.0147 | 0.0098 | 33877 |
| Conley                | 30 km  | 0.0385        | 0.0123 | 0.0017 | 33877 |
| Conley                | 50 km  | 0.0385        | 0.0127 | 0.0024 | 33877 |
| Conley (paper)        | 70 km  | 0.0385        | 0.0131 | 0.0033 | 33877 |
| Conley                | 100 km | 0.0385        | 0.0114 | 0.0007 | 33877 |
| Conley                | 150 km | 0.0385        | 0.0103 | 0.0002 | 33877 |

Table A.30: Conley Bandwidth Sweep for the Tenryo DiD (50 km)

**Note:** Tenryo DiD exit coefficient with standard errors computed under the cluster estimator (the main-text default) and Conley spatial HAC standard errors at five cutoffs. The point estimate does not depend on the inference method; only the SE and  $p$ -value vary. The hand-picked 70 km cutoff used in Table 1 sits near the middle of the sweep.

### C.15. Kelly (2020) Data-Driven Conley Bandwidth

As an alternative to the hand-picked cutoffs used in the main text (10 km at the village and grid levels, 70 km at the domain and DiD levels), I apply the Kelly (2019) data-driven rule, which selects the bandwidth from the empirical residual correlogram: the smallest distance at which the mean pairwise residual correlation falls below  $\exp(-1) \approx 0.37$ . Table A.31 reports this bandwidth alongside the hand-picked choice for three analyses: the village cross-section (Table 3 Model 5), the domain-level tax regression (Table 4 Model 4), and the tenryo DiD (Table 1 50 km). The data-driven choice is tighter than the hand-picked one in all three cases (6 km for villages, 10 km for tax, 10 km for DiD). Conley SEs change by less than 0.001 between the two choices in every regression. The village and DiD estimates remain significant at  $p < 0.01$ , while the tax estimate remains significant at  $p < 0.05$ .

| Analysis                                                    | Bandwidth rule           | Cutoff | $\hat{\beta}$ | Conley SE | Conley $p$ | $N$   |
|-------------------------------------------------------------|--------------------------|--------|---------------|-----------|------------|-------|
| <i>Village-level exit share (Table 3 Model 5 dyadic-FE)</i> |                          |        |               |           |            |       |
|                                                             | hand-picked              | 10km   | 0.0244        | 0.0085    | 0.0041     | 2739  |
|                                                             | Kelly (2020) data-driven | 6km    | 0.0244        | 0.0089    | 0.0062     | 2739  |
| <i>Domain tax regression (Table 4 Model 4)</i>              |                          |        |               |           |            |       |
|                                                             | hand-picked              | 70km   | -0.0112       | 0.0055    | 0.0432     | 230   |
|                                                             | Kelly (2020) data-driven | 10km   | -0.0112       | 0.0055    | 0.0449     | 230   |
| <i>Tenryo DiD (Table 1 50 km)</i>                           |                          |        |               |           |            |       |
|                                                             | hand-picked              | 70km   | 0.0385        | 0.0131    | 0.0033     | 33877 |
|                                                             | Kelly (2020) data-driven | 10km   | 0.0385        | 0.0126    | 0.0022     | 33877 |

Table A.31: Kelly (2020) Data-Driven vs. Hand-Picked Conley Bandwidth

**Note:** The Kelly bandwidth is the smallest distance at which the empirical residual correlogram falls below  $\exp(-1) \approx 0.37$ . Pairwise residual products are aggregated into 40 distance bins for the village and DiD regressions and 25 bins for the tax regression. The data-driven bandwidth is tighter than the hand-picked cutoff in all three analyses, especially for the tax regression (10 km vs. 70 km). Point estimates are identical across bandwidth choices; only SEs vary. The data-driven choice moves SEs by less than 0.001 in absolute terms across all three regressions. The village and DiD estimates remain significant at  $p < 0.01$ ; the tax estimate remains significant at  $p < 0.05$ .

## D. VILLAGE EXIT CHOICE ANALYSIS

### D.1. Full Table

| Outcome:                         | exit <sub>i</sub>     |                     |                    |                       |                       |                       |
|----------------------------------|-----------------------|---------------------|--------------------|-----------------------|-----------------------|-----------------------|
| Models:                          | (1)                   | (2)                 | (3)                | (4)                   | (5)                   | (6)                   |
| <i>Variables</i>                 |                       |                     |                    |                       |                       |                       |
| ln( $\rho_i$ )                   | 0.0458<br>(0.0154)*** | 0.0235<br>(0.0117)* | 0.0143<br>(0.0089) | 0.0201<br>(0.0071)*** | 0.0212<br>(0.0072)*** | 0.0244<br>(0.0068)*** |
| lat                              |                       |                     |                    |                       | -2.417***<br>(0.8551) | -2.413*<br>(1.460)    |
| lon                              |                       |                     |                    |                       | -0.5462**<br>(0.2119) | -0.5291<br>(0.3918)   |
| ln(Dist. Edo/Kyoto)              |                       |                     |                    |                       | 0.2731**<br>(0.1094)  | 0.2241<br>(0.1395)    |
| ln(Rice)                         |                       |                     |                    |                       | -0.0048<br>(0.0046)   | -0.0027<br>(0.0048)   |
| ln(Elev.)                        |                       |                     |                    |                       | -0.0216<br>(0.0212)   | -0.0291<br>(0.0214)   |
| ln(Rugg.)                        |                       |                     |                    |                       | 0.0276**<br>(0.0150)  | 0.0328**<br>(0.0152)  |
| Lat × Lon                        |                       |                     |                    |                       | 0.0174**<br>(0.0064)  | 0.0171<br>(0.0112)    |
| <i>Controls</i>                  |                       |                     |                    |                       |                       |                       |
| Home ( $j$ ) Domain FE           | No                    | Yes                 | Yes                | Yes                   | Yes                   | No                    |
| Neighbor ( $k$ ) Domain FE       | No                    | No                  | Yes                | Yes                   | Yes                   | No                    |
| Decade ( $t$ ) FE                | No                    | No                  | No                 | Yes                   | Yes                   | Yes                   |
| Home-Neighbor Dyad ( $j, k$ ) FE | No                    | No                  | No                 | No                    | No                    | Yes                   |
| <i>Fit Statistics</i>            |                       |                     |                    |                       |                       |                       |
| Num.Obs.                         | 2,739                 | 2,739               | 2,739              | 2,739                 | 2,739                 | 2,739                 |
| R2                               | 0.028                 | 0.390               | 0.543              | 0.621                 | 0.626                 | 0.678                 |

Table A.32: Villages Exposed to Alternative Authority Opt for Exit Over Direct Confrontation

(Full Table)

**Note:** The coefficients represent OLS estimates of the relationship between each village's geographic exposure and its choice of exit conditional on recorded resistance. Clustered (home ruler, nearest neighboring ruler, and decade) standard errors are in parentheses. Signif. Codes: \*\*\*: 0.01, \*\*: 0.05, \*: 0.1.

## D.2. Robustness Checks

### D.2.1. Alternative Covariates

| Outcome:                              | exit <sub>i</sub>            |                            |                              |                              |                              |                           |                             |                              |
|---------------------------------------|------------------------------|----------------------------|------------------------------|------------------------------|------------------------------|---------------------------|-----------------------------|------------------------------|
|                                       | Cal. Suit.                   |                            |                              |                              |                              | Dist. to Edo              |                             |                              |
| Models:                               | (1)                          | (2)                        | (3)                          | (4)                          | (5)                          | (6)                       | (7)                         | (8)                          |
| <i>Variables</i>                      |                              |                            |                              |                              |                              |                           |                             |                              |
| ln( $\rho_i$ )                        | <b>0.0434</b><br>(0.0150)*** | <b>0.0162</b><br>(0.0090)* | <b>0.0209</b><br>(0.0072)*** | <b>0.0234</b><br>(0.0069)*** | <b>0.0439</b><br>(0.0149)*** | <b>0.0135</b><br>(0.0092) | <b>0.0199</b><br>(0.0075)** | <b>0.0240</b><br>(0.0068)*** |
| lat                                   | −1.3685<br>(0.2726)          | −0.8994<br>(0.9761)        | −2.5470<br>(0.8580)          | −2.5127<br>(1.3351)          | −1.7680<br>(0.5527)          | −3.0658<br>(1.0682)       | −3.5340<br>(0.9003)         | −3.6532<br>(1.2780)          |
| lon                                   | −0.3040<br>(0.0711)          | −0.0596<br>(0.2402)        | −0.5621<br>(0.2012)          | −0.5293<br>(0.3450)          | −0.4533<br>(0.1633)          | −0.7529<br>(0.3249)       | −0.8904<br>(0.2710)         | −0.9021<br>(0.3792)          |
| ln(Dist. Edo/Kyoto)                   | −0.0238<br>(0.0449)          | 0.4825<br>(0.2092)         | 0.2846<br>(0.1153)           | 0.2497<br>(0.1346)           |                              |                           |                             |                              |
| ln(Cal.)                              | −0.0307<br>(0.2051)          | 0.1270<br>(0.1825)         | 0.1926<br>(0.1566)           | 0.2602<br>(0.1590)           |                              |                           |                             |                              |
| ln(Elev.)                             | 0.0132<br>(0.0250)           | −0.0137<br>(0.0193)        | −0.0268<br>(0.0188)          | −0.0369<br>(0.0180)          | 0.0009<br>(0.0196)           | −0.0204<br>(0.0215)       | −0.0262<br>(0.0208)         | −0.0329<br>(0.0212)          |
| ln(Rugg.)                             | 0.0324<br>(0.0246)           | 0.0210<br>(0.0161)         | 0.0274<br>(0.0153)           | 0.0319<br>(0.0154)           | 0.0445<br>(0.0209)           | 0.0316<br>(0.0160)        | 0.0331<br>(0.0154)          | 0.0368<br>(0.0158)           |
| Lat × Lon                             | 0.0096<br>(0.0019)           | 0.0057<br>(0.0073)         | 0.0183<br>(0.0064)           | 0.0178<br>(0.0102)           | 0.0127<br>(0.0041)           | 0.0223<br>(0.0083)        | 0.0260<br>(0.0070)          | 0.0266<br>(0.0100)           |
| ln(Dist. Edo)                         |                              |                            |                              |                              | −0.1417<br>(0.0723)          | −0.1751<br>(0.3853)       | −0.0181<br>(0.2963)         | −0.1370<br>(0.3973)          |
| ln(Rice)                              |                              |                            |                              |                              | −0.0082<br>(0.0069)          | −0.0079<br>(0.0064)       | −0.0046<br>(0.0045)         | −0.0027<br>(0.0048)          |
| Const.                                | 44.25***<br>(9.996)          |                            |                              |                              | 64.05***<br>(22.24)          |                           |                             |                              |
| <i>Controls</i>                       |                              |                            |                              |                              |                              |                           |                             |                              |
| Home ( <i>j</i> ) Domain FE           | No                           | Yes                        | Yes                          | No                           | No                           | Yes                       | Yes                         | No                           |
| Neighbor ( <i>k</i> ) Domain FE       | No                           | Yes                        | Yes                          | No                           | No                           | Yes                       | Yes                         | No                           |
| Decade ( <i>t</i> ) FE                | No                           | No                         | Yes                          | Yes                          | No                           | No                        | Yes                         | Yes                          |
| Home-Neighbor Dyad ( <i>j, k</i> ) FE | No                           | No                         | No                           | Yes                          | No                           | No                        | No                          | Yes                          |
| <i>Fit Statistics</i>                 |                              |                            |                              |                              |                              |                           |                             |                              |
| Num.Obs.                              | 2,739                        | 2,739                      | 2,739                        | 2,739                        | 2,739                        | 2,739                     | 2,739                       | 2,739                        |
| R2                                    | 0.061                        | 0.551                      | 0.627                        | 0.679                        | 0.069                        | 0.549                     | 0.626                       | 0.678                        |

Table A.33: Villages Exposed to Alternative Authority

Opt for Exit Over Direct Confrontation (Alternative Covariates)

**Note:** Clustered (Home & Neighbor Domains & Decade) standard errors in parentheses.  
Signif. Codes: \*\*\*: 0.01, \*\*: 0.05, \*: 0.1.

### D.2.2. Excluding Villages with More Than One Recorded Resistance Episode

| Outcome:                              | exit <sub><i>i</i></sub> |                      |                      |                      |                      |                       |
|---------------------------------------|--------------------------|----------------------|----------------------|----------------------|----------------------|-----------------------|
| Models:                               | (1)                      | (2)                  | (3)                  | (4)                  | (5)                  | (6)                   |
| <i>Variables</i>                      |                          |                      |                      |                      |                      |                       |
| ln( $\rho_i$ )                        | 0.0488<br>(0.0153)***    | 0.0270<br>(0.0119)** | 0.0228<br>(0.0104)** | 0.0229<br>(0.0094)** | 0.0234<br>(0.0094)** | 0.0231<br>(0.0080)*** |
| lat                                   |                          |                      |                      |                      | -2.6613<br>(0.9798)  | -2.8835<br>(1.5600)   |
| lon                                   |                          |                      |                      |                      | -0.6092<br>(0.2415)  | -0.6572<br>(0.4214)   |
| ln(Dist. Edo/Kyoto)                   |                          |                      |                      |                      | 0.2075<br>(0.0702)   | 0.1153<br>(0.0881)    |
| ln(Rice)                              |                          |                      |                      |                      | -0.0061<br>(0.0049)  | -0.0040<br>(0.0051)   |
| ln(Elev.)                             |                          |                      |                      |                      | -0.0267<br>(0.0214)  | -0.0371<br>(0.0222)   |
| ln(Rugg.)                             |                          |                      |                      |                      | 0.0309*<br>(0.0157)  | 0.0360**<br>(0.0161)  |
| Lat $\times$ Lon                      |                          |                      |                      |                      | 0.0194<br>(0.0073)   | 0.0208<br>(0.0120)    |
| Const.                                | 0.2299***<br>(0.0348)    |                      |                      |                      |                      |                       |
| <i>Controls</i>                       |                          |                      |                      |                      |                      |                       |
| Home ( <i>j</i> ) Domain FE           | No                       | Yes                  | Yes                  | Yes                  | Yes                  | No                    |
| Neighbor ( <i>k</i> ) Domain FE       | No                       | No                   | Yes                  | Yes                  | Yes                  | No                    |
| Home-Neighbor Dyad ( <i>j, k</i> ) FE | No                       | No                   | No                   | No                   | No                   | Yes                   |
| Decade ( <i>t</i> ) FE                | No                       | No                   | No                   | Yes                  | Yes                  | Yes                   |
| Geography                             | No                       | No                   | No                   | No                   | Yes                  | Yes                   |
| <i>Fit Statistics</i>                 |                          |                      |                      |                      |                      |                       |
| Num.Obs.                              | 2,295                    | 2,295                | 2,295                | 2,295                | 2,295                | 2,295                 |
| R2                                    | 0.032                    | 0.431                | 0.590                | 0.672                | 0.675                | 0.732                 |

Table A.34: Villages Exposed to Alternative Authority Opt for Exit Over Direct Confrontation  
(Excluding Repeat Villages)

**Note:** Clustered (Home & Neighbor Domains & Decade) standard errors in parentheses. Signif. Codes: .\*\*\*: 0.01, \*\*: 0.05, \*: 0.1.

### D.2.3. Absolute Distances

| Outcome:                                   | exit <sub>i</sub>      |                     |                     |                      |                       |                       |
|--------------------------------------------|------------------------|---------------------|---------------------|----------------------|-----------------------|-----------------------|
| Models:                                    | (1)                    | (2)                 | (3)                 | (4)                  | (5)                   | (6)                   |
| <i>Variables</i>                           |                        |                     |                     |                      |                       |                       |
| ln(Dist. to Home Capital <sub>i</sub> )    | 0.0399<br>(0.0223)*    | 0.0248<br>(0.0161)  | 0.0127<br>(0.0109)  | 0.0231<br>(0.0096)** | 0.0160<br>(0.0104)    | 0.0203<br>(0.0083)**  |
| ln(Dist. to Foreign Capital <sub>i</sub> ) | -0.0596<br>(0.0140)*** | -0.0203<br>(0.0218) | -0.0172<br>(0.0199) | -0.0147<br>(0.0147)  | -0.0306<br>(0.0129)** | -0.0309<br>(0.0135)** |
| lat                                        |                        |                     |                     |                      | -2.5043<br>(0.8401)   | -2.5427<br>(1.4573)   |
| lon                                        |                        |                     |                     |                      | -0.5607<br>(0.2091)   | -0.5495<br>(0.3908)   |
| ln(Dist. Edo/Kyoto)                        |                        |                     |                     |                      | 0.2839<br>(0.1131)    | 0.2405<br>(0.1354)    |
| ln(Rice)                                   |                        |                     |                     |                      | -0.0047<br>(0.0047)   | -0.0024<br>(0.0048)   |
| ln(Elev.)                                  |                        |                     |                     |                      | -0.0192<br>(0.0217)   | -0.0252<br>(0.0220)   |
| ln(Rugg.)                                  |                        |                     |                     |                      | 0.0292<br>(0.0157)    | 0.0350<br>(0.0155)    |
| Lat × Lon                                  |                        |                     |                     |                      | 0.0180<br>(0.0063)    | 0.0181<br>(0.0112)    |
| Const.                                     | 0.2692<br>(0.0683)     |                     |                     |                      |                       |                       |
| <i>Controls</i>                            |                        |                     |                     |                      |                       |                       |
| Home ( <i>j</i> ) Domain FE                | No                     | Yes                 | Yes                 | Yes                  | Yes                   | No                    |
| Neighbor ( <i>k</i> ) Domain FE            | No                     | No                  | Yes                 | Yes                  | Yes                   | No                    |
| Home-Neighbor Dyad FE                      | No                     | No                  | No                  | No                   | No                    | Yes                   |
| Decade ( <i>t</i> ) FE                     | No                     | No                  | No                  | Yes                  | Yes                   | Yes                   |
| Geography                                  | No                     | No                  | No                  | No                   | Yes                   | Yes                   |
| <i>Fit Statistics</i>                      |                        |                     |                     |                      |                       |                       |
| Num.Obs.                                   | 2,739                  | 2,739               | 2,739               | 2,739                | 2,739                 | 2,739                 |
| R2                                         | 0.030                  | 0.390               | 0.543               | 0.621                | 0.627                 | 0.678                 |

Table A.35: Villages Exposed to Alternative Authority Opt for Exit Over Direct Confrontation  
(Abs. Dist. Measures)

**Note:** Clustered (Home & Neighbor Domains & Decade) standard errors in parentheses. Signif. Codes: .\*\*\*: 0.01, \*\*: 0.05, \*: 0.1.



#### D.2.4. Water-Crossing Subsample

The main village distance-ratio regression uses straight-line great-circle distances from each village to its home-domain capital and nearest foreign-domain capital, and these lines can traverse sea straits such as Kanmon, Bungo, Naruto, or Sado. I therefore drop every resistance-event row whose village, home capital, or nearest foreign capital lies on a different main island (Honshu, Kyushu, Shikoku, Sado, or Awaji), classified from latitude and longitude. This removes 150 of 2,739 rows (5.5%). On this water-safe subsample the coefficient on  $\log(\rho_i)$  remains positive across all six specifications, including the dyadic home-neighbor fixed-effects column that mirrors Table 3 Model 5 ( $\hat{\beta} = 0.0253$ , clustered  $p < 0.001$ ; baseline 0.0244,  $p = 0.001$ ).

| Outcome:                         | exit <sub>i</sub> |            |           |             |             |             |
|----------------------------------|-------------------|------------|-----------|-------------|-------------|-------------|
| Models:                          | (1)               | (2)        | (3)       | (4)         | (5)         | (6)         |
| Distance ratio ( $\ln(\rho_i)$ ) | 0.0460            | 0.0232     | 0.0160    | 0.0209      | 0.0207      | 0.0253      |
| Cluster SE                       | (0.0156)***       | (0.0127)*  | (0.0105)  | (0.0082)**  | (0.0080)**  | (0.0063)*** |
| Conley SE (10km)                 | (0.0121)***       | (0.0113)** | (0.0094)* | (0.0079)*** | (0.0079)*** | (0.0086)*** |
| <i>Controls</i>                  |                   |            |           |             |             |             |
| Home Domain FE                   | No                | Yes        | Yes       | Yes         | Yes         | No          |
| Neighbor Domain FE               | No                | No         | Yes       | Yes         | Yes         | No          |
| Home-Nbr Dyad FE                 | No                | No         | No        | No          | No          | Yes         |
| Decade FE                        | No                | No         | No        | Yes         | Yes         | Yes         |
| Geography                        | No                | No         | No        | No          | Yes         | Yes         |
| <i>Fit Statistics</i>            |                   |            |           |             |             |             |
| Num. Obs.                        | 2,589             | 2,589      | 2,589     | 2,589       | 2,589       | 2,589       |
| R <sup>2</sup>                   | 0.029             | 0.391      | 0.552     | 0.635       | 0.639       | 0.683       |
| Mean Dep. Var.                   | 0.251             | 0.251      | 0.251     | 0.251       | 0.251       | 0.251       |

Table A.36: Village Distance-Ratio Regression: Water-Crossing Subsample

**Note:** Re-runs the six village distance-ratio specifications on the subsample that drops resistance-event rows whose straight-line village-to-home-capital or village-to-foreign-capital line crosses a sea strait (i.e., the village, the home capital, and the nearest foreign capital are not all on the same main island). Island membership (Honshu, Kyushu, Shikoku, Sado, Awaji) is assigned from lat/lon bounding boxes. 150 (5.5%) of 2,739 rows are classified as sea-crossing and dropped. The coefficient on  $\log(\rho_i)$  remains positive across all six specifications and statistical significance is preserved in Model 6 (dyadic home-neighbor FE plus geography controls), which corresponds to Table 3 Model 5. SE rows: (i) three-way clustered on (home, neighbor, decade); (ii) Conley spatial SE with a 10 km cutoff on village coordinates. Stars reflect the corresponding estimator's  $p$ -value. Signif. Codes: \*\*\*: 0.01, \*\*: 0.05, \*: 0.1.

### D.2.5. Conley Standard Error Bandwidth Sensitivity

Table A.37 reports the bandwidth sensitivity of Conley spatial standard errors for the main village distance-ratio regression (Table 3 Model 5, dyadic home-neighbor FE). The coefficient on  $\log(\rho_i)$  is stable across all bandwidths from 25 to 150 km and remains significant at the 5% level in every specification.

| Standard error                       | Estimate | SE       | $p$   |
|--------------------------------------|----------|----------|-------|
| Cluster (home + nbr + decade, paper) | 0.0244   | (0.0068) | 0.001 |
| Conley 25 km                         | 0.0244   | (0.0089) | 0.006 |
| Conley 50 km                         | 0.0244   | (0.0097) | 0.012 |
| Conley 75 km                         | 0.0244   | (0.0102) | 0.017 |
| Conley 100 km                        | 0.0244   | (0.0099) | 0.014 |
| Conley 150 km                        | 0.0244   | (0.0078) | 0.002 |

Table A.37: Conley SE Bandwidth Sensitivity: Village Distance-Ratio Regression

**Note:** The coefficient on  $\log(\rho_i)$  from Table 3 Model 5 (dyadic home-neighbor FE) is stable across all bandwidths from 25 to 150 km and remains significant at the 5% level in every specification. The first row reports the three-way cluster standard error used in the main table.

### D.2.6. Small- $G$ Cluster Sensitivity of the Decade Dimension

Table A.38 reports a small- $G$  cluster-sensitivity check for the village distance-ratio regression (Table 3 Model 5, dyadic home-neighbor FE). The decade dimension has  $G = 26$  clusters, below the conventional safe threshold for multi-way cluster-robust inference, motivating a check of whether this dimension is load-bearing for the reported significance. Dropping the decade dimension entirely and clustering only on the two higher- $G$  dimensions (home ruler and neighbor ruler) yields  $p = 0.0024$ , close to the three-way number ( $p = 0.0013$ ); clustering on decade alone yields  $p = 0.0014$ . The main-text significance is therefore preserved under every cluster specification considered.

| Cluster specification                  | Estimate | SE       | $p$   |
|----------------------------------------|----------|----------|-------|
| Three-way (home + nbr + decade, paper) | 0.0244   | (0.0068) | 0.001 |
| Two-way (home + nbr, dropping decade)  | 0.0244   | (0.0079) | 0.002 |
| One-way on decade only ( $G = 26$ )    | 0.0244   | (0.0068) | 0.001 |

Table A.38: Small- $G$  Cluster Sensitivity of the Decade Dimension

**Note:** Re-runs Table 3 Model 5 (dyadic home-neighbor FE) under three cluster specifications. The decade dimension has  $G = 26$  clusters, below the conventional safe threshold for multi-way clustering. The key finding is that the paper’s three-way cluster SE is robust to dropping the decade dimension entirely — two-way clustering on home and neighbor gives nearly identical inference — indicating the decade cluster is not load-bearing for the reported  $p$ -value.

### D.2.7. Hatamoto Kantō Exclusion

Morris (1990) documents that hatamoto (direct shogunal retainers holding small fiefs) were particularly dense in four Kantō provinces: Musashi, Sagami, Kazusa, and Shimōsa. Because the baseline distance-ratio measure in [Section 3.4.2](#) uses active daimyō and former-domain capitals but omits individual hatamoto fiefs, a village in these provinces may have had a hatamoto landlord closer than its nearest measured outside authority. Excluding the 40 resistance-event rows (1.5% of the sample) leaves the coefficient positive and statistically significant ([Table A.39](#)).

| Sample                                     | Num. Obs. | Estimate | SE       | <i>p</i> |
|--------------------------------------------|-----------|----------|----------|----------|
| Full sample (paper)                        | 2739      | 0.0244   | (0.0068) | 0.001    |
| Excluding Musashi, Sagami, Kazusa, Shimōsa | 2699      | 0.0245   | (0.0069) | 0.002    |

Table A.39: Village Distance-Ratio Regression: Excluding Dense-Hatamoto Kantō Provinces

**Note:** Re-runs [Table 3](#) Model 5 (dyadic home-neighbor FE) on the subsample that excludes resistance-event rows in the four Kantō provinces with the highest documented hatamoto density – Musashi, Sagami, Kazusa, and Shimōsa (Morris 1990) – where the distance-ratio measure may under-capture the nearest outside authority because hatamoto fiefs are not in the baseline daimyō/former-domain-capital geography. The coefficient on  $\log(\rho_i)$  remains positive and statistically significant.

### D.2.8. Desertion-Only Village Cross-Section

Table A.40 re-estimates the preferred specifications with the exit outcome restricted to desertion only (*chōsan*), excluding end-run appeals. The preferred coefficient is positive, marginal under three-way clustering ( $p = 0.051$ ), and significant with Conley inference ( $p = 0.041$ ), so the narrower definition preserves the direction and most of the pooled magnitude with some inference sensitivity.

| Outcome:                                | Desertion <sub><i>i</i></sub> |            |
|-----------------------------------------|-------------------------------|------------|
| Models:                                 | (4)                           | (5)        |
| <i>Variables</i>                        |                               |            |
| Distance ratio ( $\ln(\rho_i)$ )        | 0.0127                        | 0.0141     |
| Cluster SE                              | (0.0060)**                    | (0.0068)*  |
| Conley SE (10km)                        | (0.0062)**                    | (0.0069)** |
| <i>Controls</i>                         |                               |            |
| Home + Neighbor + Decade FE + Geography | Yes                           | No         |
| Dyadic + Decade FE + Geography          | No                            | Yes        |
| <i>Fit Statistics</i>                   |                               |            |
| Num.Obs.                                | 2,727                         | 2,727      |
| R2                                      | 0.598                         | 0.674      |

Table A.40: Desertion-Only Village Cross-Section

**Note:** Replicates Table 3 Models 4–5 with exit restricted to desertion (*chōsan*), excluding end-run appeals. Two SE rows: (i) three-way clustered; (ii) Conley spatial SE (10 km). Signif. Codes: \*\*\*: 0.01, \*\*: 0.05, \*: 0.1.

### D.2.9. Desertion-Only Grid-Year Panel

Table A.41 replicates Table 2 with exit restricted to desertion (*chōsan*) only. The desertion-specific coefficients track the pooled exit estimates closely in both sign and significance, across the full panel and the ever-active subsample, confirming that the within-cell exit response is driven by the prototypical desertion cases rather than by end-run appeals.

| Outcome:                                           | Full sample |             | Ever-active cells |             |
|----------------------------------------------------|-------------|-------------|-------------------|-------------|
|                                                    | (1)         | (2)         | (3)               | (4)         |
|                                                    | Desertion   | Pooled exit | Desertion         | Pooled exit |
| Grid exposure ratio ( $\rho_{i,t}^{\text{grid}}$ ) | 0.00024     | 0.00035     | 0.00152           | 0.00198     |
| Cluster SE                                         | (0.00014)*  | (0.00017)** | (0.00087)*        | (0.00109)*  |
| Conley SE (10km)                                   | (0.00012)** | (0.00017)** | (0.00075)**       | (0.00103)*  |
| <i>Controls</i>                                    |             |             |                   |             |
| Grid & Year FE                                     | Yes         | Yes         | Yes               | Yes         |
| <i>Fit Statistics</i>                              |             |             |                   |             |
| Num.Obs.                                           | 2,225,802   | 2,225,802   | 372,364           | 372,364     |
| R <sup>2</sup>                                     | 0.005       | 0.006       | 0.006             | 0.008       |
| Mean Dep. Var.                                     | 0.00009     | 0.00022     | 0.00054           | 0.00131     |

Table A.41: Desertion-Only Grid-Year Panel

**Note:** LPM on binary any-event-of-type indicators. Replicates Table 2 with exit restricted to desertion (*chōsan*), excluding end-run appeals. Pooled exit columns reproduce the main specification for comparison. Columns (1)–(2) use the full panel of 8,763 cells; columns (3)–(4) restrict to the 1,466 ever-active hexagons. Grid and year FE. SEs: (i) two-way clustered (cell, year); (ii) Conley (10 km). Signif. Codes: \*\*\*: 0.01, \*\*: 0.05, \*: 0.1.

### D.2.10. Grid-Panel Count Models: Poisson and Negative Binomial

Table A.42 replicates the main-text Table 2 using two count-data estimators with hexagon and year fixed effects: Poisson (columns 1–3) and the negative binomial (columns 4–6). Because both estimators use a log link ( $\log \mathbb{E}[y_{i,t} | X_{i,t}] = \beta X_{i,t} + \alpha_i + \tau_t$ ), the raw coefficient  $\hat{\beta}$  sits on the log-count scale. The *incidence rate ratio*,  $\text{IRR} \equiv \exp(\hat{\beta})$ , back-transforms this to the count scale: a one-unit increase in the regressor multiplies the expected count by IRR. Both estimators agree on the exit-versus-confrontation asymmetry. The exit coefficient is positive and corresponds to a multiplicative association of roughly 4–5 $\times$  per unit distance ratio (Poisson  $\text{IRR} = 4.020$ ; NB  $\text{IRR} = 5.357$ ). The confrontation coefficient is small and statistically indistinguishable from zero under both estimators (Poisson  $p = 0.5676$ ; NB  $p = 0.4891$ ), and the strict-confrontation coefficient turns negative (IRRs below one). The NB columns drop all-zero cells via the conditional likelihood, so sample size varies by outcome; the Poisson columns use the full 8,763-cell panel.

| Outcome:                                           | Poisson    |                          |                                      | Neg. Binom. |                          |                                      |
|----------------------------------------------------|------------|--------------------------|--------------------------------------|-------------|--------------------------|--------------------------------------|
|                                                    | (1)        | (2)                      | (3)                                  | (4)         | (5)                      | (6)                                  |
|                                                    | Exit count | Confronta-<br>tion count | Confronta-<br>tion count<br>(strict) | Exit count  | Confronta-<br>tion count | Confronta-<br>tion count<br>(strict) |
| Grid exposure ratio ( $\rho_{i,t}^{\text{grid}}$ ) | 1.3912     | 0.3240                   | −0.4231                              | 1.6784      | 0.4453                   | −0.5370                              |
| IRR                                                | 4.020      | 1.383                    | 0.655                                | 5.357       | 1.561                    | 0.584                                |
| Cluster SE                                         | (0.9599)   | (0.5668)                 | (0.5299)                             | (0.8860)*   | (0.6437)                 | (0.7985)                             |
| Conley SE (10km)                                   | (0.9043)   | (0.5257)                 | (0.6052)                             | (0.8601)*   | (0.6620)                 | (0.7938)                             |
| <i>Controls</i>                                    |            |                          |                                      |             |                          |                                      |
| Grid FE                                            | Yes        | Yes                      | Yes                                  | Yes         | Yes                      | Yes                                  |
| Year FE                                            | Yes        | Yes                      | Yes                                  | Yes         | Yes                      | Yes                                  |
| <i>Fit Statistics</i>                              |            |                          |                                      |             |                          |                                      |
| Num.Obs.                                           | 62,900     | 202,130                  | 40,810                               | 62,900      | 202,130                  | 40,810                               |
| Pseudo R <sup>2</sup>                              | 0.135      | 0.130                    | 0.127                                | 0.093       | 0.092                    | 0.087                                |
| Mean Dep. Var.                                     | 0.0003     | 0.0009                   | 0.0004                               | 0.0003      | 0.0009                   | 0.0004                               |

Table A.42: Grid-Panel Count Models: Poisson and Negative Binomial with FE

**Note:** Columns (1)–(3) estimate a Poisson model and columns (4)–(6) a negative binomial, both with hexagon and year fixed effects on the 5 km grid-year panel (1615–1868). Coefficients are on the log-count scale;  $\text{IRR} = \exp(\hat{\beta})$  gives the multiplicative effect on the expected count. Num. Obs. in the NB columns reflects the ever-event subsample retained by the conditional likelihood; the Poisson columns use the full 8,763-cell panel. SEs: (i) two-way clustered on cell and year; (ii) Conley (10 km). Signif. Codes: .\*\*\*: 0.01, .\*\*: 0.05, .\*: 0.1.

### D.2.11. Ever-Active Cells Subsample

Table A.43 replicates Table 2 on the 1,466 “ever-active” hexagons — cells with at least one exit or confrontation event between 1615 and 1868. This restriction filters out the 83% of cells that recorded neither form of resistance and sharpens within-cell variation among active locales while using the same binary-outcome LPM and fixed-effects structure. Columns (1)–(3) use the grid exposure ratio  $\rho_{i,t}^{\text{grid}}$ ; columns (4)–(6) use the closest-capital distance  $d_{i,t}^1$  (km).

The distance-ratio coefficient on exit rises to 0.00198 ( $p = 0.054$  Conley,  $p = 0.070$  clustered), while the confrontation and strict-confrontation coefficients remain small and statistically indistinguishable from zero. The closest-capital regressor remains a null predictor of exit on this subsample ( $p = 0.179$  clustered,  $p = 0.259$  Conley). It carries a marginal negative signal for strict confrontation (Conley  $p = 0.028$ ), plausibly because uprisings and castle attacks require a nearby state target. Remoteness therefore contains some information about where direct confrontation is mechanically feasible but not where exit is chosen; relative geographic exposure is associated with exit.

| Outcome:              | Grid exposure ratio ( $\rho_{i,t}^{\text{grid}}$ ) |                        |                                    | Closest-capital dist. ( $d_{i,t}^1$ , km) |                        |                                    |
|-----------------------|----------------------------------------------------|------------------------|------------------------------------|-------------------------------------------|------------------------|------------------------------------|
|                       | (1)                                                | (2)                    | (3)                                | (4)                                       | (5)                    | (6)                                |
|                       | Any exit                                           | Any con-<br>frontation | Any con-<br>frontation<br>(strict) | Any exit                                  | Any con-<br>frontation | Any con-<br>frontation<br>(strict) |
| Coefficient           | 0.00198                                            | 0.00096                | −0.00097                           | 0.00004                                   | −0.00004               | −0.00004                           |
| Cluster SE            | (0.00109)*                                         | (0.00105)              | (0.00067)                          | (0.00003)                                 | (0.00003)              | (0.00002)*                         |
| Conley SE (10km)      | (0.00103)*                                         | (0.00105)              | (0.00062)                          | (0.00004)                                 | (0.00002)*             | (0.00002)**                        |
| <i>Controls</i>       |                                                    |                        |                                    |                                           |                        |                                    |
| Grid FE               | Yes                                                | Yes                    | Yes                                | Yes                                       | Yes                    | Yes                                |
| Year FE               | Yes                                                | Yes                    | Yes                                | Yes                                       | Yes                    | Yes                                |
| <i>Fit Statistics</i> |                                                    |                        |                                    |                                           |                        |                                    |
| Num.Obs.              | 372,364                                            | 372,364                | 372,364                            | 372,364                                   | 372,364                | 372,364                            |
| R <sup>2</sup>        | 0.008                                              | 0.011                  | 0.011                              | 0.008                                     | 0.011                  | 0.011                              |
| Mean Dep. Var.        | 0.00131                                            | 0.00384                | 0.00155                            | 0.00131                                   | 0.00384                | 0.00155                            |

Table A.43: Grid-Year Panel on Ever-Active Cells: Distance Ratio vs. Closest-Capital Distance

**Note:** Replicates Table 2 on the 1,466 “ever-active” hexagons — cells with at least one exit or confrontation event between 1615 and 1868 — which sharpens within-variation among active locales while filtering out cells with no recorded resistance. Columns (1)–(3) use the time-varying grid exposure ratio  $\rho_{i,t}^{\text{grid}} = \frac{d_{i,t}^1}{d_{i,t}^2}$ . Columns (4)–(6) replace it with the raw closest-capital distance  $d_{i,t}^1$  (km). Grid and year FE. SEs: (i) two-way clustered (cell, year); (ii) Conley (10 km). Signif. Codes: \*\*\*: 0.01, \*\*: 0.05, \*: 0.1.

#### D.2.12. Least-Cost-Path Distance Ratio

Table A.44 replicates Table 2 with the great-circle grid exposure ratio  $\rho_{i,t}^{\text{grid}} = \frac{d_{i,t}^1}{d_{i,t}^2}$  replaced by a terrain-weighted least-cost-path variant  $\rho_{i,t}^{\text{grid,LCP}}$ , computed from a  $(1 + \text{slope})$  cost surface on the SRTM elevation raster. For each of the 8,763 5 km hexagon centroids and each unique capital location, I run a single-source Dijkstra over the cost surface and record the cumulative terrain-weighted travel cost;  $\rho_{i,t}^{\text{grid,LCP}}$  is the ratio of the nearest-to-second-nearest such cost among capitals active in year  $t$ . The two regressors capture overlapping but distinct information: their Pearson correlation is 0.44, and the LCP ratio has mean 0.82 and standard deviation 0.24 (vs. 0.63 and 0.26 for the great-circle ratio), reflecting the common terrain component that travel costs to multiple capitals share when paths cross the same ridges and valleys.

The exit coefficient preserves its sign and standardized magnitude: a one-standard-deviation increase in  $\rho_{i,t}^{\text{grid,LCP}}$  corresponds to  $0.00045 \times 0.24 \approx 0.011\%$ -points, close to the 0.009%-points implied by Table 2 column (1). The exit-versus-confrontation asymmetry also survives: confrontation coefficients remain small and statistically indistinguishable from zero across the full panel and ever-active subsample. Statistical significance on exit falls below conventional thresholds ( $p = 0.278$  clustered;  $p = 0.203$  Conley). Terrain friction adds a shared component to the cost of reaching any capital and compresses the contrast between the nearest two, making the LCP ratio a noisier proxy for relative geographic exposure. The alternative measure therefore preserves sign and standardized magnitude but yields less precise inference.

| Outcome:                                                   | <i>Full sample</i> |                        |                                    | <i>Ever-active cells</i> |                        |                                    |
|------------------------------------------------------------|--------------------|------------------------|------------------------------------|--------------------------|------------------------|------------------------------------|
|                                                            | (1)                | (2)                    | (3)                                | (4)                      | (5)                    | (6)                                |
|                                                            | Any exit           | Any con-<br>frontation | Any con-<br>frontation<br>(strict) | Any exit                 | Any con-<br>frontation | Any con-<br>frontation<br>(strict) |
| Grid exposure ratio (LCP, $\rho_{i,t}^{\text{grid,LCP}}$ ) | 0.00045            | 0.00053                | −0.00008                           | 0.00173                  | 0.00177                | −0.00051                           |
| Cluster SE                                                 | (0.00041)          | (0.00060)              | (0.00031)                          | (0.00169)                | (0.00221)              | (0.00122)                          |
| Conley SE (10km)                                           | (0.00035)          | (0.00068)              | (0.00026)                          | (0.00141)                | (0.00192)              | (0.00074)                          |
| <i>Controls</i>                                            |                    |                        |                                    |                          |                        |                                    |
| Grid FE                                                    | Yes                | Yes                    | Yes                                | Yes                      | Yes                    | Yes                                |
| Year FE                                                    | Yes                | Yes                    | Yes                                | Yes                      | Yes                    | Yes                                |
| <i>Fit Statistics</i>                                      |                    |                        |                                    |                          |                        |                                    |
| Num.Obs.                                                   | 2,225,802          | 2,225,802              | 2,225,802                          | 372,364                  | 372,364                | 372,364                            |
| R <sup>2</sup>                                             | 0.006              | 0.007                  | 0.006                              | 0.008                    | 0.011                  | 0.011                              |
| Mean Dep. Var.                                             | 0.00022            | 0.00064                | 0.00026                            | 0.00131                  | 0.00384                | 0.00155                            |

Table A.44: Grid-Year Panel with Least-Cost-Path Distance Ratio

**Note:** LPM on binary any-event-of-type indicators. Replicates Table 2 with the haversine grid exposure ratio replaced by a terrain-weighted least-cost-path (LCP) variant:  $\rho_{i,t}^{\text{grid,LCP}}$  equals the ratio of nearest-to-second-nearest LCP cost from each hexagon centroid to any active domain capital in year  $t$ , where cost is cumulative across pixels of a  $(1 + \text{slope})$  cost surface derived from the SRTM elevation raster. Columns (1)–(3) use the full panel; (4)–(6) restrict to ever-active hexagons. Grid and year FE. SEs: (i) two-way clustered (cell, year); (ii) Conley (10 km). Signif. Codes: \*\*\*: 0.01, \*\*: 0.05, \*: 0.1.

### D.2.13. Daikansho-Inclusive Distance Ratio (Village)

Table A.45 replicates Table 3 with  $d_i^{\text{FC}}$  redefined as the distance to the nearest non-home administrative center of any type, including shogunal intendant offices (*daikansho*). The preferred dyadic-FE specification (Model 5) is virtually unchanged ( $\hat{\beta} = 0.0268$  vs. 0.0244 in Table 3).

| Outcome:                         | Exit <sub>i</sub> |             |             |            |             |
|----------------------------------|-------------------|-------------|-------------|------------|-------------|
| Models:                          | (1)               | (2)         | (3)         | (4)        | (5)         |
| <i>Variables</i>                 |                   |             |             |            |             |
| Distance ratio ( $\ln(\rho_i)$ ) | 0.0422            | 0.0496      | 0.0201      | 0.0202     | 0.0268      |
| Cluster SE                       | (0.0148)***       | (0.0137)*** | (0.0074)**  | (0.0075)** | (0.0067)*** |
| Conley SE (10km)                 | (0.0117)***       | (0.0093)*** | (0.0077)*** | (0.0079)** | (0.0085)*** |
| <i>Controls</i>                  |                   |             |             |            |             |
| Decade FE                        | No                | Yes         | Yes         | Yes        | Yes         |
| Home Domain FE                   | No                | No          | Yes         | Yes        | No          |
| Neighbor Domain FE               | No                | No          | Yes         | Yes        | No          |
| Dyadic FE                        | No                | No          | No          | No         | Yes         |
| Geography                        | No                | No          | No          | Yes        | Yes         |
| <i>Fit Statistics</i>            |                   |             |             |            |             |
| Num.Obs.                         | 2,739             | 2,739       | 2,739       | 2,739      | 2,739       |
| R2                               | 0.024             | 0.222       | 0.633       | 0.638      | 0.686       |
| Mean Dep. Var.                   | 0.246             | 0.246       | 0.246       | 0.246      | 0.246       |

Table A.45: Village Exit Choice with Daikansho-Inclusive Distance Ratio

**Note:** Replicates Table 3 with  $d_i^{\text{FC}}$  redefined to include shogunal intendant offices (*daikansho*). Two SE rows: (i) clustered on home, nearest neighboring domain, and decade; (ii) Conley spatial standard errors (10 km). Signif. Codes: \*\*\*: 0.01, \*\*: 0.05, \*: 0.1.

#### D.2.14. Daikansho-Inclusive Distance Ratio (Grid)

Table A.46 replicates Table 2 with the daikansho-inclusive distance ratio. Adding daikansho to the capital set compresses the ratio toward unity and reduces within-cell variation; all grid-level coefficients attenuate and lose statistical significance. The village-level and tax results are unaffected by this inclusion (SI D.2.13; SI D.3.1).

|                                                       |  | Full sample |                        |                                    |           | Ever-active cells      |                                    |
|-------------------------------------------------------|--|-------------|------------------------|------------------------------------|-----------|------------------------|------------------------------------|
| Outcome:                                              |  | (1)         | (2)                    | (3)                                | (4)       | (5)                    | (6)                                |
|                                                       |  | Any exit    | Any confronta-<br>tion | Any confronta-<br>tion<br>(strict) | Any exit  | Any confronta-<br>tion | Any confronta-<br>tion<br>(strict) |
| Grid exposure<br>ratio ( $\rho_{i,t}^{\text{grid}}$ ) |  | 0.00019     | 0.00023                | −0.00006                           | 0.00095   | −0.00012               | −0.00096                           |
| Cluster SE                                            |  | (0.00019)   | (0.00021)              | (0.00012)                          | (0.00111) | (0.00111)              | (0.00068)                          |
| Conley SE<br>(10km)                                   |  | (0.00018)   | (0.00021)              | (0.00012)                          | (0.00107) | (0.00097)              | (0.00060)                          |
| <i>Controls</i>                                       |  |             |                        |                                    |           |                        |                                    |
| Grid & Year<br>FE                                     |  | Yes         | Yes                    | Yes                                | Yes       | Yes                    | Yes                                |
| <i>Fit Statistics</i>                                 |  |             |                        |                                    |           |                        |                                    |
| Num.Obs.                                              |  | 2,225,802   | 2,225,802              | 2,225,802                          | 372,364   | 372,364                | 372,364                            |
| R <sup>2</sup>                                        |  | 0.006       | 0.007                  | 0.006                              | 0.008     | 0.011                  | 0.011                              |
| Mean Dep. Var.                                        |  | 0.00022     | 0.00064                | 0.00026                            | 0.00131   | 0.00384                | 0.00155                            |

Table A.46: Grid-Year Panel with Daikansho-Inclusive Distance Ratio

**Note:** LPM on binary any-event-of-type indicators. Replicates Table 2 with the daikansho-inclusive distance ratio. Columns (1)–(3) use the full panel of 8,763 cells; columns (4)–(6) restrict to the 1,466 ever-active hexagons. See Table 2 notes for details. Signif. Codes: .\*\*\*: 0.01, .\*\*: 0.05, .\*: 0.1.

### D.3. Additional Analyses

#### D.3.1. Daikansho-Inclusive Distance Ratio (Tax)

Table A.47 replicates Table 4 with the daikansho-inclusive distance ratio. Models 1–2 remain significant at  $p < 0.01$ ; models 3–4 show modest  $p$ -value drift (from  $p < 0.05$  to  $p \approx 0.07$ ) with stable point estimates, consistent with daikansho offices compressing the distance ratio’s variance. The spatial lag models (5–6) preserve the negative sign but are not statistically significant, mirroring the pattern in the main table where the fully controlled spatial lag is only marginally significant.

| Outcome:                         | Tax Rate ( $\tau_j$ ) |             |           |           |                    |          |
|----------------------------------|-----------------------|-------------|-----------|-----------|--------------------|----------|
|                                  | <i>OLS</i>            |             |           |           | <i>Spatial Lag</i> |          |
| Models:                          | (1)                   | (2)         | (3)       | (4)       | (5)                | (6)      |
| <i>Variables</i>                 |                       |             |           |           |                    |          |
| Distance ratio ( $\ln(\rho_j)$ ) | −0.0233               | −0.0192     | −0.0098   | −0.0096   | −0.0130            | −0.0080  |
| HC SE                            | (0.0054)***           | (0.0058)*** | (0.0054)* | (0.0054)* | (0.0047)***        | (0.0050) |
| <i>Controls</i>                  |                       |             |           |           |                    |          |
| Domain Characteristics           | No                    | Yes         | Yes       | Yes       | No                 | Yes      |
| Geography                        | No                    | No          | Yes       | Yes       | No                 | Yes      |
| Neighbor Characteristics         | No                    | No          | No        | Yes       | No                 | Yes      |
| $\tau_j$ Spatial Lag             | No                    | No          | No        | No        | Yes                | Yes      |
| <i>Fit Statistics</i>            |                       |             |           |           |                    |          |
| Num.Obs.                         | 232                   | 230         | 230       | 230       | 230                | 230      |
| R2 / Pseudo R2                   | 0.071                 | 0.138       | 0.368     | 0.386     | 0.309              | 0.437    |
| Mean Dep. Var.                   | 0.391                 | 0.391       | 0.391     | 0.391     | 0.394              | 0.394    |

Table A.47: Domain Tax Rates with Daikansho-Inclusive Distance Ratio

**Note:** Replicates Table 4 with the daikansho-inclusive distance ratio. Models 1–4: OLS with heteroskedasticity-robust standard errors. Models 5–6: spatial lag (SAR) with neighborhood-weighted tax rates as spatial regressor; standard errors from the spatial likelihood. Signif. Codes: \*\*\*: 0.01, \*\*: 0.05, \*: 0.1.

### D.3.2. Controlling for Distance to the Border

| Outcome:                                                  | exit <sub>i</sub>     |                     |                    |                       |                       |                       |
|-----------------------------------------------------------|-----------------------|---------------------|--------------------|-----------------------|-----------------------|-----------------------|
| Models:                                                   | (1)                   | (2)                 | (3)                | (4)                   | (5)                   | (6)                   |
| <i>Variables</i>                                          |                       |                     |                    |                       |                       |                       |
| ln( $\rho_i$ )                                            | 0.0487<br>(0.0161)*** | 0.0225<br>(0.0124)* | 0.0152<br>(0.0098) | 0.0206<br>(0.0068)*** | 0.0206<br>(0.0065)*** | 0.0264<br>(0.0075)*** |
| ln(Shortest Distance to Border) <sub><math>t</math></sub> | 0.0060<br>(0.0103)    | 0.0073<br>(0.0085)  | 0.0010<br>(0.0106) | 0.0057<br>(0.0089)    | 0.0058<br>(0.0086)    | 0.0037<br>(0.0086)    |
| <i>Controls</i>                                           |                       |                     |                    |                       |                       |                       |
| Home ( $j$ ) Domain FE                                    | No                    | Yes                 | Yes                | Yes                   | Yes                   | No                    |
| Neighbor ( $k$ ) Domain FE                                | No                    | No                  | Yes                | Yes                   | Yes                   | No                    |
| Decade ( $t$ ) FE                                         | No                    | No                  | No                 | Yes                   | Yes                   | Yes                   |
| Home-Neighbor Dyad ( $j, k$ ) FE                          | No                    | No                  | No                 | No                    | No                    | Yes                   |
| <i>Fit Statistics</i>                                     |                       |                     |                    |                       |                       |                       |
| Num.Obs.                                                  | 2,391                 | 2,391               | 2,391              | 2,391                 | 2,391                 | 2,391                 |
| R2                                                        | 0.037                 | 0.377               | 0.535              | 0.632                 | 0.638                 | 0.676                 |

Table A.48: Villages Exposed to Alternative Authority Opt for Exit Over Direct Confrontation

(Shortest Distance to Border)

**Note:** The coefficients relate the distance ratio (home-capital distance divided by distance to the nearest active non-home capital in the baseline daimyō/former-domain-capital geography) to exit choice conditional on recorded resistance, controlling for shortest distance to another domain's border. Border data match 2,391 of the 2,739 preferred-panel rows. On this covered sample, the preferred dyadic estimate is 0.0265 without and 0.0264 with the border control; the border coefficient is 0.0037 ( $p = 0.668$ ). Clustered (home ruler, nearest neighboring ruler, and decade) standard errors are in parentheses. Signif. Codes: .\*\*\*: 0.01, .\*\*: 0.05, .\*: 0.1.

### D.3.3. Sequential Decomposition of the Village Choice

The main analysis in [Section 3.4.3](#) conditions on recorded resistance and so cannot, on its own, rule out an association between geographic exposure and resistance incidence. I therefore combine the resisting-village data with the comprehensive end-period village ledger, adding compliant villages to the analysis. For each resisting village, I identify its counterpart in the full ledger, using the geographically nearest match when names do not resolve the record, and remove matched ledger rows to avoid duplication. A village-event remains a separate observation when the home or neighboring domain differs or when the event records a distinct tactic. The combined data contain 46,322 observations; actions by resisting villages are classified as exit or direct confrontation, while unmatched villages are labeled compliance.

On this combined sample, I decompose the village's three-way choice into two binary steps: Step 1 estimates recorded resistance versus compliance over the full village sample, and Step 2 estimates exit versus direct confrontation conditional on resistance. This is the two-step structure implied by a nested logit with resistance nest = {exit, confrontation}, and it can be estimated as two internally consistent binary models without the IIA restriction imposed by a flat multinomial logit.

Formally, let  $i$  index villages and let the choice set be  $\mathcal{J} = \{C, E, F\}$  (compliance, exit, direct confrontation) with nest structure  $B_0 = \{C\}$  and  $B_1 = \{E, F\}$ . Random utility is  $U_{ij} = V_{ij} + \varepsilon_{ij}$ , with the GEV distribution

$$F(\varepsilon_i) = \exp\left\{-e^{-\varepsilon_{iC}} - \left(e^{-\frac{\varepsilon_{iE}}{\lambda}} + e^{-\frac{\varepsilon_{iF}}{\lambda}}\right)^\lambda\right\},$$

where  $\lambda \in (0, 1]$  is the dissimilarity parameter for the resistance nest ( $\lambda = 1$  collapses to multinomial logit). With covariates  $X_i$  governing resistance incidence and  $Z_i$  governing tactic composition, the linear indices are  $V_{iC} = 0$  (normalization),  $V_{iE} = X_{i'}\gamma + Z_{i'}\delta_E$ , and  $V_{iF} = X_{i'}\gamma + Z_{i'}\delta_F$ . Defining the inclusive value  $I_i = \log\left[\exp\left(\frac{V_{iE}}{\lambda}\right) + \exp\left(\frac{V_{iF}}{\lambda}\right)\right]$ , the closed-form choice probabilities are

$$\mathbb{P}(R_i = 1 \mid X_i, Z_i) = \frac{\exp(\lambda I_i)}{1 + \exp(\lambda I_i)}, \quad \mathbb{P}(E_i = 1 \mid R_i = 1, Z_i) = \Lambda\left((\delta_E - \delta_F)' \frac{Z_i}{\lambda}\right),$$

where  $\Lambda(\cdot)$  is the logistic CDF. Letting  $r_i = \mathbb{1}\{y_i \in \{E, F\}\}$  and  $e_i = \mathbb{1}\{y_i = E\}$ , the nested-logit log-likelihood factors as

$$\begin{aligned} \log \mathcal{L} = & \underbrace{\sum_i [r_i \log \mathbb{P}(R_i) + (1 - r_i) \log(1 - \mathbb{P}(R_i))]}_{\log \mathcal{L}_1(\psi_1)} \\ & + \underbrace{\sum_{i:r_i=1} [e_i \log \mathbb{P}(E_i | R_i) + (1 - e_i) \log(1 - \mathbb{P}(E_i | R_i))]}_{\log \mathcal{L}_2(\psi_2)}. \end{aligned}$$

Because  $\log \mathcal{L}_1$  depends only on the top-nest parameters  $\psi_1$  and  $\log \mathcal{L}_2$  only on the within-nest contrast  $\psi_2 = \frac{\delta_E - \delta_F}{\lambda}$ , joint MLE coincides with the concatenation of two independent maximizations. Reparameterizing the singleton-nest top level into a single binary index  $X_{i'}\beta_1$ , the two pieces actually estimated are

$$\text{Step 1: } \mathbb{P}(R_i = 1 | X_i) = \Lambda(X_{i'}\beta_1), \quad i = 1, \dots, N,$$

$$\text{Step 2: } \mathbb{P}(E_i = 1 | R_i = 1, Z_i) = \Lambda(Z_{i'}\beta_2), \quad i \in \{i : r_i = 1\}.$$

$\beta_1$  identifies correlates of resistance incidence;  $\beta_2$  identifies the within-resistance contrast  $\frac{\delta_E - \delta_F}{\lambda}$ , so its sign and significance are interpretable directly while its magnitude is identified jointly with the dissimilarity parameter  $\lambda$ . Estimating Step 1 and Step 2 separately writes the nested logit in sequential form, provided  $X_i$  and  $Z_i$  enter the linear indices as a clean partition and  $V_{iC}$  is normalized to zero.

Table A.49 reports three complementary specifications side by side. Columns (1)–(2) report a flat multinomial logit (no fixed effects) as an IIA-restricted anchor. Columns (3)–(4) and (5)–(6) estimate the sequential decomposition as binary logit and linear probability model respectively. Step 1 (Columns (3) and (5)) runs on one row per unique village and uses no fixed effects because compliant villages drawn from the full ledger have no observed neighboring ruler or event date. Step 2 (Columns (4) and (6)) runs on the full resistance-event panel and matches Table 3 Model 5: home  $\times$  neighbor dyadic fixed effects, decade fixed effects, the same geographic controls, and three-way clustering on home ruler, neighbor, and decade.

The sequential decomposition addresses selection into recorded resistance directly. Step 1 uses one row per unique village (46,296 villages, of which 2,482 appear in the resisting set). The binary logit gives  $\beta = +0.028$  ( $p = 0.58$ ), while the LPM gives  $\beta = -0.00078$  ( $p = 0.12$  heteroskedasticity-robust;  $p = 0.42$  Conley); both are small and statistically indistinguishable from zero. Step 2 is positive in both estimators: the binary logit gives  $\beta = 0.276$  (cluster  $p = 0.009$ ; Conley  $p = 0.017$ ) and the LPM gives  $\beta = 0.0244$  (cluster  $p = 0.001$ ; Conley  $p =$

0.004). The Step 2 LPM reproduces Table 3 Model 5. Geographic exposure is therefore associated with tactic composition (exit versus direct confrontation) without a detectable positive association with resistance incidence in this specification.

|                                  | <i>Multinomial logit</i> |                      | <i>Seq. binary logit</i> |                       | <i>Seq. LPM</i> |                       |
|----------------------------------|--------------------------|----------------------|--------------------------|-----------------------|-----------------|-----------------------|
| Outcome:                         | exit                     | direct confrontation | non-compliance           | exit   non-compliance | non-compliance  | exit   non-compliance |
| Models:                          | (1)                      | (2)                  | (3)                      | (4)                   | (5)             | (6)                   |
| <i>Variables</i>                 |                          |                      |                          |                       |                 |                       |
| Distance ratio ( $\ln(\rho_i)$ ) | 0.2823                   | −0.0055              | 0.0281                   | 0.2762                | −0.00078        | 0.0244                |
| Robust / 3-way cluster SE        | (0.0149)***              | (0.0167)             | (0.0514)                 | (0.1054)***           | (0.00050)       | (0.0068)***           |
| Conley SE (10km)                 | —                        | —                    | (0.0771)                 | (0.1159)**            | (0.00096)       | (0.0085)***           |
| <i>Controls</i>                  |                          |                      |                          |                       |                 |                       |
| Geo controls                     | Yes                      | Yes                  | Yes                      | Yes                   | Yes             | Yes                   |
| Home × Neighbor FE               | No                       | No                   | No                       | Yes                   | No              | Yes                   |
| Decade FE                        | No                       | No                   | No                       | Yes                   | No              | Yes                   |
| <i>Fit Statistics</i>            |                          |                      |                          |                       |                 |                       |
| Num. Obs.                        | 46,387                   | 46,387               | 46,296                   | 1,310                 | 46,296          | 2,739                 |
| Mean Dep. Var.                   | 0.014                    | 0.042                | 0.054                    | 0.246                 | 0.054           | 0.246                 |

Table A.49: Sequential Decomposition of the Village Choice: Logit and LPM

**Note:** Columns (1)–(2) report a flat multinomial logit (no fixed effects) as an IIA-restricted anchor; both coefficients come from a single regression with compliance as the baseline category. Columns (3)–(4) and (5)–(6) estimate the sequential decomposition as binary logit and linear probability model respectively. Step 1 (Columns (3) and (5)) runs on one row per unique village (non-compliance vs. compliance); it omits dyadic and decade fixed effects because compliant villages drawn from the full ledger have no observed neighboring ruler or event date. Step 2 (Columns (4) and (6)) runs on the full non-compliance-event panel (exit vs. direct confrontation conditional on non-compliance) and reproduces Table 3 Model 5: home × neighbor dyadic fixed effects, decade fixed effects, and the same geographic controls. Two SE rows are reported in parentheses. The first is heteroskedasticity-robust for the multinomial logit, Step-1 logit, and Step-1 LPM, and three-way clustered on home ruler, neighbor, and decade for Step 2. The second is Conley spatial inference with a 10 km cutoff, omitted for Columns (1)–(2) because no standard implementation exists for the multinomial likelihood. Stars reflect the corresponding estimator’s  $p$ -value. Signif. Codes: \*\*\*: 0.01, \*\*: 0.05, \*: 0.1.

#### D.3.4. Sequential Decomposition Under Time-Averaged Distance Ratio

The distance ratio used in the main Step-1 regression (Table A.49) differs across outcomes: resisting villages receive the event-time ratio, while compliant villages from the full ledger receive the end-period (1868) ratio. Table A.50 removes this asymmetry by applying to both groups each village’s mean annual home-capital distance divided by nearest-foreign-capital distance over 1615–1868, with active capitals reconstructed each year. Under this symmetric construction Step 1 is small and negative ( $\hat{\beta}_{\text{logit}} = -0.0303$ ,  $p = 0.0402$ ; LPM  $\hat{\beta} = -0.00155$ ,  $p = 0.0387$ ). Step 2, exit versus direct confrontation conditional on resistance, retains its positive sign ( $\hat{\beta} = 0.01436$ , cluster  $p = 0.0089$ , Conley  $p = 0.0524$ ). The symmetric measure therefore preserves the composition result without evidence that exposure positively selects villages into recorded resistance.

| Inference                                      | Step 1: non-compliance vs. compliance |              |                    | Step 2: exit vs. direct confrontation |                    |  |
|------------------------------------------------|---------------------------------------|--------------|--------------------|---------------------------------------|--------------------|--|
|                                                | Logit                                 | LPM (robust) | LPM (Conley 10 km) | LPM (3-way cluster)                   | LPM (Conley 10 km) |  |
| $\hat{\beta}$ on $\ln(\bar{p}_i^{\text{avg}})$ | -0.0303                               | -0.00155     | -0.00155           | 0.01436                               | 0.01436            |  |
| SE                                             | 0.0148                                | 0.00075      | 0.00072            | 0.00506                               | 0.00740            |  |
| $p$                                            | 0.0402                                | 0.0387       | 0.0311             | 0.0089                                | 0.0524             |  |
| Geo controls                                   | Yes                                   | Yes          | Yes                | Yes                                   | Yes                |  |
| Home $\times$ Neighbor FE                      | No                                    | No           | No                 | Yes                                   | Yes                |  |
| Decade FE                                      | No                                    | No           | No                 | Yes                                   | Yes                |  |
| Num. Obs.                                      | 46024                                 | 46024        | 46024              | 2740                                  | 2740               |  |
| Mean Dep. Var.                                 | 0.0543                                | 0.0543       | 0.0543             | 0.2464                                | 0.2464             |  |

Table A.50: Sequential Decomposition Under Time-Averaged Distance Ratio

**Note:** Replicates the sequential decomposition in Table A.49 with a time-averaged distance ratio applied symmetrically to resisting and compliant villages. The time-averaged ratio is the mean of the annual distance-to-home-domain capital / distance-to-nearest-foreign-domain capital across 1615–1868, with the active-domain panel recomputed each year using the augmented tenryo spells. This symmetric construction removes the measurement asymmetry in the existing Step-1 table, where the distance ratio for resisting villages used event-time capitals while that for compliant villages used end-period positions. Step 1 (non-compliance vs. compliance) is estimated on all 46,024 villages (2,497 non-compliant); Step 2 (exit vs. direct confrontation conditional on non-compliance) uses its own 2,740-event time-averaged analytic sample. Under time-averaged capitals Step 1 is a small negative association ( $\hat{\beta}_{\text{logit}} = -0.030$ ,  $p = 0.040$ ; LPM  $\hat{\beta} = -0.00155$ ,  $p = 0.039$ ), and Step 2 retains its positive sign ( $\hat{\beta} = 0.014$ ,  $p = 0.009$  cluster,  $p = 0.052$  Conley). The symmetric specification therefore provides no evidence that exposure positively selects villages into recorded non-compliance while preserving the positive composition estimate conditional on non-compliance.

### D.3.5. Linear Hypothesis Testing

In the village-level analysis for validating the village mobility measure, I use the log-ratio of home-capital distance to nearby-capital distance:

$$\ln\left(\frac{\text{Dist. to Home Cap.}}{\text{Dist. to Foreign Cap.}}\right) = \ln(\text{Dist. to Home Cap.}) - \ln(\text{Dist. to Foreign Cap.})$$

Implicit in this choice is the restriction:  $\beta_{\ln(\text{Dist. to Home Cap.})} + \beta_{\ln(\text{Dist. to Foreign Cap.})} = 0$

To check whether the data are consistent with this restriction, I estimate the unrestricted model including both  $\ln(\text{Dist. to Home Cap.})$  and  $\ln(\text{Dist. to Foreign Cap.})$  separately, then conduct a Wald test of the linear hypothesis  $\beta_{\ln(\text{Dist. to Home Cap.})} + \beta_{\ln(\text{Dist. to Foreign Cap.})} = 0$ .

| Corresponding Model | Outcome         | Restriction                                             | Df | $\chi^2$ | p-value |
|---------------------|-----------------|---------------------------------------------------------|----|----------|---------|
| Table 3 (5)         | $\text{exit}_i$ | $\beta_{\ln(\text{DHC})} + \beta_{\ln(\text{DFC})} = 0$ | 1  | 0.009    | 0.9244  |

Table A.51: Wald Test of the Log-Ratio Restriction

The Wald tests indicates that the log-ratio specification is consistent with the data; the high p-values (0.92 and 0.69, respectively) indicate no significant evidence against the restriction.

## E. DOMAIN TAXATION ANALYSIS

### E.1. Full Tables

#### E.1.1. Main Model

| Outcome:                      | $\tau_j$ |             |         |             |         |             |         |             |                      |             |                      |            |
|-------------------------------|----------|-------------|---------|-------------|---------|-------------|---------|-------------|----------------------|-------------|----------------------|------------|
| Models:                       | (1)      |             | (2)     |             | (3)     |             | (4)     |             | (5: SPL)             |             | (6: SPL)             |            |
| <i>Variables</i>              |          |             |         |             |         |             |         |             |                      |             |                      |            |
| $\ln(\rho_j)$                 | −0.0241  | (0.0053)*** | −0.0203 | (0.0058)*** | −0.0110 | (0.0055)**  | −0.0112 | (0.0055)**  | −0.0140              | (0.0047)*** | −0.0094              | (0.0051)*  |
| $\ln(\text{Rice})$            |          |             | −0.0010 | (0.0158)    | 0.0076  | (0.0157)    | 0.0116  | (0.0163)    |                      |             | 0.0004               | (0.0135)   |
| $\ln(\text{Area})$            |          |             | −0.0260 | (0.0112)**  | −0.0318 | (0.0144)**  | −0.0340 | (0.0151)**  |                      |             | −0.0227              | (0.0130)*  |
| $\ln(\text{Pop.})$            |          |             | 0.0392  | (0.0167)**  | 0.0263  | (0.0144)*   | 0.0232  | (0.0150)    |                      |             | 0.0222               | (0.0147)   |
| New FL                        |          |             | 0.0002  | (0.0003)    | 0.0005  | (0.0004)    | 0.0003  | (0.0004)    |                      |             | 0.0004               | (0.0002)   |
| Shog. Ally                    |          |             | −0.0316 | (0.0177)*   | 0.0059  | (0.0161)    | 0.0011  | (0.0164)    |                      |             | 0.0025               | (0.0143)   |
| Goshi                         |          |             | 0.0217  | (0.0256)    | 0.0196  | (0.0261)    | 0.0274  | (0.0271)    |                      |             | 0.0242               | (0.0215)   |
| lat                           |          |             |         |             | −0.5279 | (0.2934)*   | −0.4747 | (0.3077)    |                      |             | −0.3535              | (0.2855)   |
| lon                           |          |             |         |             | −0.1758 | (0.0754)**  | −0.1639 | (0.0796)**  |                      |             | −0.1252              | (0.0743)*  |
| $\ln(\text{Elev.})$           |          |             |         |             | −0.0118 | (0.0160)    | −0.0137 | (0.0165)    |                      |             | −0.0100              | (0.0134)   |
| $\ln(\text{Rugg.})$           |          |             |         |             | 0.0398  | (0.0160)**  | 0.0373  | (0.0158)**  |                      |             | 0.0238               | (0.0152)   |
| $\ln(\text{Dist. Edo/Kyoto})$ |          |             |         |             | −0.0370 | (0.0158)**  | −0.0441 | (0.0171)**  |                      |             | −0.0335              | (0.0167)** |
| Rainy                         |          |             |         |             | −1.6629 | (0.8969)*   | −1.5887 | (0.9078)*   |                      |             | −1.0143              | (0.7369)   |
| Snowy                         |          |             |         |             | 0.7322  | (1.8580)    | 0.6920  | (1.8597)    |                      |             | −0.2419              | (1.4193)   |
| Lat × Lon                     |          |             |         |             | 0.0042  | (0.0021)*   | 0.0038  | (0.0023)*   |                      |             | 0.0029               | (0.0021)   |
| Kin Nbs.                      |          |             |         |             |         |             | 0.0396  | (0.0296)    |                      |             | 0.0188               | (0.0257)   |
| Nb. FL                        |          |             |         |             |         |             | 0.0002  | (0.0001)    |                      |             | 0.0001               | (0.0001)   |
| $\ln(\text{Nb. Rice})$        |          |             |         |             |         |             | −0.0244 | (0.0189)    |                      |             | −0.0221              | (0.0172)   |
| $\ln(\text{Nb. Area})$        |          |             |         |             |         |             | 0.0180  | (0.0176)    |                      |             | 0.0116               | (0.0159)   |
| $\ln(\text{Nb. Pop.})$        |          |             |         |             |         |             | −0.0047 | (0.0207)    |                      |             | −0.0015              | (0.0196)   |
| Const.                        | 0.4152   | (0.0091)*** | 0.1760  | (0.1247)    | 23.1744 | (10.2659)** | 21.8425 | (10.7990)** | 0.1761               | (0.0274)*** | 16.4558              | (10.0938)  |
| <i>Fit Statistics</i>         |          |             |         |             |         |             |         |             |                      |             |                      |            |
| Num.Obs.                      | 232      |             | 230     |             | 230     |             | 230     |             | 230                  |             | 230                  |            |
| R2                            | 0.074    |             | 0.141   |             | 0.370   |             | 0.389   |             | 0.312 <sup>[a]</sup> |             | 0.439 <sup>[a]</sup> |            |

Table A.52: Geographic Exposure and Reported Domain Tax Rates

**Note:** OLS estimates (maximum likelihood in Models 5–6) of reported effective tax rates on the log domain-average distance ratio. HC1 standard errors appear in Models 1–4 and spatial-likelihood standard errors in Models 5–6. Signif. Codes: \*\*\*: 0.01, \*\*: 0.05, \*: 0.1. <sup>[a]</sup> Nagelkerke pseudo-R2

### E.1.2. Interaction Model

| Outcome:                                                         | $\tau_j$               |                        |                       |                       |
|------------------------------------------------------------------|------------------------|------------------------|-----------------------|-----------------------|
| Models:                                                          | (1)                    | (2)                    | (3)                   | (4)                   |
| <i>Variables</i>                                                 |                        |                        |                       |                       |
| Distance ratio $(\ln(\rho_j) \times \hat{y}_{-j})$ : <b>High</b> | -0.0350<br>(0.0084)*** | -0.0286<br>(0.0089)*** | -0.0143<br>(0.0070)** | -0.0158<br>(0.0069)** |
| Distance ratio $(\ln(\rho_j) \times \hat{y}_{-j})$ : <b>Low</b>  | -0.0161<br>(0.0053)*** | -0.0151<br>(0.0058)*** | -0.0092<br>(0.0063)   | -0.0085<br>(0.0064)   |
| <i>Controls</i>                                                  |                        |                        |                       |                       |
| Main                                                             | No                     | Yes                    | Yes                   | Yes                   |
| Geography                                                        | No                     | No                     | Yes                   | Yes                   |
| Neighbors                                                        | No                     | No                     | No                    | Yes                   |
| <i>Fit Statistics</i>                                            |                        |                        |                       |                       |
| Num.Obs.                                                         | 232                    | 230                    | 230                   | 230                   |
| R2                                                               | 0.091                  | 0.150                  | 0.371                 | 0.391                 |
| Mean Dep. Var.                                                   | 0.392                  | 0.394                  | 0.394                 | 0.394                 |

Table A.53: Geographic Exposure and Tax Rates, Interacted with Outside-Option Quality

**Note:** The coefficients represent OLS estimates of the relationship between the average distance ratio of each domain interacted with  $\hat{y}_{-j}$  (indicator of whether the weighted average labor-to-potential-yield ratio of neighboring domains is above its median) and the domain tax rate. Heteroskedasticity-robust standard errors in parentheses. Signif. Codes: .\*\*\*: 0.01, .\*\*: 0.05, .\*: 0.1.

### E.1.3. Interaction Model (Full Table)

| Outcome:                         | $\tau_j$ |             |         |             |         |             |         |             |
|----------------------------------|----------|-------------|---------|-------------|---------|-------------|---------|-------------|
| Models:                          | (1)      |             | (2)     |             | (3)     |             | (4)     |             |
| <i>Variables</i>                 |          |             |         |             |         |             |         |             |
| $\ln(\rho_j) \times \text{High}$ | −0.0350  | (0.0084)*** | −0.0286 | (0.0089)*** | −0.0143 | (0.0070)**  | −0.0158 | (0.0069)**  |
| $\ln(\rho_j) \times \text{Low}$  | −0.0161  | (0.0053)*** | −0.0151 | (0.0058)*** | −0.0092 | (0.0063)    | −0.0085 | (0.0064)    |
| $\ln(\text{Rice})$               |          |             | 0.0014  | (0.0156)    | 0.0083  | (0.0153)    | 0.0122  | (0.0159)    |
| $\ln(\text{Area})$               |          |             | −0.0256 | (0.0111)**  | −0.0314 | (0.0144)**  | −0.0336 | (0.0151)**  |
| $\ln(\text{Pop.})$               |          |             | 0.0360  | (0.0171)**  | 0.0253  | (0.0145)*   | 0.0221  | (0.0151)    |
| New FL                           |          |             | 0.0002  | (0.0003)    | 0.0005  | (0.0004)    | 0.0003  | (0.0004)    |
| Shog. Ally                       |          |             | −0.0300 | (0.0180)*   | 0.0064  | (0.0163)    | 0.0016  | (0.0166)    |
| Goshi                            |          |             | 0.0221  | (0.0257)    | 0.0197  | (0.0262)    | 0.0278  | (0.0271)    |
| lat                              |          |             |         |             | −0.5362 | (0.2946)*   | −0.4835 | (0.3095)    |
| lon                              |          |             |         |             | −0.1785 | (0.0758)**  | −0.1668 | (0.0802)**  |
| $\ln(\text{Elev.})$              |          |             |         |             | −0.0111 | (0.0160)    | −0.0125 | (0.0165)    |
| $\ln(\text{Rugg.})$              |          |             |         |             | 0.0381  | (0.0163)**  | 0.0348  | (0.0161)**  |
| $\ln(\text{Dist. Edo/Kyoto})$    |          |             |         |             | −0.0373 | (0.0159)**  | −0.0445 | (0.0172)**  |
| Rainy                            |          |             |         |             | −1.6210 | (0.9021)*   | −1.5333 | (0.9156)*   |
| Snowy                            |          |             |         |             | 0.6224  | (1.8655)    | 0.5548  | (1.8808)    |
| Lat $\times$ Lon                 |          |             |         |             | 0.0042  | (0.0022)*   | 0.0038  | (0.0023)*   |
| Kin Nbs.                         |          |             |         |             |         |             | 0.0404  | (0.0297)    |
| Nb. FL                           |          |             |         |             |         |             | 0.0002  | (0.0001)    |
| $\ln(\text{Nb. Rice})$           |          |             |         |             |         |             | −0.0224 | (0.0190)    |
| $\ln(\text{Nb. Area})$           |          |             |         |             |         |             | 0.0191  | (0.0174)    |
| $\ln(\text{Nb. Pop.})$           |          |             |         |             |         |             | −0.0081 | (0.0206)    |
| Const.                           | 0.4181   | (0.0092)*** | 0.1754  | (0.1211)    | 23.4988 | (10.3154)** | 22.2086 | (10.8672)** |
| <i>Fit Statistics</i>            |          |             |         |             |         |             |         |             |
| Num.Obs.                         | 232      |             | 230     |             | 230     |             | 230     |             |
| R2                               | 0.091    |             | 0.150   |             | 0.371   |             | 0.391   |             |

Table A.54: Geographic Exposure and Reported Tax Rates by Outside-Option Quality  
(Full Table)

**Note:** The coefficients represent OLS estimates of the relationship between the average distance ratio of each domain interacted with  $h@(\mathbf{y})_{\{-j\}}$  (indicator of whether the weighted average labor-to-potential-yield ratio of neighboring domains is above its median) and the domain tax rate. Heteroskedasticity-robust standard errors in parentheses. Signif. Codes: \*\*\*: 0.01, \*\*: 0.05, \*: 0.1.

## E.2. Robustness Checks

### E.2.1. Alternative Covariates

| Outcome:                        | $\tau_j$ |             |         |            |         |             |         |             |         |            |
|---------------------------------|----------|-------------|---------|------------|---------|-------------|---------|-------------|---------|------------|
| Models:                         | (1)      |             | (2)     |            | (3)     |             | (4)     |             | (5)     |            |
| <i>Variables</i>                |          |             |         |            |         |             |         |             |         |            |
| $\ln(\rho_j)$                   | −0.0099  | (0.0049)**  | −0.0106 | (0.0057)*  | −0.0112 | (0.0058)*   | −0.0108 | (0.0055)**  | −0.0131 | (0.0063)** |
| $\ln(\text{Kokudaka})$          | −0.0788  | (0.0160)*** |         |            |         |             |         |             |         |            |
| $\ln(\text{Pop.})$              | 0.0820   | (0.0168)*** |         |            |         |             |         |             | 0.0282  | (0.0156)*  |
| $\ln(\text{Area})$              | −0.0050  | (0.0127)    | −0.0328 | (0.0128)** | −0.0251 | (0.0142)*   | −0.0052 | (0.0064)    | −0.0350 | (0.0166)** |
| New FL                          | 0.0003   | (0.0004)    | 0.0003  | (0.0004)   | 0.0003  | (0.0004)    | 0.0004  | (0.0004)    | 0.0003  | (0.0003)   |
| Shog. Ally                      | 0.0068   | (0.0154)    | −0.0057 | (0.0165)   | 0.0009  | (0.0171)    | 0.0032  | (0.0167)    | 0.0069  | (0.0160)   |
| Goshi                           | 0.0245   | (0.0260)    | 0.0261  | (0.0272)   | 0.0309  | (0.0269)    | 0.0267  | (0.0273)    | 0.0321  | (0.0248)   |
| lat                             | −0.3791  | (0.2791)    | −0.4525 | (0.3160)   | −0.5680 | (0.3505)    | −0.3697 | (0.3181)    | 0.1399  | (0.2658)   |
| lon                             | −0.1412  | (0.0729)*   | −0.1581 | (0.0828)*  | −0.1879 | (0.0909)**  | −0.1392 | (0.0834)*   | 0.0166  | (0.0673)   |
| $\ln(\text{Elev.})$             | −0.0064  | (0.0157)    | −0.0144 | (0.0161)   | −0.0155 | (0.0165)    | −0.0203 | (0.0163)    | −0.0081 | (0.0167)   |
| $\ln(\text{Rugg.})$             | 0.0143   | (0.0156)    | 0.0352  | (0.0159)** | 0.0338  | (0.0164)**  | 0.0296  | (0.0157)*   | 0.0250  | (0.0152)   |
| $\ln(\text{Dist. Edo/Kyoto})$   | −0.0418  | (0.0165)**  | −0.0423 | (0.0173)** | −0.0435 | (0.0172)**  | −0.0481 | (0.0168)*** |         |            |
| Snowy                           | 1.8194   | (1.7508)    | 0.4931  | (1.8551)   | 0.9999  | (1.9673)    | 1.0110  | (1.9217)    | −0.8507 | (2.0177)   |
| Rainy                           | −2.0637  | (0.8820)**  | −1.4692 | (0.8923)   | −1.7103 | (0.9270)*   | −1.6807 | (0.9375)*   | −1.0287 | (0.9469)   |
| Kin Nbs.                        | 0.0266   | (0.0301)    | 0.0500  | (0.0283)*  | 0.0418  | (0.0292)    | 0.0468  | (0.0290)    | 0.0177  | (0.0304)   |
| Nb. FL                          | 0.0001   | (0.0001)    | 0.0002  | (0.0001)   | 0.0002  | (0.0001)    | 0.0002  | (0.0001)    | 0.0002  | (0.0001)*  |
| $\ln(\text{Nb. Rice})$          | −0.0050  | (0.0174)    | −0.0296 | (0.0189)   | −0.0291 | (0.0199)    | −0.0222 | (0.0186)    | −0.0190 | (0.0187)   |
| $\ln(\text{Nb. Area})$          | −0.0019  | (0.0170)    | 0.0176  | (0.0167)   | 0.0177  | (0.0176)    | 0.0100  | (0.0163)    | 0.0073  | (0.0182)   |
| $\ln(\text{Nb. Pop.})$          | −0.0073  | (0.0186)    | 0.0018  | (0.0203)   | 0.0005  | (0.0215)    | 0.0011  | (0.0202)    | −0.0039 | (0.0206)   |
| Lat × Lon                       | 0.0031   | (0.0021)    | 0.0036  | (0.0023)   | 0.0044  | (0.0026)*   | 0.0030  | (0.0023)    | −0.0008 | (0.0020)   |
| $\ln(\text{Rice})$              |          |             | −0.0017 | (0.0250)   | 0.0239  | (0.0151)    |         |             | 0.0047  | (0.0181)   |
| $\ln(\text{Num. Vils.})$        |          |             | 0.0373  | (0.0250)   |         |             |         |             |         |            |
| % Samurai Pop                   |          |             |         |            | −0.1259 | (0.1834)    |         |             |         |            |
| Population/wet-rice suitability |          |             |         |            |         |             | −0.0040 | (0.2763)    |         |            |
| $\ln(\text{Dist. Edo})$         |          |             |         |            |         |             |         |             | 0.0677  | (0.0261)** |
| Const.                          | 18.8458  | (9.8989)*   | 21.3229 | (11.1659)* | 25.2642 | (12.3202)** | 18.7213 | (11.2921)*  | −3.1359 | (8.9956)   |
| <i>Fit Statistics</i>           |          |             |         |            |         |             |         |             |         |            |
| Num.Obs.                        | 230      |             | 231     |            | 226     |             | 230     |             | 230     |            |
| R2                              | 0.446    |             | 0.397   |            | 0.383   |             | 0.374   |             | 0.399   |            |

Table A.55: Geographic Exposure and Reported Domain Tax Rates  
(Alternative Covariates)

**Note:** OLS estimates relating each domain's average distance ratio to its reported tax rate. The distance ratio measures geographic exposure, not observed competition or bargaining between rulers. Heteroskedasticity-robust standard errors are in parentheses. Signif. Codes: .\*\*\*: 0.01, .\*\*: 0.05, .\*: 0.1.

### E.2.2. Sensitivity Analysis

I assess the robustness of the domain-level results to omitted variable bias using the breakdown frontier framework of Diegert, Masten, and Poirier (2022). This method improves upon the widely used approach of Oster (2019) in a critical respect: it remains valid when the included control variables are correlated with the omitted confounders—a condition Diegert, Masten, and Poirier (2022) term *endogenous controls*. Standard sensitivity methods assume that the controls are exogenous, an assumption that is difficult to defend in observational cross-sections where controls such as population, productivity, and daimyō characteristics may themselves be influenced by unobserved factors. As Diegert, Masten, and Poirier (2022) show, residualization-based methods can produce incorrect robustness conclusions when this assumption is violated.

The key sensitivity parameter is  $\bar{r}_X$ , the ratio of the standard deviation of unobservables' influence on geographic exposure to that of observables. Figure A.9 displays the admissible set for the exposure coefficient as  $\bar{r}_X$  varies under the most conservative assumptions. The negative sign remains identified for  $\bar{r}_X < 0.213$ : at that threshold, the upper endpoint of the DMP identified set first includes zero. This is an SD ratio, not a share of predictive power. The calculation evaluates the observational coefficient; it does not convert the cross-section into a causal design.

When the impact of unobservables on the outcome is also bounded ( $\bar{r}_Y \leq \bar{r}_X$ , the “equal-importance” assumption), the identified set tightens considerably. Figure A.10 plots the breakdown frontier in the  $(\bar{r}_X, \bar{r}_Y)$  plane. The diagonal sign-identification breakdown is  $\bar{r} = 0.554$ : both SD ratios must reach about this level before the identified set admits a nonnegative coefficient.

Those breakdowns concern population sign identification, not sampling significance. To answer the latter question transparently, Table A.56 also reports auxiliary thresholds that hold the baseline HC1 standard error fixed and ask when the upper DMP endpoint plus 1.96 standard errors reaches zero. The thresholds are  $\bar{r}_X = 0.007$  with no outcome-side bound and  $\bar{r}_X = \bar{r}_Y = 0.096$  under equal importance. These are fixed-SE diagnostics rather than fully bias-adjusted DMP confidence sets. They show that the observational association is much easier to render statistically imprecise than to make its negative sign unidentified.

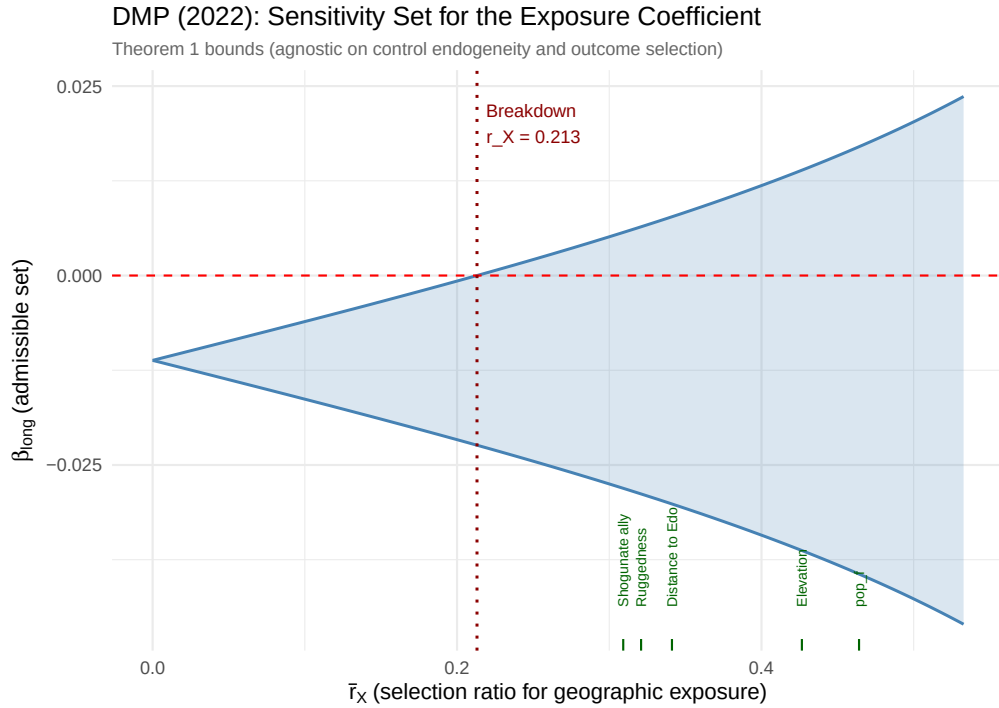


Figure A.9: DMP Sensitivity Set for the Exposure Coefficient

**Note:** The shaded region shows the range of  $\beta_{\text{long}}$  consistent with a given level of selection on unobservables ( $\bar{r}_X$ ). The dashed red line marks  $\beta = 0$ ; the dotted vertical line marks the breakdown point ( $\bar{r}_X = 0.213$ ). Green ticks calibrate  $\bar{r}_X$  to each labeled covariate's importance for predicting geographic exposure.

| Sensitivity criterion                          | $\bar{r}$ | Interpretation                                                |
|------------------------------------------------|-----------|---------------------------------------------------------------|
| Sign identification: $\bar{r}_X$ only          | 0.213     | Upper endpoint of the DMP identified set first includes zero  |
| Sign identification: $\bar{r}_X = \bar{r}_Y$   | 0.554     | Same criterion under the equal-importance restriction         |
| Fixed-SE significance: $\bar{r}_X$ only        | 0.007     | Upper endpoint plus 1.96 baseline HC1 SEs first includes zero |
| Fixed-SE significance: $\bar{r}_X = \bar{r}_Y$ | 0.096     | Same auxiliary criterion under equal importance               |

Table A.56: Diegert-Masten Sensitivity Thresholds for the Tax Regression

**Note:** Thresholds for Table 4 Model 4 (fully controlled).  $\bar{r}_X$  and  $\bar{r}_Y$  are ratios of standard deviations, not shares of explained variance. The first two rows are DMP population sign-identification breakdowns: the observed estimate remains fixed while the identified set widens to include zero. The last two rows answer the separate significance question by adding and subtracting 1.96 times the baseline HC1 SE from the DMP set while holding that SE fixed. They are transparent auxiliary thresholds, not fully bias-adjusted DMP confidence sets.

As a complementary sample-based check, I re-estimate Model 4 after excluding Chōshū, Satsuma, Mito, Tosa, and Aizu, whose 1850s–1860s rates may reflect acute bakumatsu fiscal and political distress. Excluding them changes the point estimate from  $-0.0112$  to  $-0.0106$ , with the  $p$ -value moving from 0.045 to 0.055 (Table A.57). The sign and magnitude are therefore not attributable to these five high-profile domains.

| Sample                                      | Num. Obs. | Estimate | SE       | $p$   |
|---------------------------------------------|-----------|----------|----------|-------|
| Full sample (paper)                         | 230       | −0.0112  | (0.0055) | 0.045 |
| Excluding Chōshū, Satsuma, Mito, Tosa, Aizu | 225       | −0.0106  | (0.0055) | 0.055 |

Table A.57: Domain Tax Regression: Excluding Bakumatsu Political Heavyweights

**Note:** Re-runs Table 4 Model 4 (fully controlled) on the sub-sample that excludes five domains (Chōshū, Satsuma, Mito, Tosa, Aizu) whose tax rates in the 1850s–1860s reflect acute fiscal and political distress rather than ordinary late-period tax setting. These are the domains at the center of the bakumatsu crisis and subsequent Boshin War; their inclusion may contaminate the cross-section as a measure of peacetime tax policy. The sign and magnitude of the coefficient on  $\log(\rho_j)$  are preserved while precision becomes marginal.

For completeness, Table A.58 reports the Oster (2019) coefficient-stability parameter. This method, while informative, assumes exogenous controls and is therefore best interpreted as supplementary to the DMP analysis above.

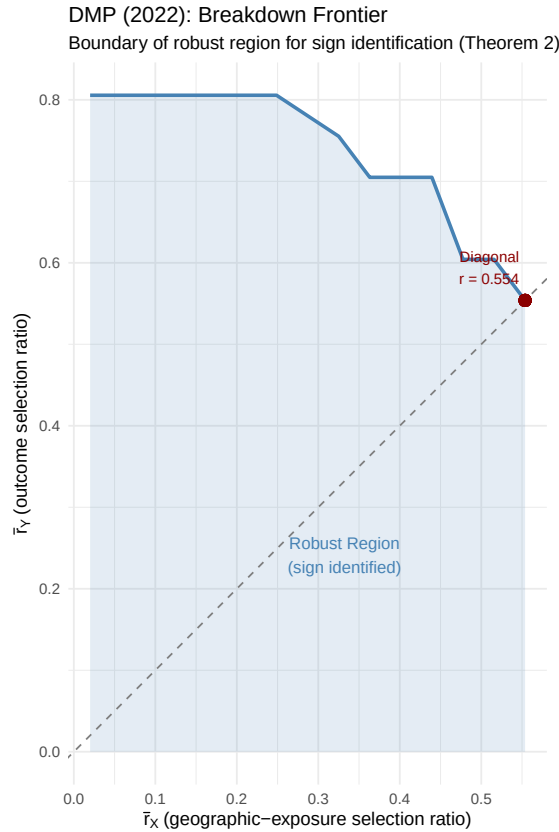


Figure A.10: DMP Breakdown Frontier

**Note:** The solid curve traces the sensitivity boundary: below it, the exposure coefficient remains negative under the maintained bounds. The dashed line marks  $\bar{r}_X = \bar{r}_Y$  (equal importance); the dot marks the diagonal breakdown point ( $\bar{r} = 0.554$ ).

| Oster (2019) Coefficient Stability                                        |                              |                             |                      |                     |          |
|---------------------------------------------------------------------------|------------------------------|-----------------------------|----------------------|---------------------|----------|
|                                                                           | $\hat{\beta}_{\text{short}}$ | $\hat{\beta}_{\text{full}}$ | $R_{\text{short}}^2$ | $R_{\text{full}}^2$ | $\delta$ |
| $R_{\text{max}} = 1.3\tilde{R}^2$                                         | -0.0245                      | -0.0112                     | 0.077                | 0.389               | 2.245    |
| $R_{\text{max}} = 1$                                                      | -0.0245                      | -0.0112                     | 0.077                | 0.389               | 0.428    |
| $\hat{\beta}^*(\delta = 1) = -0.006$ at $R_{\text{max}} = 1.3\tilde{R}^2$ |                              |                             |                      |                     |          |

Table A.58: Oster Sensitivity Analysis (Supplementary)

**Note:** Based on Model 4 in [Table 4](#). The Oster  $\delta$  is the proportional selection ratio needed to drive  $\hat{\beta}^*$  to zero;  $\delta > 1$  indicates robustness under  $R_{\text{max}} = 1.3\tilde{R}^2$ . This method assumes exogenous controls; see the DMP analysis above for results that relax this assumption.

### E.2.3. Data-Driven Control Variables Selection

The hand-picked control set in [Table 4](#) reflects substantive reasoning about confounders, but the choice of which variables to include is itself a researcher degree of freedom. I assess this concern using double/debiased machine learning (DML), which replaces manual nuisance-function specification with cross-fitted prediction ([Chernozhukov et al. 2018](#)). DML estimates a partially linear model

$$\tau_j = \beta \ln(\rho_j) + g(X_j) + \varepsilon_j, \quad \ln(\rho_j) = m(X_j) + \nu_j \quad (1)$$

where  $g(\cdot)$  and  $m(\cdot)$  predict the outcome and geographic exposure, and  $\beta$  is recovered after partialling both variables on  $X_j$ . Five-fold cross-fitting reduces overfitting bias in these nuisance estimates. I report two feature dictionaries: the 19 columns that reproduce the controls in [Table 4](#) Model 4, and an enriched 38-feature dictionary that adds quadratic terms and selected pairwise interactions. Within the exact dictionary I compare ridge and random-forest nuisance learners; within the enriched dictionary I compare ridge with a stack of LASSO, ridge, elastic net, and random forest.

The results are mixed. Ridge estimates remain negative: the exact-dictionary estimate is statistically distinguishable from zero at the 5% level, while the enriched-dictionary estimate is marginal at the 10% level. The random-forest and stacked estimates are close to zero and imprecise. DML therefore does not uniformly corroborate the hand-picked OLS association; instead, it shows that the pooled estimate is sensitive to highly flexible nuisance learning in this modest domain-level sample ( $N = 230$ ).

| Outcome: $\tau_j$ . Treatment: $\ln(\rho_j)$                                                                                                                       |          |          |       |       |           |           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|-------|-------|-----------|-----------|
| Method                                                                                                                                                             | Estimate | S.E.     | $t$   | $p$   | 95% CI lo | 95% CI hi |
| DML-Ridge (19 exact)                                                                                                                                               | −0.0114  | (0.0056) | −2.03 | 0.042 | −0.0225   | −0.0004   |
| DML-Random Forest (19 exact)                                                                                                                                       | −0.0007  | (0.0054) | −0.13 | 0.895 | −0.0113   | 0.0099    |
| DML-Ridge (38 enriched)                                                                                                                                            | −0.0098  | (0.0057) | −1.74 | 0.082 | −0.0209   | 0.0013    |
| DML-Stacked Ensemble (38 enriched)                                                                                                                                 | 0.0018   | (0.0055) | 0.32  | 0.745 | −0.0090   | 0.0126    |
| Hand-picked OLS (HC1)                                                                                                                                              | −0.0112  | (0.0055) | −2.02 | 0.045 | −0.0221   | −0.0003   |
| <i>Exact dictionary: <math>p = 19</math> Model 4 columns. Enriched dictionary: <math>p = 38</math> (17 base + 8 quadratics + 10 interactions + 3 kin dummies).</i> |          |          |       |       |           |           |
| <i>DML: 5 folds and 10 repetitions, except 5 repetitions for the enriched stack. Stack: LASSO + Ridge + Elastic Net + random forest.</i>                           |          |          |       |       |           |           |

Table A.59: Data-Driven Control Selection: DML Estimates

**Note:** Each DML row estimates the partial association between  $\ln(\rho_j)$  and  $\tau_j$  using the indicated nuisance learner and feature dictionary. Reported standard errors incorporate repeated cross-fitting. The OLS row reproduces Model 4 from Table 4 with HC1 standard errors.

### E.2.4. Controlling for Average Distance to Home Capital

| Outcome:                        | $\tau_j$ |             |         |            |         |             |         |             |                      |            |
|---------------------------------|----------|-------------|---------|------------|---------|-------------|---------|-------------|----------------------|------------|
| Models:                         | (1)      |             | (2)     |            | (3)     |             | (4)     |             | (5: SPL)             |            |
| <i>Variables</i>                |          |             |         |            |         |             |         |             |                      |            |
| $\ln(\rho_j)$                   | -0.0336  | (0.0116)*** | -0.0200 | (0.0132)   | -0.0148 | (0.0135)    | -0.0171 | (0.0143)    | -0.0231              | (0.0136)*  |
| $\ln(\text{Avg. Dist to HC}_j)$ | 0.0132   | (0.0156)    | -0.0004 | (0.0169)   | 0.0049  | (0.0165)    | 0.0077  | (0.0178)    | 0.0178               | (0.0163)   |
| $\ln(\text{Rice})$              |          |             | -0.0011 | (0.0160)   | 0.0083  | (0.0154)    | 0.0121  | (0.0161)    | 0.0010               | (0.0135)   |
| $\ln(\text{Area})$              |          |             | -0.0259 | (0.0123)** | -0.0328 | (0.0138)**  | -0.0353 | (0.0147)**  | -0.0254              | (0.0132)*  |
| $\ln(\text{Pop.})$              |          |             | 0.0393  | (0.0167)** | 0.0261  | (0.0145)*   | 0.0231  | (0.0150)    | 0.0219               | (0.0146)   |
| New FL                          |          |             | 0.0002  | (0.0003)   | 0.0005  | (0.0004)    | 0.0003  | (0.0004)    | 0.0004               | (0.0002)   |
| Shog. Ally                      |          |             | -0.0316 | (0.0178)*  | 0.0059  | (0.0162)    | 0.0010  | (0.0165)    | 0.0023               | (0.0143)   |
| Goshi                           |          |             | 0.0217  | (0.0257)   | 0.0196  | (0.0262)    | 0.0274  | (0.0270)    | 0.0241               | (0.0214)   |
| lat                             |          |             |         |            | -0.5324 | (0.2940)*   | -0.4736 | (0.3085)    | -0.3473              | (0.2845)   |
| lon                             |          |             |         |            | -0.1771 | (0.0755)**  | -0.1637 | (0.0798)**  | -0.1236              | (0.0740)*  |
| $\ln(\text{Elev.})$             |          |             |         |            | -0.0117 | (0.0159)    | -0.0134 | (0.0164)    | -0.0094              | (0.0134)   |
| $\ln(\text{Rugg.})$             |          |             |         |            | 0.0397  | (0.0160)**  | 0.0372  | (0.0158)**  | 0.0232               | (0.0151)   |
| $\ln(\text{Dist. Edo/Kyoto})$   |          |             |         |            | -0.0383 | (0.0172)**  | -0.0451 | (0.0177)**  | -0.0356              | (0.0168)** |
| Rainy                           |          |             |         |            | -1.6816 | (0.9046)*   | -1.6114 | (0.9179)*   | -1.0491              | (0.7358)   |
| Snowy                           |          |             |         |            | 0.7952  | (1.9057)    | 0.7810  | (1.9144)    | -0.0637              | (1.4268)   |
| Lat $\times$ Lon                |          |             |         |            | 0.0042  | (0.0021)*   | 0.0038  | (0.0023)*   | 0.0028               | (0.0021)   |
| Kin Nbs.                        |          |             |         |            |         |             | 0.0391  | (0.0299)    | 0.0171               | (0.0256)   |
| Nb. FL                          |          |             |         |            |         |             | 0.0002  | (0.0001)    | 0.0001               | (0.0001)   |
| $\ln(\text{Nb. Rice})$          |          |             |         |            |         |             | -0.0234 | (0.0188)    | -0.0196              | (0.0172)   |
| $\ln(\text{Nb. Area})$          |          |             |         |            |         |             | 0.0170  | (0.0177)    | 0.0089               | (0.0160)   |
| $\ln(\text{Nb. Pop.})$          |          |             |         |            |         |             | -0.0060 | (0.0208)    | -0.0044              | (0.0197)   |
| Const.                          | 0.3815   | (0.0424)*** | 0.1773  | (0.1374)   | 23.3407 | (10.2900)** | 21.8218 | (10.8253)** | 16.2389              | (10.0560)  |
| <i>Fit Statistics</i>           |          |             |         |            |         |             |         |             |                      |            |
| Num.Obs.                        | 232      |             | 230     |            | 230     |             | 230     |             | 230                  |            |
| R2                              | 0.077    |             | 0.141   |            | 0.371   |             | 0.389   |             | 0.442 <sup>[a]</sup> |            |

Table A.60: Controlling for Average Distance to Home Capital

**Note:** OLS estimates (maximum likelihood in Model 5) relating each domain's average distance ratio to its reported tax rate while controlling for average distance to the home capital. HC1 standard errors appear in Models 1–4 and spatial-likelihood standard errors in Model 5. Signif. Codes: .\*\*\*: 0.01, .\*\*: 0.05, .\*: 0.1. <sup>[a]</sup> Nagelkerke pseudo-R2

### E.2.5. Least-Costly Path Distance

| Outcome:                      | $\tau_j$ |             |         |             |         |             |         |            |                      |            |
|-------------------------------|----------|-------------|---------|-------------|---------|-------------|---------|------------|----------------------|------------|
| Models:                       | (1)      |             | (2)     |             | (3)     |             | (4)     |            | (5: SPL)             |            |
| <i>Variables</i>              |          |             |         |             |         |             |         |            |                      |            |
| $\ln(\rho_j^{\text{LCP}})$    | -0.0116  | (0.0035)*** | -0.0097 | (0.0037)*** | -0.0058 | (0.0030)*   | -0.0060 | (0.0031)*  | -0.0053              | (0.0032)   |
| $\ln(\text{Rice})$            |          |             | -0.0033 | (0.0147)    | 0.0071  | (0.0150)    | 0.0110  | (0.0157)   | -0.0003              | (0.0135)   |
| $\ln(\text{Area})$            |          |             | -0.0242 | (0.0112)**  | -0.0308 | (0.0139)**  | -0.0333 | (0.0147)** | -0.0220              | (0.0130)*  |
| $\ln(\text{Pop.})$            |          |             | 0.0419  | (0.0165)**  | 0.0268  | (0.0143)*   | 0.0239  | (0.0149)   | 0.0228               | (0.0147)   |
| New FL                        |          |             | 0.0002  | (0.0003)    | 0.0005  | (0.0004)    | 0.0003  | (0.0004)   | 0.0004               | (0.0002)   |
| Shog. Ally                    |          |             | -0.0355 | (0.0178)**  | 0.0052  | (0.0162)    | 0.0005  | (0.0165)   | 0.0022               | (0.0144)   |
| Goshi                         |          |             | 0.0210  | (0.0257)    | 0.0198  | (0.0263)    | 0.0279  | (0.0272)   | 0.0246               | (0.0215)   |
| lat                           |          |             |         |             | -0.4938 | (0.2899)*   | -0.4356 | (0.3059)   | -0.3222              | (0.2840)   |
| lon                           |          |             |         |             | -0.1682 | (0.0746)**  | -0.1547 | (0.0792)*  | -0.1177              | (0.0740)   |
| $\ln(\text{Elev.})$           |          |             |         |             | -0.0101 | (0.0162)    | -0.0120 | (0.0167)   | -0.0087              | (0.0134)   |
| $\ln(\text{Rugg.})$           |          |             |         |             | 0.0391  | (0.0159)**  | 0.0366  | (0.0157)** | 0.0232               | (0.0152)   |
| $\ln(\text{Dist. Edo/Kyoto})$ |          |             |         |             | -0.0371 | (0.0160)**  | -0.0440 | (0.0174)** | -0.0334              | (0.0167)** |
| Rainy                         |          |             |         |             | -1.6610 | (0.8949)*   | -1.6083 | (0.9031)*  | -1.0184              | (0.7381)   |
| Snowy                         |          |             |         |             | 0.7901  | (1.8375)    | 0.7935  | (1.8372)   | -0.1818              | (1.4196)   |
| Lat $\times$ Lon              |          |             |         |             | 0.0039  | (0.0021)*   | 0.0035  | (0.0022)   | 0.0026               | (0.0021)   |
| Kin Nbs.                      |          |             |         |             |         |             | 0.0408  | (0.0296)   | 0.0195               | (0.0257)   |
| Nb. FL                        |          |             |         |             |         |             | 0.0002  | (0.0001)   | 0.0001               | (0.0001)   |
| $\ln(\text{Nb. Rice})$        |          |             |         |             |         |             | -0.0228 | (0.0189)   | -0.0208              | (0.0172)   |
| $\ln(\text{Nb. Area})$        |          |             |         |             |         |             | 0.0197  | (0.0177)   | 0.0128               | (0.0158)   |
| $\ln(\text{Nb. Pop.})$        |          |             |         |             |         |             | -0.0079 | (0.0204)   | -0.0041              | (0.0195)   |
| Const.                        | 0.4083   | (0.0092)*** | 0.1663  | (0.1205)    | 22.1064 | (10.1745)** | 20.5929 | (10.7617)* | 15.4412              | (10.0485)  |
| <i>Fit Statistics</i>         |          |             |         |             |         |             |         |            |                      |            |
| Num.Obs.                      | 232      |             | 230     |             | 230     |             | 230     |            | 230                  |            |
| R2                            | 0.037    |             | 0.119   |             | 0.367   |             | 0.385   |            | 0.437 <sup>[a]</sup> |            |

Table A.61: Geographic Exposure and Reported Tax Rates

(Least-Cost Paths)

**Note:** OLS estimates (maximum likelihood in Model 5) relating each domain's stored least-cost-path distance ratio to its reported tax rate. HC1 standard errors appear in Models 1–4 and spatial-likelihood standard errors in Model 5. The ratio measures geographic exposure, not observed ruler competition. Signif. Codes: .\*\*\*: 0.01, .\*\*: 0.05, .\*: 0.1. <sup>[a]</sup> Nagelkerke pseudo-R2

### E.2.6. Conley Standard Error Bandwidth Sensitivity

Table A.62 reports the bandwidth sensitivity of Conley spatial standard errors for the fully-controlled domain tax regression (Table 4 Model 4). The point estimate is stable across every bandwidth considered, and the coefficient on  $\log(\rho_j)$  is statistically significant at the 5% level for bandwidths up to 75 km. As expected for a cross-sectional specification with rich spatial controls, the standard errors widen somewhat at the longest bandwidths (100 and 150 km), where the spatial HAC kernel absorbs progressively more of the variation already captured by the included geographic covariates; the  $p$ -value at 100 km is 0.102 and at 150 km is 0.145. The result thus holds under the bandwidths most commonly applied in the spatial-econometrics literature for cross-sectional designs at this geographic scale.

| Standard error                    | Estimate            | SE                    | $p$                |
|-----------------------------------|---------------------|-----------------------|--------------------|
| Heteroskedasticity-robust (paper) | -0.0111952548949787 | (0.00554317773276672) | 0.0446969310445375 |
| Conley 25 km                      | -0.0111952548949787 | (0.00486874644301278) | 0.0224693058430259 |
| Conley 50 km                      | -0.0111952548949787 | (0.00557498706242249) | 0.0459175666096826 |
| Conley 75 km                      | -0.0111952548949787 | (0.00539253446894661) | 0.0391111847436515 |
| Conley 100 km                     | -0.0111952548949787 | (0.00682291828207967) | 0.102336570113632  |
| Conley 150 km                     | -0.0111952548949787 | (0.00765342600008831) | 0.145030590653059  |

Table A.62: Conley SE Bandwidth Sensitivity: Domain Tax Regression

**Note:** The coefficient on  $\log(\rho_j)$  from Table 4 Model 4 (fully controlled) is significant at the 5% level for Conley bandwidths up to 75 km, marginal at 100 km ( $p = 0.091$ ), and not significant at 150 km ( $p = 0.130$ ). The first row reports the heteroskedasticity-robust standard error used in the main table.

## E.3. Additional Analyses

### E.3.1. Average Distance to Home Capital

| Outcome:                               | $\tau_j$       |                    |                |                    |                |                  |                |                 |                      |                 |
|----------------------------------------|----------------|--------------------|----------------|--------------------|----------------|------------------|----------------|-----------------|----------------------|-----------------|
| Models:                                | (1)            |                    | (2)            |                    | (3)            |                  | (4)            |                 | (5: SPL)             |                 |
| <i>Variables</i>                       |                |                    |                |                    |                |                  |                |                 |                      |                 |
| <b>ln(Avg. Dist to HC<sub>j</sub>)</b> | <b>−0.0234</b> | <b>(0.0072)***</b> | <b>−0.0226</b> | <b>(0.0072)***</b> | <b>−0.0114</b> | <b>(0.0068)*</b> | <b>−0.0114</b> | <b>(0.0069)</b> | <b>−0.0081</b>       | <b>(0.0061)</b> |
| ln(Rice)                               |                |                    | −0.0065        | (0.0151)           | 0.0059         | (0.0154)         | 0.0108         | (0.0162)        | −0.0002              | (0.0135)        |
| ln(Area)                               |                |                    | −0.0194        | (0.0111)*          | −0.0291        | (0.0140)**       | −0.0320        | (0.0149)**      | −0.0213              | (0.0130)        |
| ln(Pop.)                               |                |                    | 0.0421         | (0.0167)**         | 0.0269         | (0.0144)*        | 0.0233         | (0.0151)        | 0.0223               | (0.0147)        |
| New FL                                 |                |                    | 0.0001         | (0.0003)           | 0.0004         | (0.0004)         | 0.0003         | (0.0004)        | 0.0003               | (0.0002)        |
| Shog. Ally                             |                |                    | −0.0341        | (0.0175)*          | 0.0049         | (0.0160)         | 0.0003         | (0.0164)        | 0.0013               | (0.0144)        |
| Goshi                                  |                |                    | 0.0209         | (0.0258)           | 0.0196         | (0.0263)         | 0.0276         | (0.0274)        | 0.0244               | (0.0216)        |
| lat                                    |                |                    |                |                    | −0.4960        | (0.2905)*        | −0.4513        | (0.3081)        | −0.3219              | (0.2861)        |
| lon                                    |                |                    |                |                    | −0.1680        | (0.0747)**       | −0.1581        | (0.0797)**      | −0.1174              | (0.0745)        |
| ln(Elev.)                              |                |                    |                |                    | −0.0111        | (0.0158)         | −0.0131        | (0.0163)        | −0.0090              | (0.0135)        |
| ln(Rugg.)                              |                |                    |                |                    | 0.0391         | (0.0159)**       | 0.0365         | (0.0157)**      | 0.0227               | (0.0152)        |
| ln(Dist. Edo/Kyoto)                    |                |                    |                |                    | −0.0333        | (0.0161)**       | −0.0418        | (0.0172)**      | −0.0315              | (0.0167)*       |
| Rainy                                  |                |                    |                |                    | −1.6381        | (0.8959)*        | −1.5879        | (0.9058)*       | −1.0377              | (0.7409)        |
| Snowy                                  |                |                    |                |                    | 0.6554         | (1.8687)         | 0.6559         | (1.8770)        | −0.2013              | (1.4338)        |
| Lat × Lon                              |                |                    |                |                    | 0.0039         | (0.0021)*        | 0.0036         | (0.0023)        | 0.0026               | (0.0021)        |
| Kin Nbs.                               |                |                    |                |                    |                |                  | 0.0407         | (0.0295)        | 0.0201               | (0.0258)        |
| Nb. FL                                 |                |                    |                |                    |                |                  | 0.0002         | (0.0001)        | 0.0001               | (0.0001)        |
| ln(Nb. Rice)                           |                |                    |                |                    |                |                  | −0.0256        | (0.0190)        | −0.0228              | (0.0173)        |
| ln(Nb. Area)                           |                |                    |                |                    |                |                  | 0.0207         | (0.0176)        | 0.0142               | (0.0158)        |
| ln(Nb. Pop.)                           |                |                    |                |                    |                |                  | −0.0040        | (0.0207)        | −0.0018              | (0.0198)        |
| Const.                                 | 0.4679         | (0.0248)***        | 0.2420         | (0.1226)**         | 22.0932        | (10.1723)**      | 21.0544        | (10.8064)*      | 15.4092              | (10.1175)       |
| <i>Fit Statistics</i>                  |                |                    |                |                    |                |                  |                |                 |                      |                 |
| Num.Obs.                               | 232            |                    | 230            |                    | 230            |                  | 230            |                 | 230                  |                 |
| R2                                     | 0.046          |                    | 0.135          |                    | 0.367          |                  | 0.385          |                 | 0.435 <sup>[a]</sup> |                 |

Table A.63: Average Distance to Home Capital Weakly Predicts Tax Rates

**Note:** OLS estimates (maximum likelihood in Model 5) relating each domain's average absolute distance to its home capital (log) to its reported tax rate. HC1 standard errors appear in Models 1–4 and spatial-likelihood standard errors in Model 5. Signif. Codes: \*\*\*: 0.01, \*\*: 0.05, \*: 0.1. <sup>[a]</sup> Nagelkerke pseudo-R2

### E.3.2. Controlling for the Shortest Distance to the Border

| Outcome:                          | $\tau_j$ |             |         |             |         |             |         |             |                      |            |
|-----------------------------------|----------|-------------|---------|-------------|---------|-------------|---------|-------------|----------------------|------------|
| Models:                           | (1)      |             | (2)     |             | (3)     |             | (4)     |             | (5: SPL)             |            |
| <i>Variables</i>                  |          |             |         |             |         |             |         |             |                      |            |
| $\ln(\rho_j)$                     | -0.0240  | (0.0053)*** | -0.0200 | (0.0058)*** | -0.0111 | (0.0055)**  | -0.0113 | (0.0055)**  | -0.0095              | (0.0051)*  |
| $\ln(\text{Avg. Dist. Border}_j)$ | 0.0014   | (0.0067)    | -0.0106 | (0.0084)    | -0.0056 | (0.0071)    | -0.0038 | (0.0067)    | -0.0024              | (0.0075)   |
| $\ln(\text{Rice})$                |          |             | -0.0004 | (0.0152)    | 0.0083  | (0.0155)    | 0.0121  | (0.0162)    | 0.0007               | (0.0135)   |
| $\ln(\text{Area})$                |          |             | -0.0198 | (0.0121)    | -0.0286 | (0.0148)*   | -0.0318 | (0.0156)**  | -0.0214              | (0.0136)   |
| $\ln(\text{Pop.})$                |          |             | 0.0379  | (0.0165)**  | 0.0254  | (0.0145)*   | 0.0226  | (0.0151)    | 0.0218               | (0.0147)   |
| New FL                            |          |             | 0.0001  | (0.0003)    | 0.0005  | (0.0004)    | 0.0003  | (0.0004)    | 0.0004               | (0.0002)   |
| Shog. Ally                        |          |             | -0.0319 | (0.0178)*   | 0.0055  | (0.0162)    | 0.0010  | (0.0165)    | 0.0024               | (0.0143)   |
| Goshi                             |          |             | 0.0261  | (0.0262)    | 0.0210  | (0.0263)    | 0.0282  | (0.0272)    | 0.0247               | (0.0215)   |
| lat                               |          |             |         |             | -0.5551 | (0.2901)*   | -0.4979 | (0.3056)    | -0.3688              | (0.2893)   |
| lon                               |          |             |         |             | -0.1820 | (0.0746)**  | -0.1693 | (0.0791)**  | -0.1287              | (0.0751)*  |
| $\ln(\text{Elev.})$               |          |             |         |             | -0.0130 | (0.0160)    | -0.0143 | (0.0166)    | -0.0105              | (0.0135)   |
| $\ln(\text{Rugg.})$               |          |             |         |             | 0.0408  | (0.0159)**  | 0.0380  | (0.0157)**  | 0.0243               | (0.0153)   |
| $\ln(\text{Dist. Edo/Kyoto})$     |          |             |         |             | -0.0370 | (0.0159)**  | -0.0444 | (0.0172)**  | -0.0337              | (0.0167)** |
| Rainy                             |          |             |         |             | -1.6537 | (0.8933)*   | -1.5857 | (0.9065)*   | -1.0138              | (0.7370)   |
| Snowy                             |          |             |         |             | 0.7125  | (1.8486)    | 0.6839  | (1.8577)    | -0.2447              | (1.4195)   |
| Lat $\times$ Lon                  |          |             |         |             | 0.0043  | (0.0021)**  | 0.0039  | (0.0022)*   | 0.0030               | (0.0021)   |
| Kin Nbs.                          |          |             |         |             |         |             | 0.0396  | (0.0296)    | 0.0189               | (0.0257)   |
| Nb. FL                            |          |             |         |             |         |             | 0.0002  | (0.0001)    | 0.0001               | (0.0001)   |
| $\ln(\text{Nb. Rice})$            |          |             |         |             |         |             | -0.0238 | (0.0187)    | -0.0218              | (0.0172)   |
| $\ln(\text{Nb. Area})$            |          |             |         |             |         |             | 0.0180  | (0.0176)    | 0.0116               | (0.0159)   |
| $\ln(\text{Nb. Pop.})$            |          |             |         |             |         |             | -0.0045 | (0.0206)    | -0.0014              | (0.0196)   |
| Const.                            | 0.4130   | (0.0138)*** | 0.1620  | (0.1210)    | 24.0420 | (10.1575)** | 22.5897 | (10.7386)** | 16.9504              | (10.2058)* |
| <i>Fit Statistics</i>             |          |             |         |             |         |             |         |             |                      |            |
| Num.Obs.                          | 232      |             | 230     |             | 230     |             | 230     |             | 230                  |            |
| R2                                | 0.074    |             | 0.147   |             | 0.372   |             | 0.389   |             | 0.439 <sup>[a]</sup> |            |

Table A.64: Geographic Exposure and Reported Tax Rates  
(Controlling for Border Distance)

**Note:** OLS estimates (maximum likelihood in Model 5) relating each domain's average distance ratio to its reported tax rate while controlling for average distance to the nearest border. HC1 standard errors appear in Models 1–4 and spatial-likelihood standard errors in Model 5. Signif. Codes: \*\*\*: 0.01, \*\*: 0.05, \*: 0.1. <sup>[a]</sup> Nagelkerke pseudo-R2

### E.3.3. Outside-Option Quality Using Population Density

| Outcome:                            | $\tau_j$ |             |         |             |         |             |         |             |         |             |
|-------------------------------------|----------|-------------|---------|-------------|---------|-------------|---------|-------------|---------|-------------|
| Models:                             | (1)      |             | (2)     |             | (3)     |             | (4)     |             | (5)     |             |
| <i>Variables</i>                    |          |             |         |             |         |             |         |             |         |             |
| $\ln(\rho_j) \times$<br><b>Low</b>  | -0.0299  | (0.0097)*** | -0.0262 | (0.0092)*** | -0.0197 | (0.0073)*** | -0.0265 | (0.0094)*** | -0.0204 | (0.0071)*** |
| $\ln(\rho_j) \times$<br><b>High</b> | -0.0216  | (0.0056)*** | -0.0176 | (0.0062)*** | -0.0074 | (0.0059)    | -0.0176 | (0.0063)*** | -0.0067 | (0.0062)    |
| ln(Rice)                            |          |             | -0.0016 | (0.0156)    | 0.0071  | (0.0152)    | -0.0019 | (0.0164)    | 0.0114  | (0.0160)    |
| ln(Area)                            |          |             | -0.0241 | (0.0114)**  | -0.0300 | (0.0142)**  | -0.0229 | (0.0119)*   | -0.0334 | (0.0148)**  |
| ln(Pop.)                            |          |             | 0.0383  | (0.0168)**  | 0.0254  | (0.0143)*   | 0.0355  | (0.0169)**  | 0.0228  | (0.0150)    |
| New FL                              |          |             | 0.0002  | (0.0003)    | 0.0005  | (0.0004)    | 0.0002  | (0.0003)    | 0.0004  | (0.0004)    |
| Shog. Ally                          |          |             | -0.0313 | (0.0179)*   | 0.0071  | (0.0161)    | -0.0298 | (0.0181)    | 0.0025  | (0.0164)    |
| Goshi                               |          |             | 0.0210  | (0.0257)    | 0.0181  | (0.0261)    | 0.0219  | (0.0261)    | 0.0267  | (0.0269)    |
| lat                                 |          |             |         |             | -0.5827 | (0.2993)*   |         |             | -0.5190 | (0.3153)    |
| lon                                 |          |             |         |             | -0.1909 | (0.0774)**  |         |             | -0.1758 | (0.0819)**  |
| ln(Elev.)                           |          |             |         |             | -0.0125 | (0.0160)    |         |             | -0.0147 | (0.0166)    |
| ln(Rugg.)                           |          |             |         |             | 0.0406  | (0.0159)**  |         |             | 0.0379  | (0.0158)**  |
| ln(Dist.<br>Edo/Ky-<br>oto)         |          |             |         |             | -0.0379 | (0.0160)**  |         |             | -0.0450 | (0.0174)**  |
| Rainy                               |          |             |         |             | -1.6127 | (0.9138)*   |         |             | -1.6134 | (0.9218)*   |
| Snowy                               |          |             |         |             | 0.5457  | (1.8746)    |         |             | 0.6470  | (1.8827)    |
| Lat $\times$ Lon                    |          |             |         |             | 0.0046  | (0.0022)**  |         |             | 0.0041  | (0.0023)*   |
| Kin Nbs.                            |          |             |         |             |         |             | 0.0421  | (0.0304)    | 0.0422  | (0.0296)    |
| Nb. FL                              |          |             |         |             |         |             | 0.0000  | (0.0001)    | 0.0002  | (0.0001)    |
| ln(Nb.<br>Rice)                     |          |             |         |             |         |             |         |             | -0.0210 | (0.0187)    |
| ln(Nb.<br>Area)                     |          |             |         |             |         |             | -0.0015 | (0.0083)    | 0.0245  | (0.0181)    |
| ln(Nb.<br>Pop.)                     |          |             |         |             |         |             |         |             | -0.0135 | (0.0210)    |
| Const.                              | 0.4155   | (0.0091)*** | 0.1846  | (0.1241)    | 25.1677 | (10.5108)** | 0.2119  | (0.1509)    | 23.4706 | (11.0924)** |
| <i>Fit Statistics</i>               |          |             |         |             |         |             |         |             |         |             |
| Num.Obs.                            | 232      |             | 230     |             | 230     |             | 230     |             | 230     |             |
| R2                                  | 0.077    |             | 0.144   |             | 0.376   |             | 0.152   |             | 0.395   |             |

Table A.65: Geographic Exposure and Reported Tax Rates by Neighbor Population Density

**Note:** The coefficients represent OLS estimates of the relationship between the average distance ratio measure of each domain interacted with the population density of its neighbors (low vs. high) and the domain tax rate. Heteroskedasticity-robust standard errors in parentheses. Signif. Codes: .\*\*\*: 0.01, .\*\*: 0.05, .\*: 0.1.

#### E.3.4. Fixed-Neighbor-Set and HMX Kernel Diagnostics

The outside-option interaction in [Figure 7](#) and [Table A.53](#) weights  $\hat{y}_{-j}$  toward neighbors whose capitals are closest to home-domain villages, the same geometry that defines  $\ln(\rho_j)$ . The moderator may therefore vary mechanically with exposure: domains with high  $\ln(\rho_j)$  can have a different weighted neighbor set than domains with low  $\ln(\rho_j)$ . The interaction could reflect that compositional shift rather than moderation by outside-option quality.

[Table A.66](#) addresses this concern with two complementary checks. The first breaks the mechanical link. Panel A reconstructs  $\hat{y}_{-j}$  as an unweighted simple mean of neighbor labor-absorption capacity ( $p_j^{\text{labor}}$ ) over all domains within 100 km of  $j$ 's capital, regardless of the nearest-foreign-capital weights used in the main text. Under this construction the asymmetry survives in the expected direction: in the fully-controlled Model 4, the  $\ln(\rho_j)$  slope is -0.0172 ( $p = 0.0077$ ) where neighbors have high labor demand, compared with -0.0074 ( $p = 0.3153$ ) where they do not, consistent with the main-text result.

The second check asks whether the asymmetry is an artifact of the linear interaction form. Following [Hainmueller, Mummolo, and Xu \(2019\)](#), Panels B and C estimate local-linear associations between  $\ln(\rho_j)$  and  $\tau_j$  across the continuous moderator support, using an Epanechnikov kernel with 95% confidence intervals. Under both the paper's  $\hat{y}_{-j}$  and the fixed-100 km measure, the point estimate is negative at all three reported moderator quartiles, but every 95% confidence interval includes zero. The curves show no precisely estimated monotonic pattern, and the small sample ( $N \approx 230$ ) limits resolution. This diagnostic preserves the negative sign in point estimates without establishing continuous moderation or a causal relationship. [Figure A.11](#) plots the curves.

| Panel A: Fixed-Neighbor-Set $\hat{y}_{-j}$ (100 km Unweighted Mean)                       |                                             |                   |          |          |
|-------------------------------------------------------------------------------------------|---------------------------------------------|-------------------|----------|----------|
| Moderator level                                                                           | (1)                                         | (2)               | (3)      | (4)      |
| $\ln(\rho_j) \times \hat{y}_{-j}^{\text{Low}}$                                            | -0.0258                                     | -0.0220           | -0.0079  | -0.0074  |
| Robust SE                                                                                 | (0.0064)                                    | (0.0069)          | (0.0071) | (0.0073) |
| $p$                                                                                       | 0.0001                                      | 0.0015            | 0.2621   | 0.3153   |
| $\ln(\rho_j) \times \hat{y}_{-j}^{\text{High}}$                                           | -0.0197                                     | -0.0165           | -0.0159  | -0.0172  |
| Robust SE                                                                                 | (0.0072)                                    | (0.0069)          | (0.0066) | (0.0064) |
| $p$                                                                                       | 0.0067                                      | 0.0180            | 0.0167   | 0.0077   |
| Num. Obs.                                                                                 | 231                                         | 229               | 229      | 229      |
| $R^2$                                                                                     | 0.075                                       | 0.142             | 0.374    | 0.393    |
| Panel B: HMX Kernel Marginal Effect at Moderator Quartiles (paper's $\hat{y}_{-j}$ )      |                                             |                   |          |          |
| Moderator quartile                                                                        | $\hat{\partial}\tau_j/\partial \ln(\rho_j)$ | 95% CI            | $p$      |          |
| 25th                                                                                      | -0.0119                                     | [-0.0249, 0.0011] | 0.0722   |          |
| 50th                                                                                      | -0.0123                                     | [-0.0261, 0.0014] | 0.0785   |          |
| 75th                                                                                      | -0.0139                                     | [-0.0321, 0.0042] | 0.1324   |          |
| Panel C: HMX Kernel Marginal Effect at Moderator Quartiles (fixed-100 km $\hat{y}_{-j}$ ) |                                             |                   |          |          |
| 25th                                                                                      | -0.0099                                     | [-0.0227, 0.0029] | 0.1288   |          |
| 50th                                                                                      | -0.0102                                     | [-0.0251, 0.0048] | 0.1813   |          |
| 75th                                                                                      | -0.0142                                     | [-0.0409, 0.0125] | 0.2984   |          |

Table A.66: Tax-Exposure Interaction: Fixed-Neighbor-Set and HMX Kernel Marginal Effects

**Note:** Panel A re-estimates the interaction specifications 1-4 of Table A.53 after replacing the nearest-foreign-weighted outside-option proxy  $\hat{y}_{-j}$  with an unweighted mean of  $p_j^{\text{labor}}$  over all domains within 100 km of  $j$ 's capital. This breaks the mechanical link between the capital geometry that defines  $\ln(\rho_j)$  and the neighbor set that defines  $\hat{y}_{-j}$  flagged by the methodologist referee. Under the fully-controlled Model 4, the  $\hat{y}_{-j}^{\text{High}}$  slope is -0.0172 ( $p = 0.0077$ ), versus -0.0074 ( $p = 0.3153$ ) for  $\hat{y}_{-j}^{\text{Low}}$ . Panel B and Panel C report the Hainmueller-Mummolo-Xu (2019) kernel-smoothed marginal effect of  $\ln(\rho_j)$  on  $\tau_j$  at the 25th, 50th, and 75th percentiles of the continuous moderator support, estimated by local linear regression with an Epanechnikov kernel after residualizing against the Model 4 control set. The estimated marginal effect is negative through the middle of the moderator support under either definition (Figure A.11 plots the full curve). The kernel estimates do not establish a monotonic gradient.

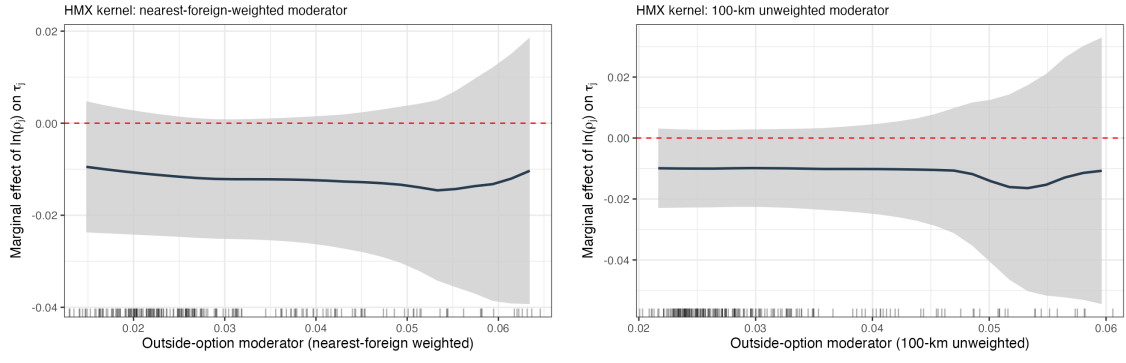


Figure A.11: HMX Kernel-Smoothed Marginal Effect of  $\ln(\rho_j)$  on  $\tau_j$

**Note:** Local-linear-kernel marginal effect of  $\ln(\rho_j)$  on the tax rate  $\tau_j$ , plotted from the 5th through the 95th percentile of the outside-option moderator  $\hat{y}_{-j}$ . Epanechnikov kernel with bandwidth equal to the interquartile range of the moderator; covariates residualized using the Model 4 control set. Shaded band: 95% CI. Red dashed line: zero. Rug: moderator-value density within the plotted range. Left: paper's nearest-foreign-weighted  $\hat{y}_{-j}$ . Right: fixed-100 km unweighted  $\hat{y}_{-j}$ . Both curves are negative over the middle of the moderator range, with the CI narrowest around the median where the support is densest.